IGCSE Physics Workbook Topic 1-4

2024 · Paper 2 (Multiple Choice) + Paper 4 (Theory) mixed by topic

Name: _____

Class: _____

Cambridge IGCSE Physics 0625 — 2024 (Feb/Mar, May/Jun, Oct/Nov)

练习册

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1 MOTION, FORCES AND ENERGY

85 questions · Paper 2 & Paper 4

Vocabulary · Topic 1 MOTION, FORCES AND ENERGY

词汇表（中英对照）

speed	速度	**moment**	力矩
velocity	速度（矢量）	**pivot**	支点
acceleration	加速度	**centre of gravity**	重心
distance	距离	**equilibrium**	平衡
displacement	位移	**momentum**	动量
scalar	标量	**impulse**	冲量
vector	矢量	**conservation of momentum**	动量守恒
mass	质量	**energy**	能量
weight	重量	**kinetic energy**	动能
gravitational field strength	重力场强度	**gravitational potential energy**	重力势能
density	密度	**elastic potential energy**	弹性势能
force	力	**work done**	做功
resultant force	合力	**power**	功率
friction	摩擦力	**efficiency**	效率
air resistance	空气阻力	**pressure**	压强
drag	阻力	**renewable**	可再生
spring constant	弹簧常数	**non-renewable**	不可再生
extension	伸长量		

1.1 Physical quantities and measurement techniques

8 MCQ + 0 theory · Paper 2 & Paper 4

Date: _____ Score: _____

1 In which row are quantities correctly categorised into scalar quantities and vector quantities?

	scalar quantities	vector quantities
A	mass and energy	weight and acceleration
B	gravitational field strength and time	force and electric field strength
C	speed and momentum	distance and force
D	distance and energy	velocity and temperature

0625/21/M/J/24 Q1

1 A student uses a metre ruler to measure the length of a sheet of paper.

Which measurement is shown to the nearest millimetre?

A 0.2932 m **B** 0.293 m **C** 0.29 m **D** 0.3 m

0625/21/O/N/24 Q1

1 A student has a measuring cylinder containing water and also has a balance.

Which of these could she use to find the volume of a small metal sphere?

She has no other apparatus.

A either the measuring cylinder containing water or the balance

B the measuring cylinder containing water only

C the balance only

D neither the measuring cylinder nor the balance

0625/22/F/M/24 Q1

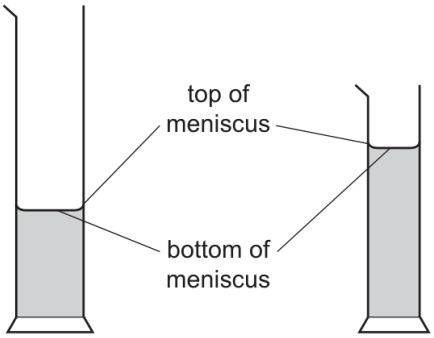
Date: _____ Score: _____

4 Which row contains one scalar quantity and one vector quantity?

	quantity 1	quantity 2
A	energy	velocity
B	mass	time
C	momentum	weight
D	distance	temperature

0625/22/F/M/24 Q4

1 A student wishes to measure accurately the volume of approximately 40 cm^3 of water. She has two measuring cylinders, a larger one that can hold 100 cm^3, and a smaller one that can hold 50 cm^3. The water forms a meniscus where it touches the glass.



Which cylinder and which water level does the student use to ensure an accurate result?

	cylinder	water level
A	larger one	bottom of meniscus
B	larger one	top of meniscus
C	smaller one	bottom of meniscus
D	smaller one	top of meniscus

0625/22/M/J/24 Q1

1 A student measures the volume of a small, irregularly shaped stone.

Which apparatus must be used?

A a ruler and a measuring cylinder containing water

B a measuring cylinder containing water only

C a ruler and an empty measuring cylinder

D a ruler only

0625/23/M/J/24 Q1

Date: _____ Score: _____

2 Which quantity is a vector?

A electric field strength

B energy

C mass

D temperature

0625/23/M/J/24 Q2

1 Which list contains only vector quantities?

A acceleration, distance, speed

B electric field strength, momentum, weight

C energy, mass, temperature

D force, time, velocity

0625/23/O/N/24 Q1

1.2 Motion

10 MCQ + 6 theory · Paper 2 & Paper 4

Date: _____ Score: _____

7 A stone is dropped from a tall tower and falls a distance of 50 m to the ground.

The stone has a mass of 3.0 kg.

At which speed does the stone hit the ground?

A 17 m / s **B** 31 m / s **C** 54 m / s **D** 150 m / s

0625/21/M/J/24 Q7

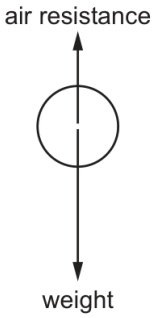
2 A rocket travels with an average speed of 6 km / s for 2 minutes.

What is the distance travelled by the rocket?

A 12 km **B** 50 km **C** 720 km **D** 12 000 km

0625/21/O/N/24 Q2

3 The diagram shows the vertical forces acting on a ball as it falls vertically through the air. The ball does not reach terminal velocity.

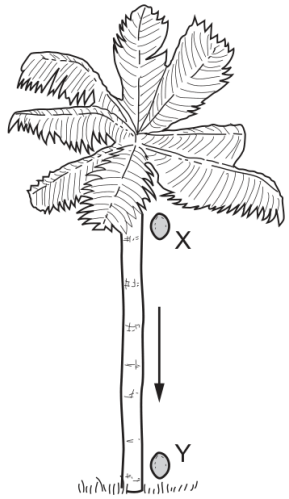


Which row describes what happens to the resultant force on the ball and what happens to the acceleration of the ball as it falls through the air?

	resultant force	acceleration
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

0625/21/O/N/24 Q3

4 A coconut falls from a palm tree. At X, it has just started falling. Y is the point just before it hits the ground.



What is the acceleration of the coconut at X?

(Air resistance can be ignored.)

A zero

B less than that at Y

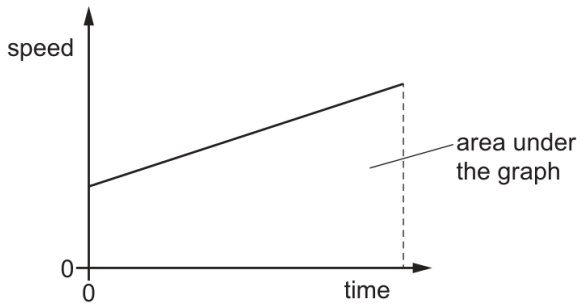
C the same as that at Y

D more than that at Y

0625/21/O/N/24 Q4

Date: _____ Score: _____

2 The motion of an object is represented by the speed–time graph shown.



Which quantity is equal to the area under the graph?

A acceleration

B average speed

C distance travelled

D kinetic energy

0625/22/M/J/24 Q2

2 A boy takes 30 minutes to cycle a distance of 8.0 km. He then walks a further distance of 2.0 km in 15 minutes.

What is his average speed?

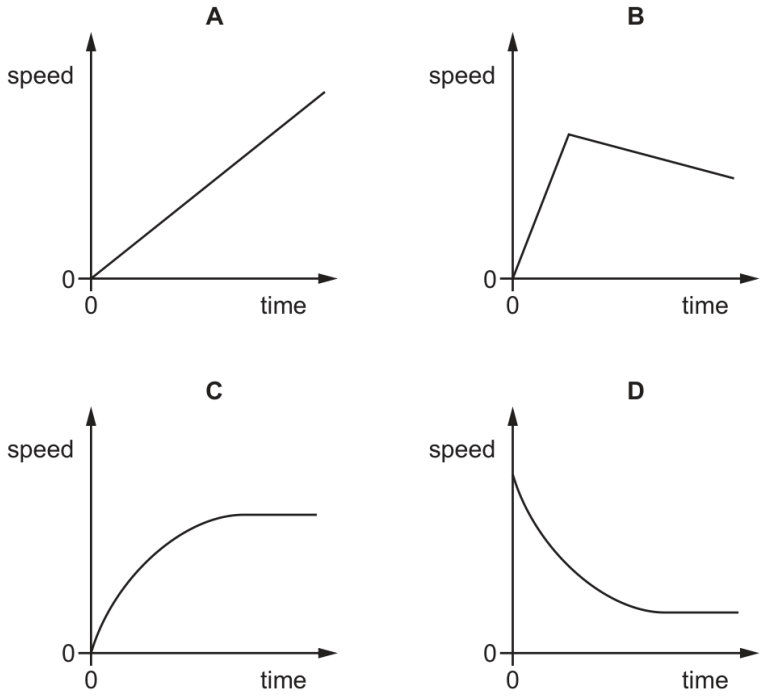
A 4.5 km / h **B** 5.6 km / h **C** 12 km / h **D** 13 km / h

0625/22/O/N/24 Q2

Date: _____ Score: _____

3 A small, light ball is dropped from the top of a tall building.

Which graph shows how the speed of the ball changes with time?



0625/22/O/N/24 Q3

Date: _____ Score: _____

3 An athlete runs 2.4 km in 12 minutes.

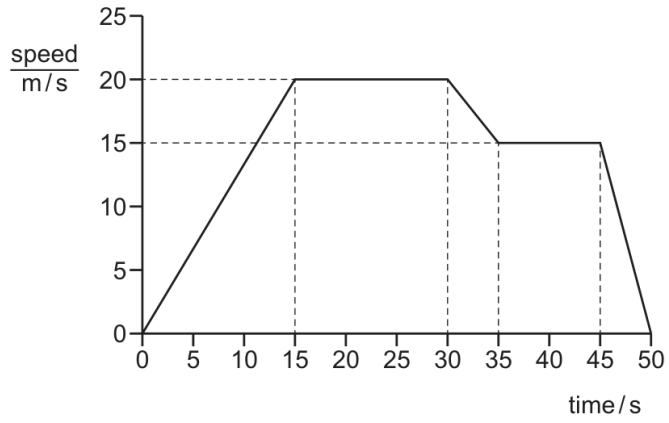
What is the average speed of the athlete?

A 0.20 m / s **B** 3.3 m / s **C** 29 m / s **D** 200 m / s

0625/23/M/J/24 Q3

Date: _____ Score: _____

4 The graph shows how the speed of a car changes during a period of 50 s.

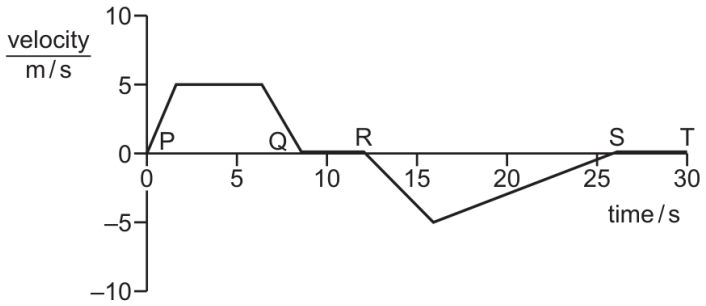


Which row gives the car's greatest acceleration and the car's greatest deceleration?

	greatest acceleration m/s^2	greatest deceleration m/s^2
A	0.75	1.0
B	0.75	3.0
C	1.3	1.0
D	1.3	3.0

0625/23/M/J/24 Q4

2 The graph shows how the velocity of an elevator varies with time. The elevator starts from the third floor of a tall building. Point P indicates the start of the elevator's journey.



When is the elevator on the third floor again?

A Q to R and S to T

B Q to R only

C S to T only

D The lift does not return to the third floor.

0625/23/O/N/24 Q2

1 A long tube contains oil. A small ball is held at rest at the surface of the oil. At time $t = 0$, the ball is released and begins to fall vertically through the oil.

Fig. 1.1 shows the ball falling through the oil.

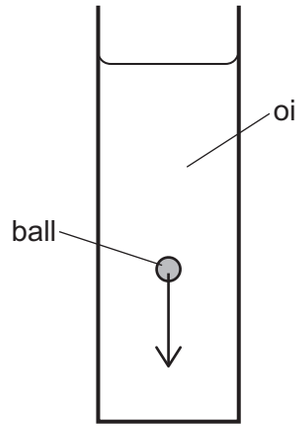


Fig. 1.1

As the ball begins to fall through the oil, it accelerates.

(a) Define acceleration.

.....

..... [1]

(b) The mass of the ball is 0.0075 kg.

Calculate the resultant force acting on the ball when it is accelerating downwards at $2.8\,m/s^2$.

resultant force = ... [2]

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(c) As the ball falls, its speed v is recorded. Fig. 1.2 is the speed–time graph for the falling ball.

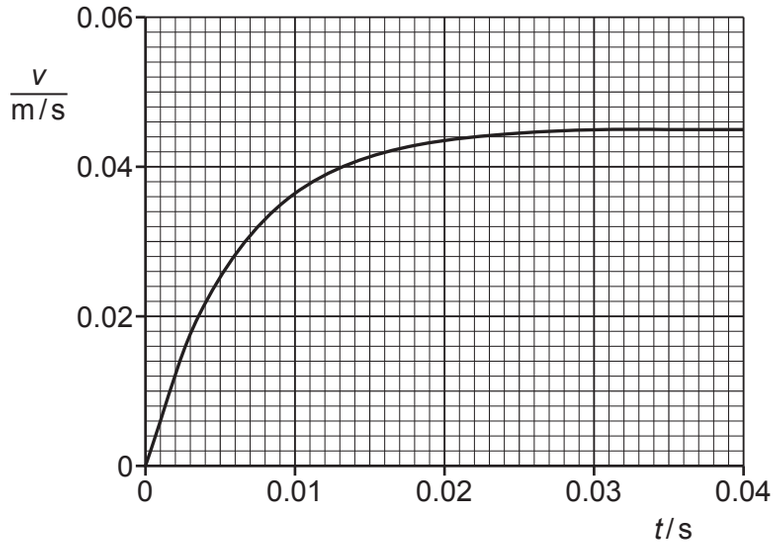

Fig. 1.2

(i) Describe what happens to the acceleration between $t = 0$ and $t = 0.040$ s.

Explain why this happens.

.....

.....

.....

..... [4]

(ii) By drawing a tangent on Fig. 1.2, determine a value for the acceleration of the ball at $t = 0.010$ s.

acceleration = ... [3]

[Total: 10]

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[Turn over

2 A drag car is a racing car that is powered by a rocket engine.

A drag car accelerates uniformly from rest until it reaches the finishing line. The engine is then switched off and a parachute opens. The car decelerates until it stops.

Fig. 2.1 shows a drag car decelerating after a race.

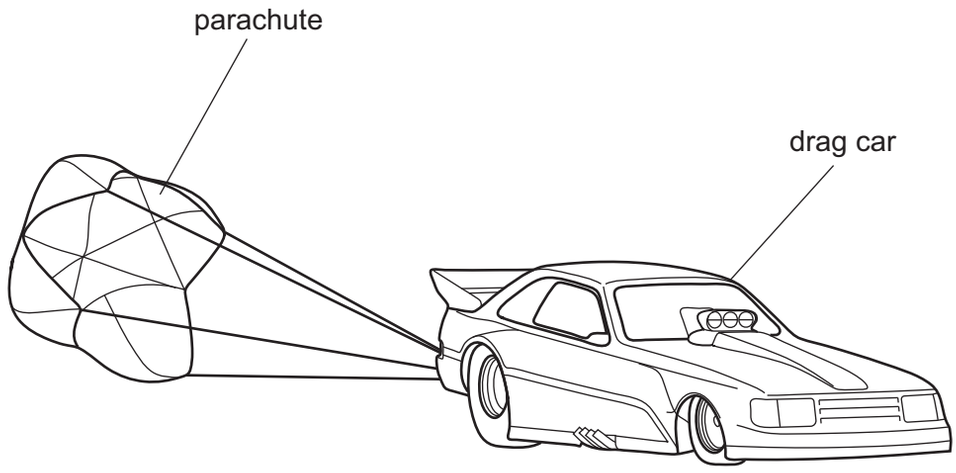

Fig. 2.1

This drag car has a mass of 1400 kg.

Fig. 2.2 is the speed–time graph for the car during a race on a straight horizontal track.

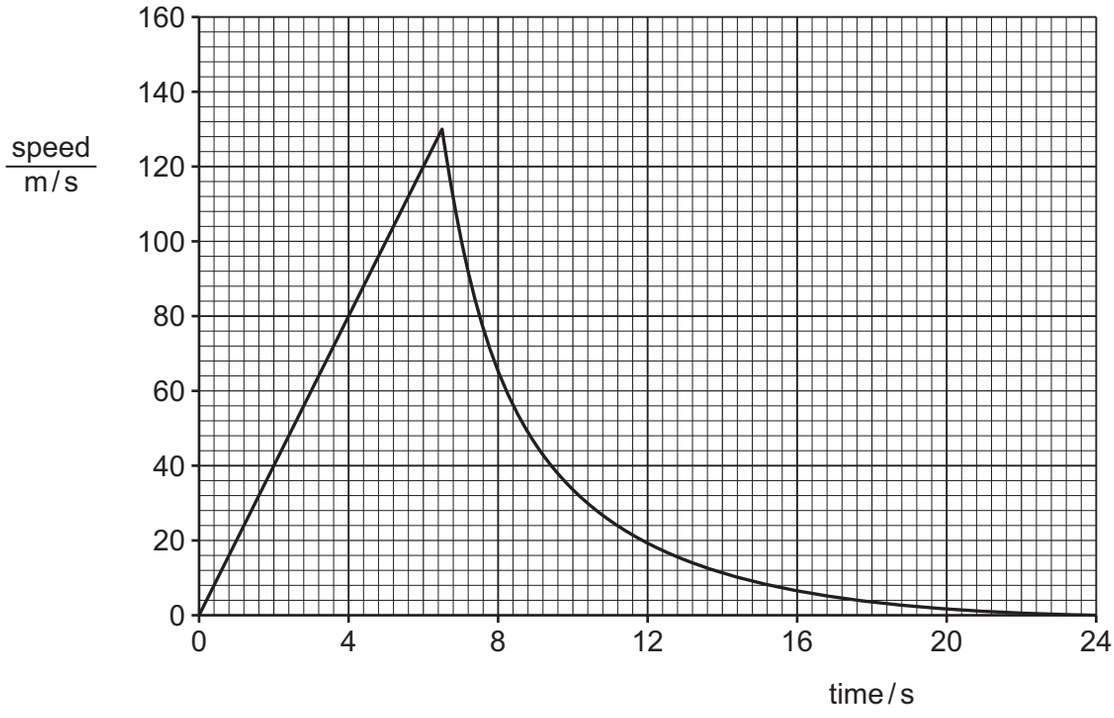

Fig. 2.2

The car reaches its maximum speed of 130 m / s at a time of 6.5 s.

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(a) **(i)** Calculate the maximum momentum of the car during the race.

maximum momentum = ... [2]

(ii) State the feature of Fig. 2.2 that represents the distance travelled by the car.

.....

..... [1]

(iii) Determine the distance travelled by the car in the first 6.5 s.

distance = ... [2]

(b) The parachute opens at 6.5 s and the car decelerates.

Describe how Fig. 2.2 shows that, after 6.5 s:

(i) the car decelerates

.....

..... [1]

(ii) the deceleration of the car is **not** constant.

.....

..... [1]

(c) Describe the energy transfer that takes place as the car slows down.

.....

..... [2]

[Total: 9]

1 **(a)** Fig. 1.1. is a speed–time graph for the first 5 minutes of a bus journey.

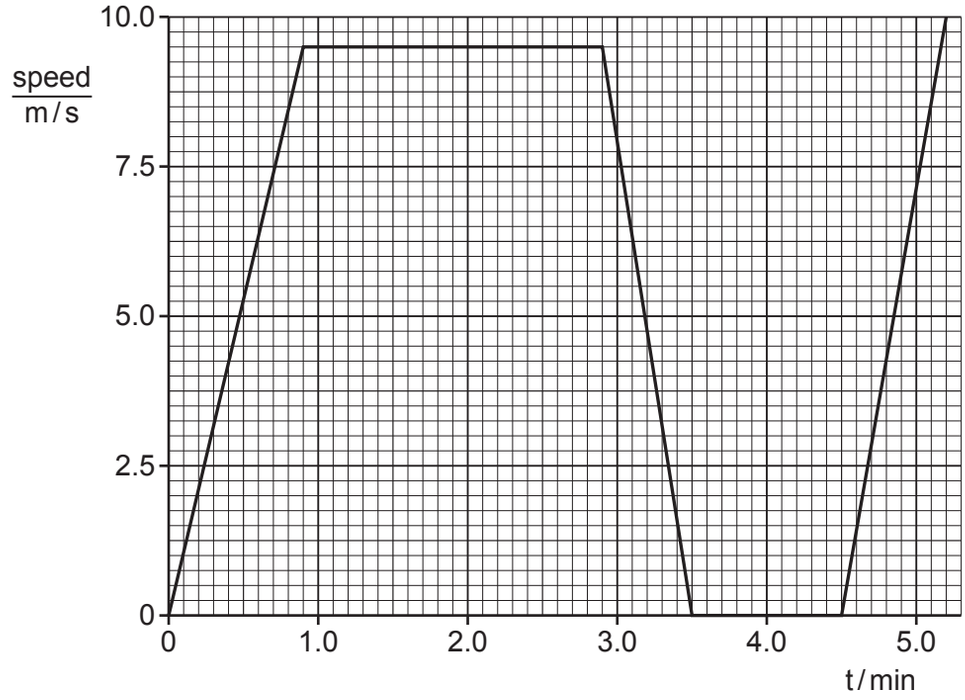

Fig. 1.1

Describe the motion between:

1. t = 0.90 min and t = 2.9 min .....

2. t = 2.9 min and t = 3.5 min .....

3. t = 3.5 min and t = 4.5 min .....

[3]

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(b) Another bus travels at a speed of 8.9 m / s. The brakes apply a constant force and the bus stops in a distance of 23 m. This bus has a mass of 18 000 kg.

(i) Calculate the kinetic energy of the bus before the brakes are applied.

kinetic energy = ... [2]

(ii) Calculate the force applied to stop the bus.

force = ... [3]

[Total: 8]

* 0019655392604 *

2 **(a)** Define acceleration.

.....

..... [1]

(b) A train has a total mass of 520 000 kg. The train accelerates at 1.1 m / s^2.

(i) Calculate the time taken for the train to increase its speed from 15 m / s to 28 m / s.

time = ... [2]

(ii) Calculate the force required to produce an acceleration of 1.1 m / s^2 for this train.

force = ... [2]

(iii) The train uses electric motors.

Explain why the force on the train due to the motors is greater than the value calculated in **(ii)**.

.....

..... [1]

[Total: 6]

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1 **(a)** A rocket has an initial mass of 7.4×10^6 kg.

(i) Calculate the initial weight of the rocket.

weight = ... [1]

(ii) Define, in words, the term weight.

.....

..... [1]

(b) Fig. 1.1 shows part of the speed-time graph for the rocket as it leaves the ground and travels into space.

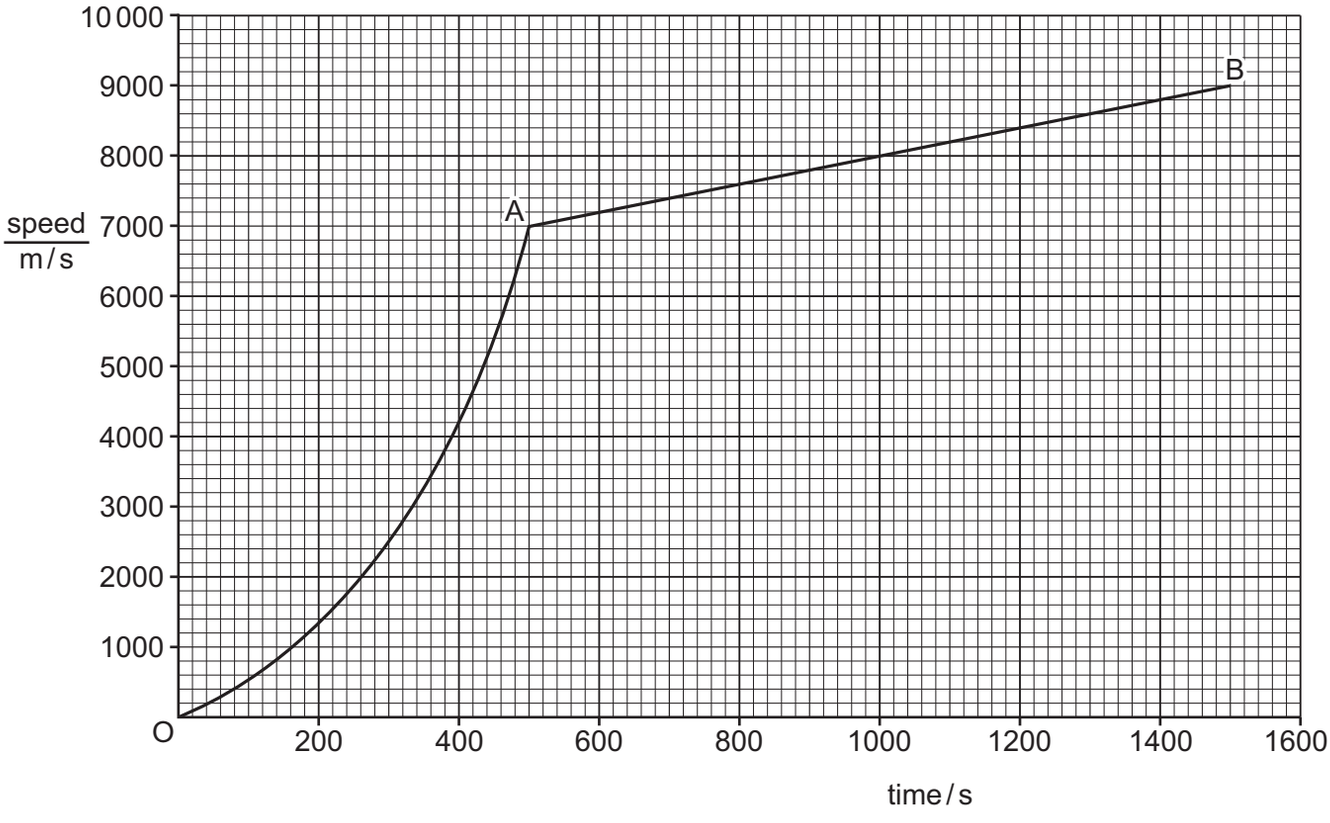

Fig. 1.1

(i) Describe the motion of the rocket:

From O to A .....

From A to B .....
[2]

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(ii) Draw a tangent to the graph at time = 400 s and use this to calculate the acceleration of the rocket at this time. Show your working.

acceleration = ... [2]

(c) Rockets are used to launch satellites into space. When the satellite is released, the rocket returns to the Earth.

Explain in terms of forces why the rocket reaches terminal velocity as it travels through the atmosphere back to the Earth.

.....

.....

..... [2]

[Total: 8]

2

1 A ball of mass 130 g is launched from the ground at an initial velocity of 14 m / s vertically upwards. It decelerates until it is at rest momentarily at a height *h* above the ground.

(a) Define deceleration.

.....

..... [2]

(b) The acceleration of free fall is 9.8 m / s^2.

Show that the time taken for the ball to reach height *h* is 1.4 s. Ignore the effect of air resistance.

[1]

(c) Calculate *h*. Ignore the effect of air resistance.

h = ... [3]

(d) The ball is dropped from the top of a tall building.

Describe and explain the motion of the ball as it falls. Consider the effect of air resistance in your answer.

.....

.....

.....

..... [3]

[Total: 9]

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1.3 Mass and weight

6 MCQ + 0 theory · Paper 2 & Paper 4

Date: _____ Score: _____

2 Which name is given to the quantity of matter in an object?

A density

B mass

C volume

D weight

0625/21/M/J/24 Q2

3 A body is moved from place X to place Y where the gravitational field strength is different.

What happens to its mass and to its weight due to the move?

	mass	weight
A	changes	changes
B	changes	stays the same
C	stays the same	changes
D	stays the same	stays the same

0625/21/M/J/24 Q3

3 An astronaut of mass 80 kg is standing on a planet with gravitational field strength $g = 3.8\,\text{N/kg}$.

What is the weight of the astronaut on this planet?

A 780 N **B** 300 N **C** 210 N **D** 21 N

0625/22/M/J/24 Q3

Date: _____ Score: _____

4 Which statement about gravitational field strength is **not** correct?

A It changes the mass of an object.

B It is equivalent to the acceleration of free fall.

C It is measured in N / kg.

D Its magnitude is different on other planets in our Solar system.

0625/22/O/N/24 Q4

5 Which statement about gravitational field strength g is correct?

A $g = \frac{\text{mass}}{\text{weight}}$

B $g = \text{mass} \times \text{weight}$

C g is defined as the mass per unit force

D g is equal to the acceleration of free fall

0625/23/M/J/24 Q5

3 An object of mass 1.0 kg is at rest on the Earth. An identical object is at rest on a planet with a gravitational field strength of twice that on the Earth.

Which row correctly compares the object on the planet to the object on the Earth?

	its weight	its acceleration when the same horizontal resultant force is applied
A	double	equal to that on the Earth
B	double	half that on the Earth
C	half	equal to that on the Earth
D	half	half that on the Earth

0625/23/O/N/24 Q3

1.4 Density

5 MCQ + 0 theory · Paper 2 & Paper 4

Date: _____ Score: _____

4 The mass of an object is 2.5 kg. The volume of the object is 480 cm^3.

What is the density of the object?

A $5.2 \times 10^{-3}\,g/cm^3$

B $5.2\,g/cm^3$

C $190\,g/cm^3$

D $1.2 \times 10^{3}\,g/cm^3$

0625/21/M/J/24 Q4

5 The table gives data about four liquids, W, X, Y and Z.

liquid	mass / g	volume / cm^3
W	10	10
X	20	15
Y	30	50
Z	40	50

None of the liquids mixes with any of the other liquids.

All the liquids are put in the same container and settle to form four separate layers.

Which statement is correct?

A W forms the layer below X.

B X forms the top layer.

C Y forms the bottom layer.

D Z forms the layer below Y.

0625/22/F/M/24 Q5

4 A sealed container of volume 2000 cm^3 contains air at high pressure.

The container is placed on a top-pan balance.

The balance reads 200.00 g.

All the air is removed by a vacuum pump and the balance reading changes to 196.00 g.

What was the density of the pressurised air?

A 0.00200 g/cm^3

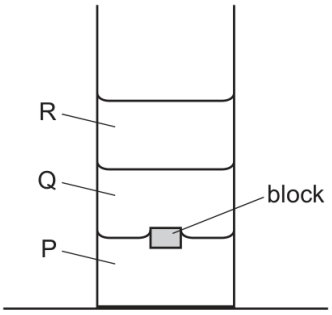
B 0.098 g/cm^3

C 4.00 g/cm^3

D 10.2 g/cm^3

0625/22/M/J/24 Q4

5 Three liquids P, Q and R have different densities and do not mix. The liquids are placed in a measuring cylinder and allowed to settle. A small block is then dropped into the measuring cylinder and comes to rest, as shown.



Which statement about the density of the block is correct?

A It is equal to the density of Q.

B It is greater than the density of P.

C It is greater than the density of R.

D It is less than the density of Q.

0625/22/O/N/24 Q5

Date: _____ Score: _____

4 An engineer researched the density of different liquids being used as coolants.

liquid	density / kg/m^3
X	785
Y	1020
Z	1250

None of the liquids mix with water.

Each liquid is poured, in turn, into a container of water. The density of water is 1000 kg/m^3.

Which liquids rise to the surface?

A X only **B** Y and Z **C** Y only **D** Z only

0625/23/O/N/24 Q4

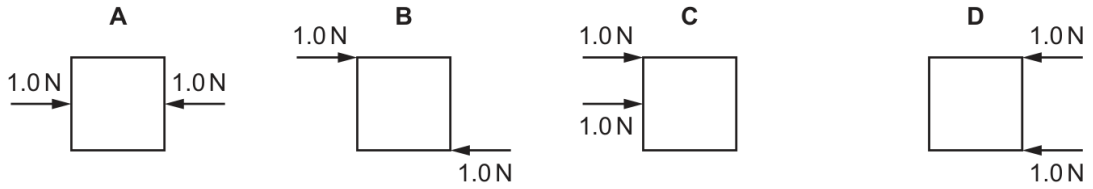
1.5 Forces

11 MCQ + 6 theory · Paper 2 & Paper 4

Date: _____ Score: _____

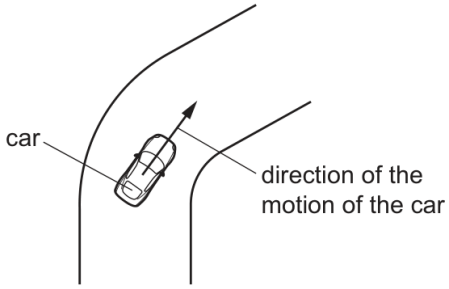
5 Four objects each have two forces acting on them.

Which object is in equilibrium?



0625/21/M/J/24 Q5

5 A car is driven round a bend in the road at a constant speed.



What is the direction of the resultant force on the car when it is going round the bend?

A parallel to the motion and in the same direction as the motion

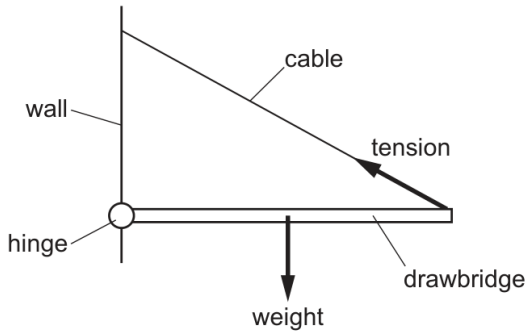
B parallel to the motion and in the opposite direction to the motion

C perpendicular to the motion and towards the inside of the bend

D perpendicular to the motion and towards the outside of the bend

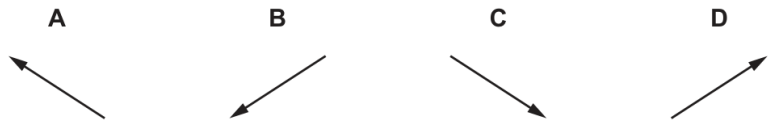
0625/21/O/N/24 Q5

6 The diagram shows a drawbridge that is attached to a wall by a hinge at one end and a cable at the other.



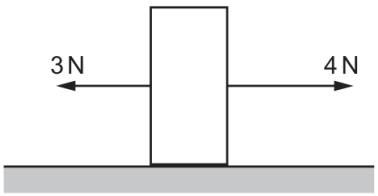
The weight of the drawbridge and the tension in the cable are represented by the labelled arrows in the diagram. There is a third force at the hinge, which is not shown. The drawbridge is in equilibrium.

Which arrow shows a possible direction for the force at the hinge?



0625/21/O/N/24 Q6

2 The diagram shows a solid object on a flat surface, with two forces acting on the object.



What is the resultant force on the object?

A 1 N to the left

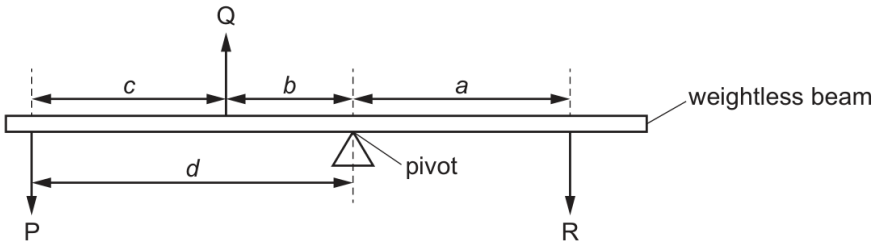
B 1 N to the right

C 7 N to the left

D 7 N to the right

0625/22/F/M/24 Q2

6 A weightless beam is balanced on a pivot as shown. Forces P, Q and R act on the beam.



Which equation is correct?

A $(P \times c) + (Q \times b) = (R \times a)$

B $(P \times c) - (Q \times b) = (R \times a)$

C $(P \times d) + (Q \times b) = (R \times a)$

D $(P \times d) - (Q \times b) = (R \times a)$

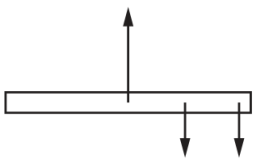
0625/22/F/M/24 Q6

Date: _____ Score: _____

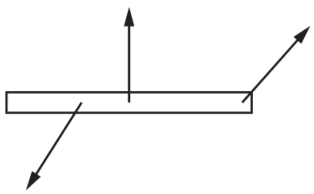
5 Each diagram shows three forces on a beam.

In which situation is it possible for the three forces shown to be in equilibrium?

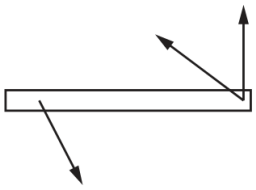
A



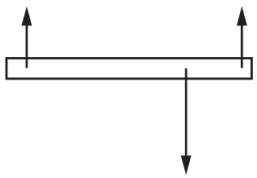
B



C



D



0625/22/M/J/24 Q5

1 Two forces with magnitudes F_1 and F_2 act on an object at right angles to each other.

What is the magnitude of the resultant force?

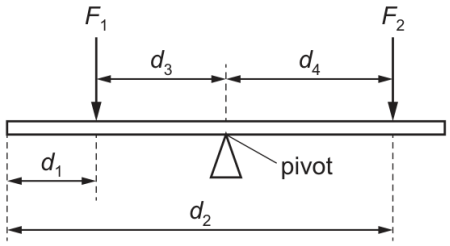
A $\sqrt{F_1^2 + F_2^2}$ **B** $\sqrt{F_1} + \sqrt{F_2}$ **C** $\sqrt{F_1^2} + \sqrt{F_2^2}$ **D** $\sqrt{F_1 + F_2}$

0625/22/O/N/24 Q1

Date: _____ Score: _____

6 The diagram shows a uniform beam pivoted at its centre.

A student applies two forces, F_1 and F_2, as shown.



Which row is correct?

	clockwise moment about the pivot	anticlockwise moment about the pivot
A	F_2d_2	F_1d_1
B	F_2d_2	F_1d_3
C	F_2d_4	F_1d_1
D	F_2d_4	F_1d_3

0625/22/O/N/24 Q6

6 Which property of an object **cannot** be changed by applying forces?

A mass

B shape

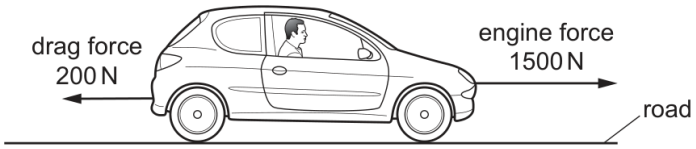
C speed

D volume

0625/23/M/J/24 Q6

5 The diagram shows a car moving along a road.

The force due to the engine is 1500 N and the total drag force is 200 N.



What is the motion of the car?

A decreasing speed forward

B increasing speed forward

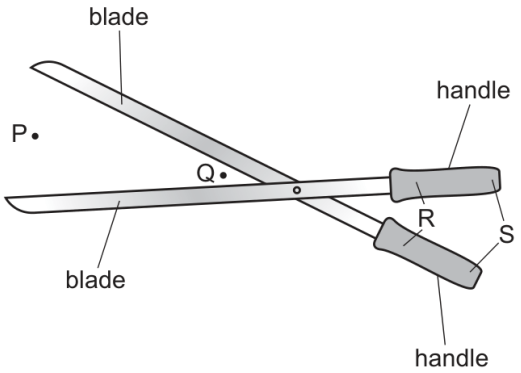
C reversing

D constant speed

0625/23/O/N/24 Q5

Date: _____ Score: _____

6 A pair of cutters is used to cut a rope.



Where should the rope be positioned and at which labelled points should the hands be positioned to produce the greatest cutting force?

	rope positioned	hands positioned
A	P	R
B	P	S
C	Q	R
D	Q	S

0625/23/O/N/24 Q6

* 0000800000002 *

1 A spring is suspended from a clamp. Fig. 1.1 shows a pointer attached to the lower end of the spring.

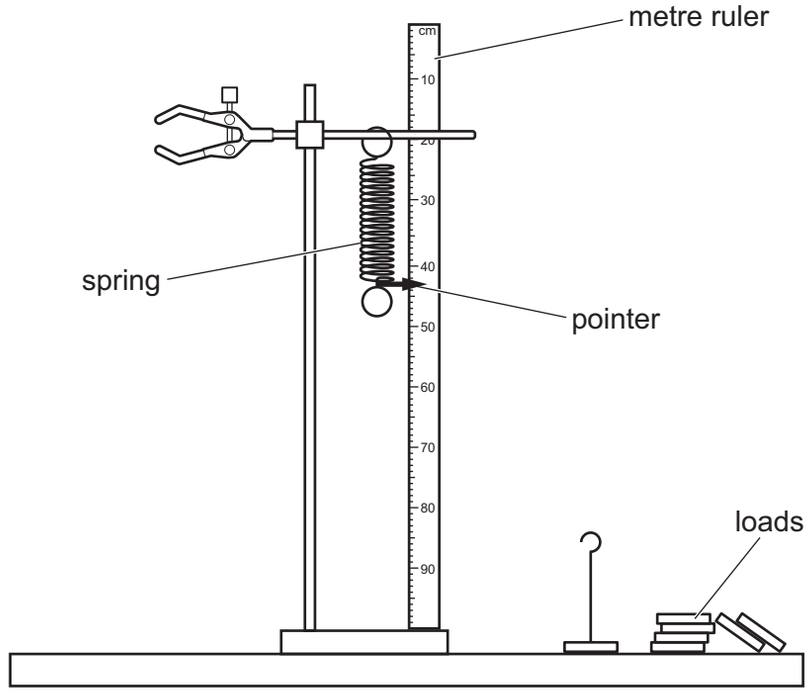


Fig. 1.1

A student suspends loads of different weights from the spring and records the readings on the metre ruler.

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Fig. 1.2 is the reading–weight graph that the student obtains.

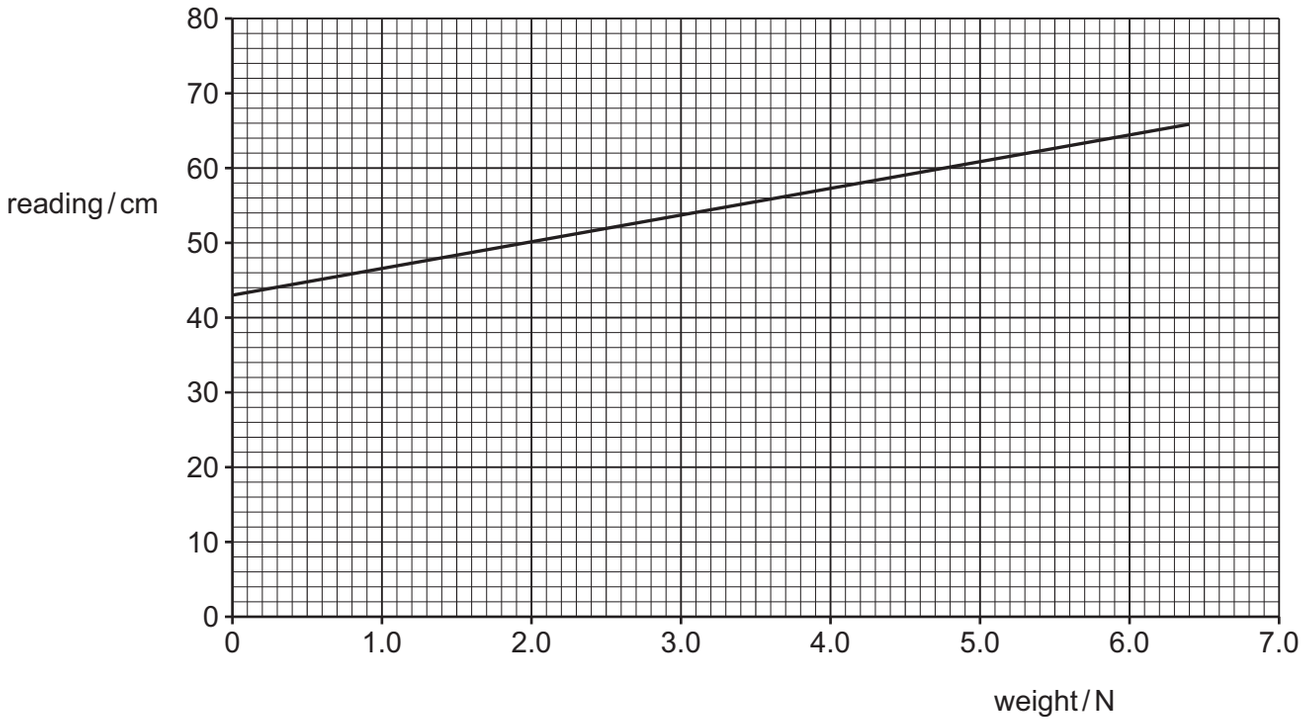

Fig. 1.2

(a) (i) Using Fig. 1.2, determine the reading on the metre ruler when

1. no weight is attached to the spring .....

2. a weight of 5.6 N is attached to the spring .....

[1]

(ii) Calculate the **extension** of the spring when the weight attached is 5.6 N.

extension = ... [1]

(b) Using the values found in **(a)**, calculate the spring constant of the spring.

spring constant = ... [2]

(c) An object of mass 0.50 kg is attached to the spring.

(i) Calculate the weight of the object.

weight = ... [1]

(ii) The object is pulled downwards until the tension in the spring is 6.5 N.

The object is released.

Calculate the acceleration of the object immediately after it is released.

acceleration = ... [3]

[Total: 8]

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* 0000800000008 *

3 **(a)** Define the moment of a force and describe the effect that it measures.

.....

.....

.....

..... [3]

(b) A large rectangular block of stone has a square base of side 3.4 m. Fig. 3.1 shows the block at rest on a horizontal surface.

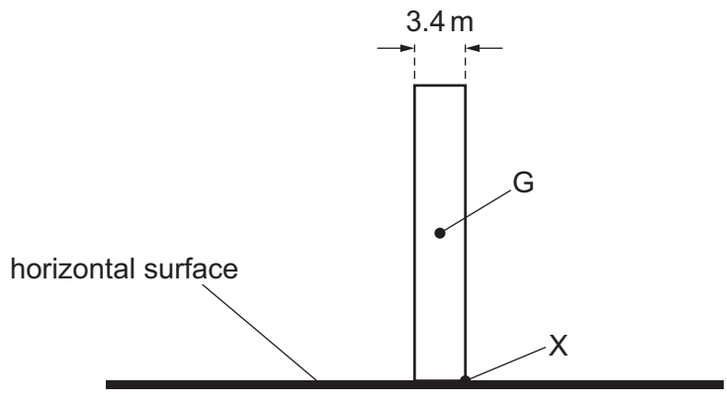

Fig. 3.1

The block is of uniform density and the centre of gravity G is at its centre.

(i) Explain what is meant by centre of gravity.

.....

..... [1]

(ii) The weight of the block is 1.3×10^7 N.

Calculate the moment of the weight of the block about corner X.

moment of weight = ... [2]

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(c) The block shown in Fig. 3.1 is in equilibrium.

State the **two** different conditions that apply when an object is in equilibrium.

1 .....

2 .....

[2]

[Total: 8]

* 0019655392602 *

1 A load is suspended from a thread. The vertical force on the thread due to the load is 0.75 N.

(a) Calculate the mass of the load.

mass = ... [2]

(b) Fig. 1.1 shows the load suspended from the thread.

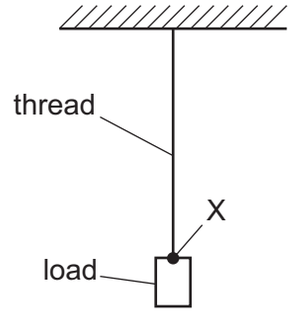

Fig. 1.1

A wire is attached to the load at point X and pulled horizontally to the right.

The tension in the horizontal wire is 1.2 N.

By drawing a scale diagram or by calculation, determine:

- the magnitude of the resultant of the force at X due to the load and due to the tension in the wire
- the direction of the resultant relative to the vertical direction.

Show your working.

magnitude of resultant force = ... N

direction of resultant relative to vertical = ... °

[4]

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(c) Forces may produce changes in the size and the shape of an object.

State **two** other changes that forces may produce.

1 .....

2 .....

[2]

[Total: 8]

3 Fig. 3.1 shows two children balanced on a seesaw. A seesaw is a length of wood which rotates about a central pivot.

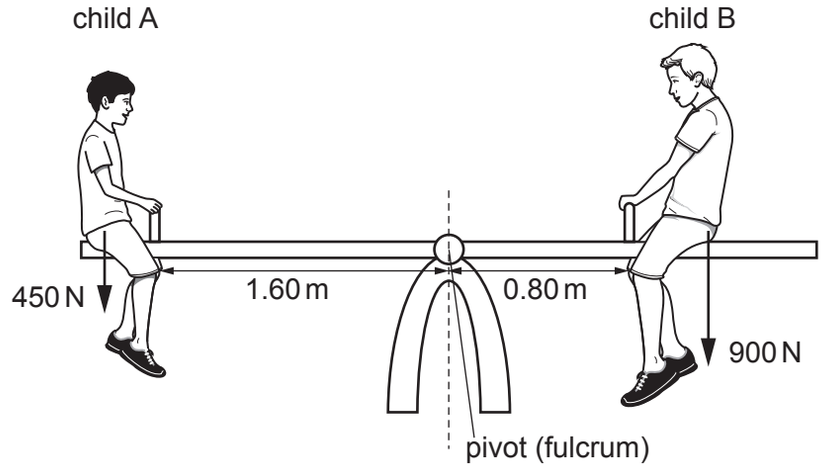

Fig. 3.1

(a) Child B moves 0.050 m further away from the pivot.

(i) Explain why the seesaw rotates clockwise.

.....

..... [1]

(ii) Child A puts on a backpack and the seesaw now balances.

Calculate the mass of the backpack.

mass = ... kg [3]

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(b) The concrete floor under the seesaw is replaced with a rubber floor. A child falls from the seesaw and experiences an impulse when they hit the floor.

(i) Define impulse.

.....

..... [1]

(ii) Explain how the rubber floor reduces injury to the child.
Use ideas about impulse, force, momentum and time in your answer.

.....

.....

.....

..... [3]

[Total: 8]

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[Turn over

* 0000800000004 *

2 **(a)** Describe an experiment to determine the spring constant of a spring.

State:

- the apparatus you need
- details of how to take measurements
- how to calculate the spring constant

You may use the space below to draw a **labelled** diagram as part of your answer.

.....

.....

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..... [4]

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(b) Fig. 2.1 shows a baby in a baby bouncer. The baby bouncer consists of a holder suspended from a spring. The baby pushes his feet on the ground and bounces gently up and down.

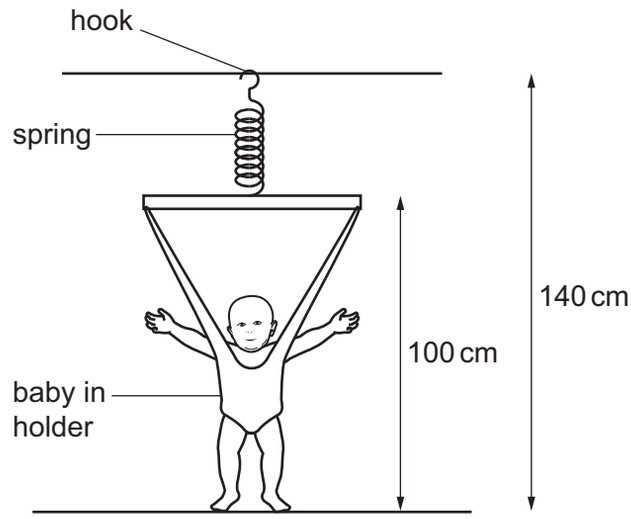

Fig. 2.1

(i) Two springs Q and R are tested to determine their spring constants.

Each spring is tested up to its limit of proportionality.

Define 'limit of proportionality'.

.....

..... [1]

(ii) Table 2.1 shows the results of the tests.

spring	spring constant / N/cm
Q	7.8
R	1.1

Table 2.1

The total weight of the baby and the holder is 120 N.

Calculate the extension of each spring for this weight.

extension of spring Q = ...

extension of spring R = ...
[1]

(iii) The unstretched length of each spring is 25 cm.

State and explain which spring would be more suitable for the baby bouncer in Fig. 2.1.

spring .....

explanation .....

.....

[1]

[Total: 7]

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7 Fig. 7.1 shows a barrier at the entrance to a car park. The wooden barrier arm has a weight of 60 N which acts through the centre of gravity at the position shown on Fig. 7.1.

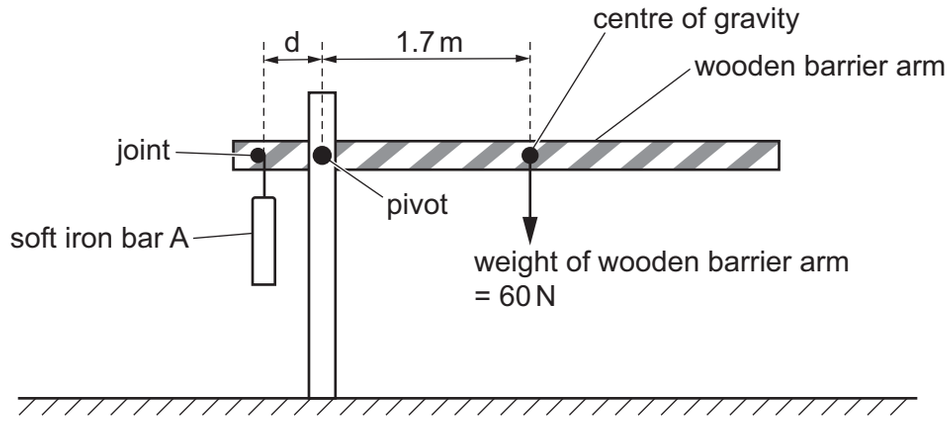

Fig. 7.1

(a) Initially the wooden barrier arm is horizontal.

(i) Using Fig. 7.1, calculate the clockwise moment of the weight of the wooden arm about the pivot.

clockwise moment = ... Nm [1]

(ii) The wooden barrier arm is in equilibrium. The mass of the soft iron bar A is 23 kg.

Calculate the distance d between the pivot and the joint holding the soft iron bar A.

distance d = ... [3]

* 0000800000016 *

(b) Fig. 7.2 shows a coil attached to a power supply placed below the soft iron bar A.

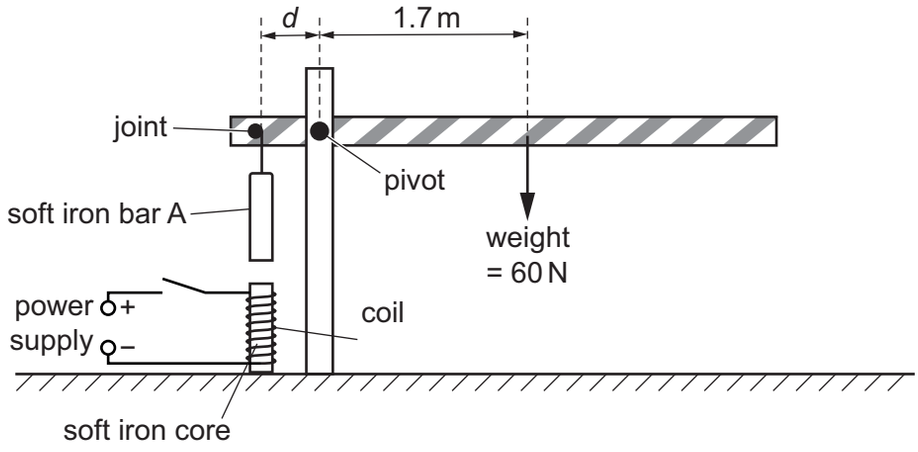

Fig. 7.2

(i) State and explain what happens to the wooden barrier arm when the switch in the coil circuit is closed.

statement .....

explanation .....

.....

.....

.....

[3]

(ii) The switch is opened. An operator decreases the potential difference across the coil and the switch is closed.

State and explain how the effect on the wooden barrier arm compares with the effect in **(b)(i)**.

statement .....

explanation .....

.....

[2]

(iii) A student suggests that the soft iron bar A is replaced by a steel bar. Explain why a steel bar is less effective than a soft iron bar in the barrier.

.....

.....

..... [2]

[Total: 11]

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1.6 Momentum

7 MCQ + 4 theory · Paper 2 & Paper 4

Date: _____ Score: _____

6 A ball of mass m falls vertically and hits a hard surface.

Its speed on hitting the surface is v_1.

It rebounds vertically upwards with speed v_2.

What is the change in momentum of the ball?

A mv_1 **B** mv_2 **C** $m(v_1 + v_2)$ **D** $m(v_2 - v_1)$

0625/21/M/J/24 Q6

7 Which equation gives the momentum change of an object?

A momentum change = $\frac{\text{force}}{\text{area}}$

B momentum change = force × distance

C momentum change = force × $\frac{\text{distance}}{\text{time}}$

D momentum change = force × time

0625/21/O/N/24 Q7

7 An object is moving at +3.0 m / s.

A force acts on the object.

After a time, the object is moving at –4.0 m / s.

The mass of the object is 5.0 kg.

What is the change in momentum of the body?

A –35 kg m / s **B** –5.0 kg m / s **C** +5.0 kg m / s **D** +35 kg m / s

0625/22/F/M/24 Q7

6 A gymnast jumps down from a high piece of apparatus.

She gradually bends her knees as she lands.

Which effect does this have?

A She will have a smaller change of momentum as she lands.

B She will lose less kinetic energy as she lands.

C She will exert a smaller impulse on the ground as she lands.

D She will experience a smaller force from the ground as she lands.

0625/22/M/J/24 Q6

7 Which expression can be used to determine the impulse on a tennis ball?

A force × time

B $\frac{\text{momentum}}{\text{time}}$

C mass × initial velocity

D mass × acceleration

0625/22/O/N/24 Q7

7 A small ball of mass 0.10 kg travels horizontally at a speed of 600 m/s. It strikes a stationary wooden block of mass 1.9 kg resting on a frictionless horizontal surface.

The ball stays in the block.

What is the speed of the ball and the block immediately after the impact?

A 30 m/s **B** 32 m/s **C** 60 m/s **D** 130 m/s

0625/23/M/J/24 Q7

7 An object has a mass of 20 kg.

Its velocity changes from 6 m/s to 10 m/s.

Which impulse has acted on the object?

A 80 Ns **B** 120 Ns **C** 200 Ns **D** 320 Ns

0625/23/O/N/24 Q7

2 Fig. 2.1 shows two identical trolleys, P and Q, held at rest on a frictionless horizontal surface. A load is fixed to trolley P.

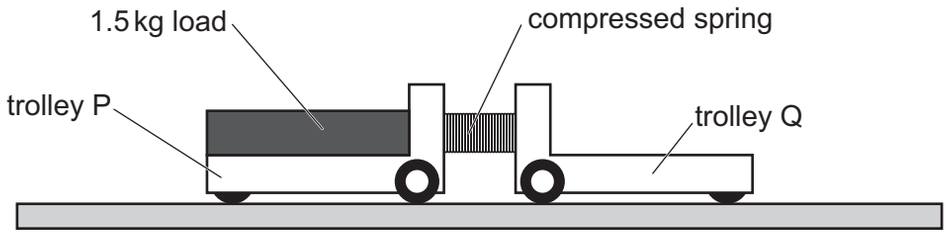

Fig. 2.1

There is a compressed spring between trolley P and trolley Q.

The trolleys are released. As the spring expands, it pushes the trolleys apart.

Trolley Q moves to the right at a constant speed of 0.36 m / s.

The mass of each trolley is 1.2 kg. The mass of the load on trolley P is 1.5 kg.

The spring has negligible mass.

(a) Calculate:

(i) the speed at which trolley P moves to the left

speed of P = ... [3]

(ii) the kinetic energy of trolley Q when it moves at 0.36 m / s.

kinetic energy of Q = ... [3]

(b) State the energy transfer that takes place as the spring expands.

.....

.....

..... [2]

[Total: 8]

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2 **(a)** Define impulse.

.....

..... [1]

(b) Fig. 2.1 shows a rocket and its exhaust gases.

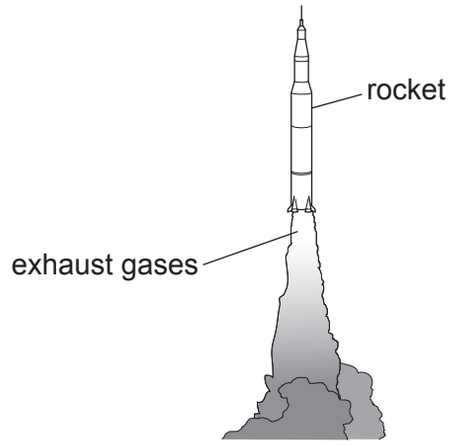

Fig. 2.1

The exhaust gases are emitted from the rocket with a velocity of 1400 m / s and at a rate of 2800 kg / s.

(i) Show that the force exerted on the rocket by the exhaust gases is 3900 kN.

State the equation you use.

[2]

(ii) Calculate the maximum mass that this force can lift from the ground. Ignore air resistance.

maximum mass = ... [3]

[Total: 6]

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* 0000800000004 *

2 Fig. 2.1 shows a golfer about to hit a golf ball with a golf club. The initial momentum of the golf ball is zero.

Fig. 2.1

(a) Define momentum.

..... [1]

(b) The golf club is in contact with the ball for 5.0×10^{-4} s. The velocity of the golf ball as it leaves the golf club is 41 m / s. The golf ball has a mass of 0.046 kg.

(i) Calculate the impulse on the golf ball.

impulse = ... [2]

(ii) Calculate the force applied to the ball by the golf club.

force = ... [2]

[Total: 5]

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1 (a) (i) State the difference between a scalar quantity and a vector quantity.

.....

..... [1]

(ii) Define momentum.

.....

..... [1]

(b) A test car crashes into a barrier to test the safety features. The test car has a total mass of 950 kg. It is moving with constant velocity from time $t = 0$ for 4.0 s. At $t = 4.0$ s, the car hits the barrier.

Fig. 1.1 shows the car as it hits the barrier.

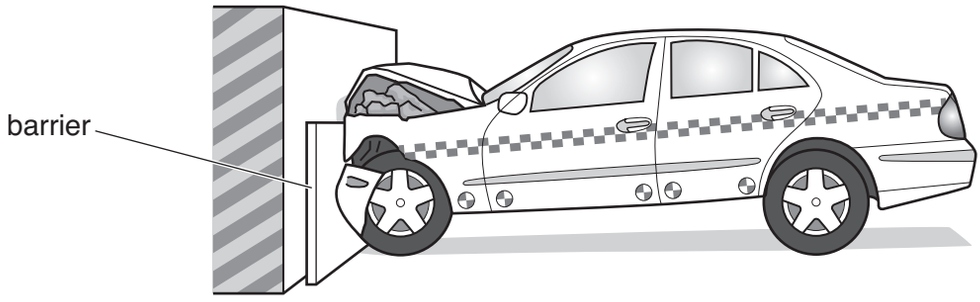

Fig. 1.1

(i) During the test crash, the resultant force acting on the car is 27 000 N. The car takes 1.5 s to come to rest. The deceleration is uniform.

Calculate the initial velocity of the car.

initial velocity = ... [3]

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(ii) On Fig. 1.2, sketch a speed–time graph to show the motion of the car from time $t = 0$ until the car becomes stationary.

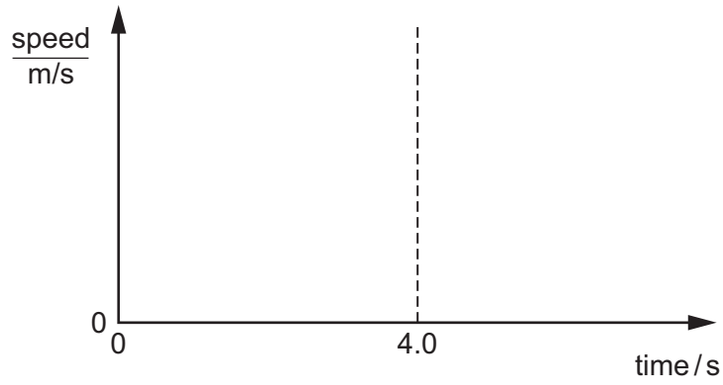


Fig. 1.2

[2]

[Total: 7]

1.7 Energy, work and power

12 MCQ + 3 theory · Paper 2 & Paper 4

Date: _____ Score: _____

8 A coal-fired power station generates electricity. Coal is burned and the energy released is used to boil water. The steam from the water makes the generator move and this produces electricity.

Which row gives the name of the energy store in the coal and the energy store of the moving generator?

	coal	generator
A	chemical	hydroelectric
B	chemical	kinetic
C	geothermal	hydroelectric
D	geothermal	kinetic

0625/21/M/J/24 Q8

8 Which statement about the use of nuclear fuel as an energy resource is correct?

A It obtains its energy from the Sun.

B It has **no** environmental impact.

C Nuclear power stations do **not** need turbines.

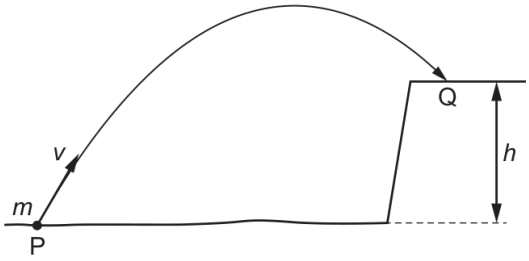
D The energy is released by nuclear fission.

0625/21/O/N/24 Q8

Date: _____ Score: _____

9 The diagram shows an object of mass m being projected with speed v from a point P to a point Q on top of a cliff height h.

The object does work W against air resistance as it travels from P to Q.



What is the kinetic energy of the object when it reaches point Q?

A $\frac{1}{2}mv^2 + mgh + W$

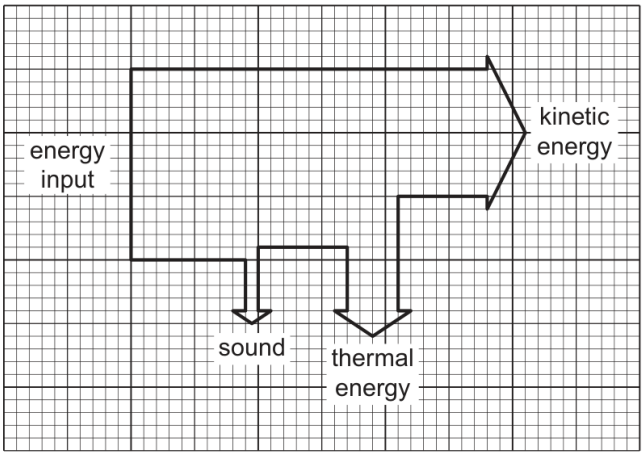
B $\frac{1}{2}mv^2 + mgh - W$

C $\frac{1}{2}mv^2 - mgh + W$

D $\frac{1}{2}mv^2 - mgh - W$

0625/21/O/N/24 Q9

10 The diagram shows a Sankey diagram for an electric drill.



scale: distance between two lines = 40 J

What is the total wasted energy shown by the diagram?

A 120 J **B** 200 J **C** 360 J **D** 560 J

0625/21/O/N/24 Q10

Date: _____ Score: _____

3 A ball is falling towards the ground.

Which transfer between energy stores is taking place?

A elastic energy to kinetic energy

B gravitational potential energy to elastic energy

C gravitational potential energy to kinetic energy

D kinetic energy to gravitational potential energy

0625/22/F/M/24 Q3

7 The table gives four energy resources and states whether the main source of energy for the resource is the Sun.

Which row is correct?

	energy resource	main source of energy is the Sun
A	geothermal	yes
B	oil	no
C	water held behind a dam	yes
D	wind	no

0625/22/M/J/24 Q7

8 Which row shows the process by which energy in the Sun is released, the process by which it is transferred to the Earth and a way in which it is stored once it reaches the Earth?

	process by which energy is released in the Sun	process by which the energy is transferred from the Sun to the Earth	a method by which the energy is stored on the Earth
A	fission	convection	biofuel
B	fusion	radiation	nuclear fuel
C	fusion	convection	nuclear fuel
D	fusion	radiation	biofuel

0625/22/O/N/24 Q8

9 An object has a mass of 20 kg.

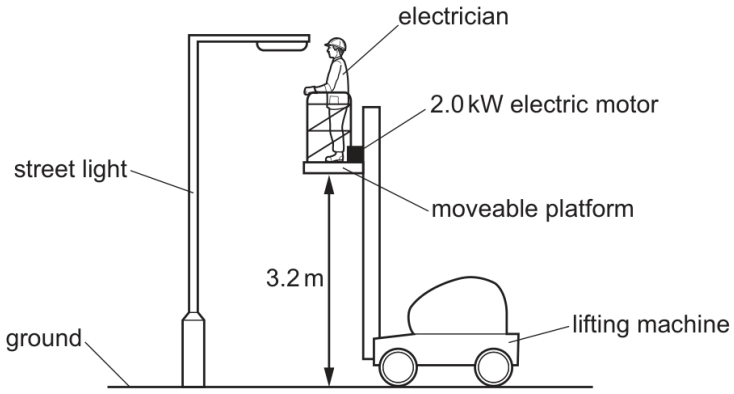
It is taken up stairs through a height of 4.0 m.

What is the increase in the store of gravitational potential energy?

A 5.0 J **B** 49 J **C** 80 J **D** 780 J

0625/22/O/N/24 Q9

10 An electrician uses a lifting machine, as shown. The lifting machine takes 4.5 s to lift the electrician a vertical height of 3.2 m.



The mass of the electrician is 72 kg.

The energy for the lifting machine is supplied by a 2.0 kW electric motor.

What is the efficiency of the lifting machine?

A 0.026 **B** 0.25 **C** 0.50 **D** 0.75

0625/22/O/N/24 Q10

8 Useful energy is obtained from different energy resources.

Four examples of energy resources are listed.

- water waves
- geothermal
- wind
- water behind hydroelectric dams

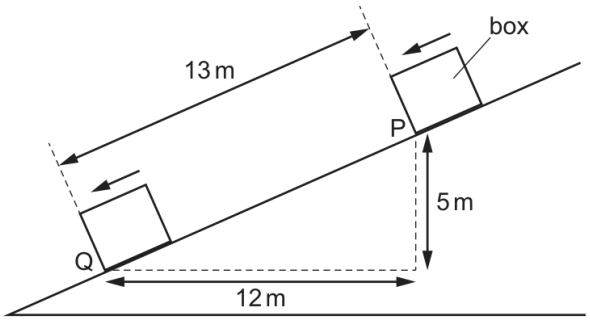
How many of these energy resources depend on radiation from the Sun?

A 1 **B** 2 **C** 3 **D** 4

0625/23/O/N/24 Q8

Date: _____ Score: _____

9 A box of mass 2.0 kg slides down a slope. The gravitational field strength is 9.8 N / kg.



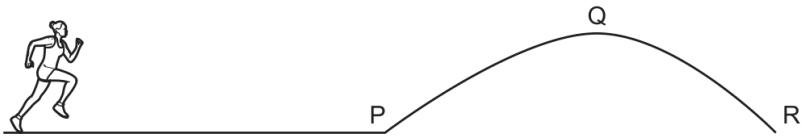
What is the decrease in the gravitational potential energy stored as the box moves from P to Q?

A 9.8 J **B** 98 J **C** 240 J **D** 250 J

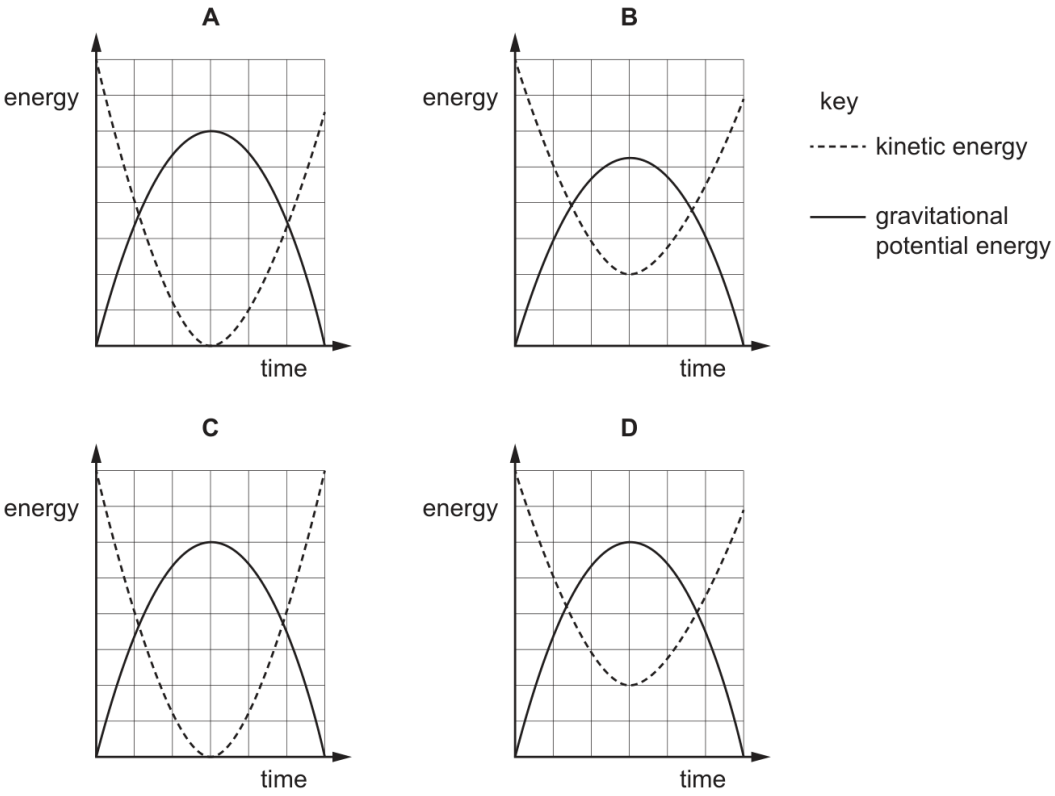
0625/23/O/N/24 Q9

Date: _____ Score: _____

10 The diagram shows the path of a long-jumper. Point R, where she lands, is on the same horizontal level as point P, from where she takes off. The long-jumper experiences air resistance as she runs and jumps.



Which graph shows how the store of kinetic energy and the store of gravitational potential energy of the long-jumper vary with time between points P and R?



0625/23/O/N/24 Q10

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3 A student drops a heavy ball from a vertical height of 1.8 m above the ground. The ball then falls to the ground. It does **not** bounce after hitting the ground.

(a) Describe the transfers of energy of the ball between stores from when the ball begins to fall to when it reaches the ground.

.....

.....

.....

.....

..... [3]

(b) Calculate the maximum speed of the ball. Ignore air resistance.

Show your working.

maximum speed = ... [3]

[Total: 6]

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3 **(a)** State **two** energy resources for which radiation from the Sun is the main source of energy.

1 .....

2 .....

[2]

(b) A wind turbine is used to generate electricity.
The useful output from the turbine in 1.0 s is 6000 J. The kinetic energy of the wind hitting the turbine in 1.0 s is 11 000 J. The velocity of the wind hitting the turbine is 6.3 m / s.

(i) Show that the mass of air hitting the turbine each second is approximately 550 kg.

[2]

(ii) Calculate the efficiency of the turbines. You may assume that all the kinetic energy stored in the wind is transferred to the turbine.

efficiency = ...% [2]

(c) Tidal energy and wind energy are both renewable energy resources.
Suggest **one** reason why tidal energy is a more useful energy resource than wind energy.

Ignore the costs of construction and maintenance.

.....

..... [1]

[Total: 7]

2 Fig. 2.1 shows solar-powered traffic warning lights.

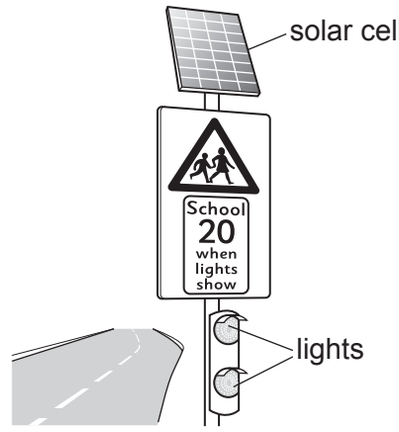

Fig. 2.1

The energy from the solar cell is stored in a battery.

(a) Name the energy store in the battery.

..... [1]

(b) The two lights in Fig. 2.1 are connected in parallel.

State **one** advantage of a parallel connection in a lighting circuit.

.....

..... [1]

(c) The efficiency of the solar cell is 22%. The power supplied to the lights by the cell is 15 W.

(i) State what is meant by 22% efficiency.

.....

..... [1]

(ii) Calculate the solar power input to the solar cell.

power = ... [2]

(d) Suggest **two** advantages of using a solar cell to power the traffic warning lights in Fig. 2.1 compared to using mains electricity.

1 .....

2 .....

[2]

[Total: 7]

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[Turn over

1.8 Pressure

6 MCQ + 1 theory · Paper 2 & Paper 4

Date: _____ Score: _____

11 An elephant of weight 45 000 N stands with all four feet on the floor.

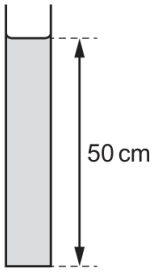
The average pressure on the floor due to the elephant foot in contact with it is 34 000 Pa.

What is the area of each foot of the elephant?

A 0.19 m^2 **B** 0.33 m^2 **C** 0.76 m^2 **D** 1.3 m^2

0625/21/O/N/24 Q11

8 The diagram shows a column of liquid. The pressure, due to the liquid, at the bottom of the liquid is 6000 N/m^2.



What is the density of the liquid?

A 310 kg/m^3

B 1200 kg/m^3

C 29 000 kg/m^3

D 120 000 kg/m^3

0625/22/F/M/24 Q8

8 The equation $\Delta p = \rho g \Delta h$ can be used for a liquid.

What is the meaning of the term ρ?

A pressure due to the liquid

B density of the liquid

C total pressure due to the liquid and the air above the liquid

D density of an object placed in the liquid

0625/22/M/J/24 Q8

Date: _____ Score: _____

11 Some altimeters use the change in air pressure to measure height.

If the pressure is 100 kPa at sea level on a particular day, what will the pressure be at the top of an 830 m hill?

(Take the density of air to be constant at 1.3 kg / m^3.)

A 1.1 kPa **B** 11 kPa **C** 89 kPa **D** 110 kPa

0625/22/O/N/24 Q11

8 An object taken underwater will be damaged if the total pressure acting on it is greater than 2.1 MPa.

What is the maximum depth that the object can be taken underwater before it breaks?

atmospheric pressure = 101 000 Pa

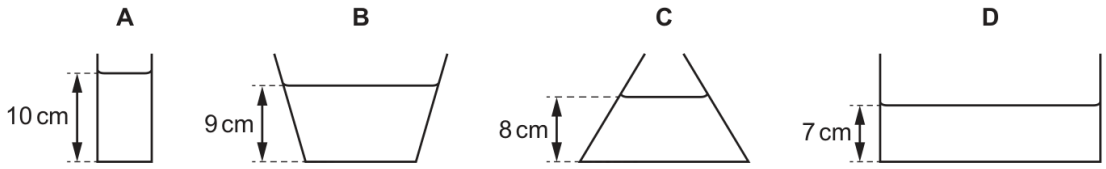
density of water = 1000 kg / m^3

A 10.3 m **B** 204 m **C** 214 m **D** 225 m

0625/23/M/J/24 Q8

11 Four tanks contain water to the depths shown.

At the bottom of which tank is the pressure due to the water greatest?



0625/23/O/N/24 Q11

3 **(a)** A car has a weight of 13 000 N. The car is supported by 4 tyres. The area of each tyre in contact with the road is 0.016 m^2.

(i) Calculate the pressure on the road due to the weight of the car.

pressure = ... [2]

(ii) Explain, in terms of particles, why the air pressure in the tyres increases when the car travels along the road.

.....

.....

.....

.....

..... [4]

(b) A gas cylinder contains helium gas at a pressure of 2.0×10^6 Pa. A volume of 0.026 m^3 of the compressed gas is released from the cylinder into balloons. Each balloon contains 0.015 m^3 of helium at atmospheric pressure (1.0×10^5 Pa). The temperature remains constant.

Calculate the maximum number of balloons that can be filled.

maximum number of balloons = ... [3]

[Total: 9]

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[Turn over

2 THERMAL PHYSICS

53 questions · Paper 2 & Paper 4

Vocabulary · Topic 2 THERMAL PHYSICS

词汇表（中英对照）

temperature	温度	**convection**	对流
thermal energy	热能	**radiation**	辐射
internal energy	内能	**infrared**	红外线
specific heat capacity	比热容	**conductor**	导体
melting	熔化	**insulator**	绝缘体
boiling	沸腾	**Brownian motion**	布朗运动
evaporation	蒸发	**particle model**	粒子模型
condensation	冷凝	**absolute zero**	绝对零度
solidification / freezing	凝固	**kelvin**	开尔文
conduction	传导	**thermal expansion**	热膨胀

2.1 Kinetic particle model of matter

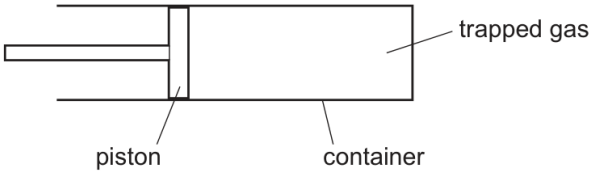
15 MCQ + 0 theory · Paper 2 & Paper 4

Date: _____ Score: _____

10 Gas of volume 200 cm^3 is trapped inside a container by a piston.

The piston is pushed to the right and the volume of the gas decreases to 100 cm^3.

The temperature of the gas remains constant.

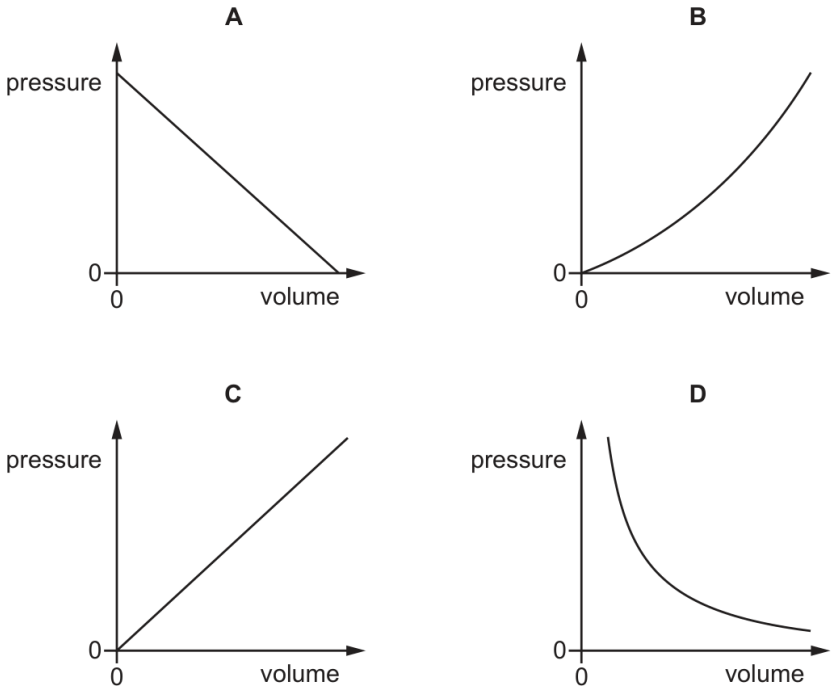


Which row states the effect that this has on the kinetic energy of the gas particles and the force per unit area exerted by the particles colliding with the inside walls of the container?

	kinetic energy	force per unit area
A	doubles	doubles
B	doubles	stays the same
C	stays the same	doubles
D	stays the same	stays the same

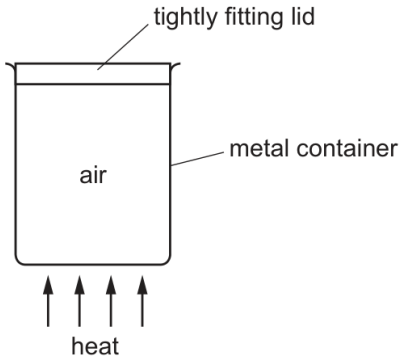
0625/21/M/J/24 Q10

11 Which graph shows the relationship between the pressure and volume of a fixed mass of gas at constant temperature?



0625/21/M/J/24 Q11

12 Some air is trapped inside a metal container with a tightly fitting lid.



The container is heated strongly behind a safety screen. The lid is blown off by the increased pressure of the air inside the container.

What causes the increase in pressure of the air inside the container?

A Each air particle expands and takes up more room.

B The air particles move more quickly.

C The number of particles inside the container increases.

D The volume occupied by the air decreases.

0625/21/O/N/24 Q12

12 When pollen grains in water are viewed through a microscope, they are seen to be in continuous, rapid, random motion.

What causes a pollen grain to move in this way?

A convection currents in the water

B bombardment by a single molecule of water

C uneven bombardment on different sides by water molecules

D collision with another pollen grain

0625/22/F/M/24 Q12

14 Matter consists of very small particles in a continual state of motion.

Which row describes the behaviour of particles in a liquid?

	force between particles	motion of particles
A	strong	move randomly at high speed
B	strong	vibrate, but change positions
C	weak	vibrate in fixed positions
D	weak	move randomly at high speed

0625/22/F/M/24 Q14

10 Extremely small pollen grains in water are viewed through a microscope. The grains are seen to move continually and randomly.

What is the reason for this random movement?

A The grains are moved by randomly moving water molecules.

B The grains are moved by random convection currents in the water.

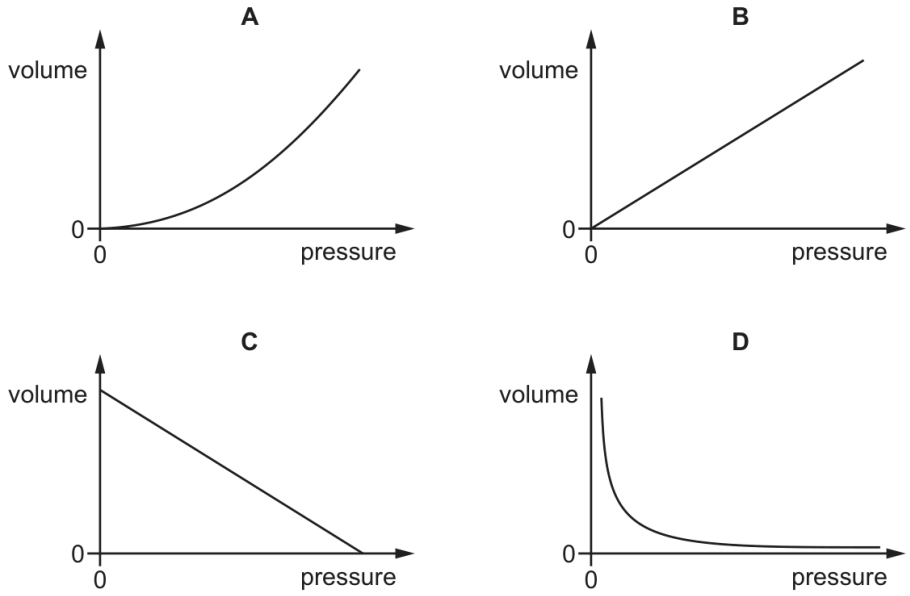
C The grains are moved by random rays of light reflecting off them.

D The grains are moved by the random motion of their own atoms.

0625/22/M/J/24 Q10

Date: _____ Score: _____

11 Which graph shows how the volume of a fixed mass of gas at constant temperature varies with its pressure?



0625/22/M/J/24 Q11

12 Water can exist in three states: solid (ice), liquid (liquid water) and gas (steam).

In which states is the motion of the water particles **only** vibrational?

A liquid water and ice

B liquid water only

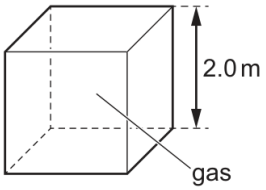
C ice only

D steam only

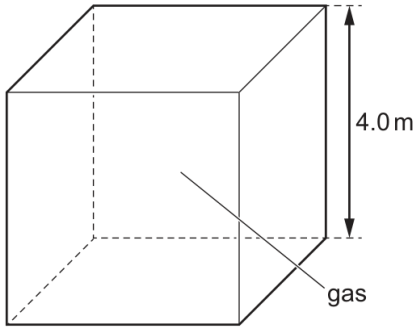
0625/22/O/N/24 Q12

Date: _____ Score: _____

13 A cube has sides 2.0 m long. The cube contains a gas at a pressure of 8000 N / m^2.



A second cube has sides 4.0 m long. This cube contains the same mass of the same gas as the first cube, at the same temperature.



What is the pressure of the gas in the second cube?

A 1000 N / m^2 **B** 4000 N / m^2 **C** 16 000 N / m^2 **D** 64 000 N / m^2

0625/22/O/N/24 Q13

9 A piece of solid metal melts to become a liquid.

How do the particles of metal or their behaviour change?

A They increase in size.

B They move around each other.

C They move much further apart.

D They vibrate more about their fixed positions.

0625/23/M/J/24 Q9

10 Brownian motion of particles is observed.

Which statements describe the movement of the particles?

1 The particles all travel along a curved path.

2 The particles move randomly.

3 The particles all travel in the same direction.

A 1 and 3 **B** 1 only **C** 2 and 3 **D** 2 only

0625/23/M/J/24 Q10

11 A fixed mass of air in a cylinder is compressed by a piston so that the volume of the air decreases at constant temperature.

How do the air particles now collide with the cylinder walls?

A less often with a greater velocity

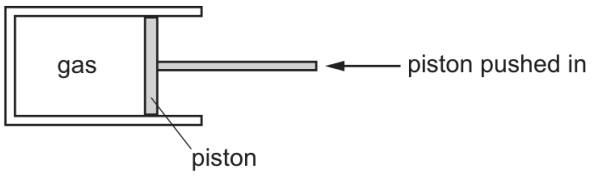
B less often with the same velocity

C more often with a greater velocity

D more often with the same velocity

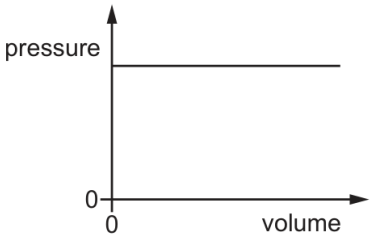
0625/23/M/J/24 Q11

12 The diagram shows a quantity of gas trapped in a cylinder. The piston is pushed in slowly and the gas is compressed. The temperature of the gas does **not** change.

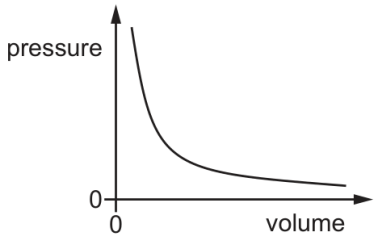


Which graph shows the relationship between the pressure and the volume of the gas?

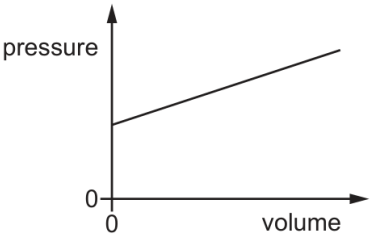
A



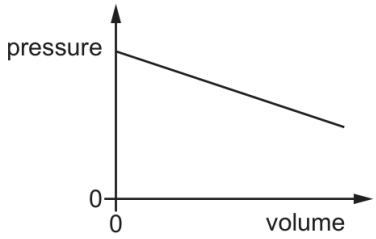
B



C



D



0625/23/M/J/24 Q12

12 A gas is heated in a sealed container.

The volume of the container does not change.

What happens to the particles of the gas?

A The average distance between particles increases.

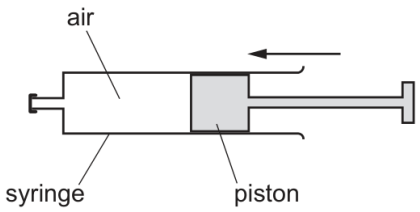
B The average kinetic energy of the particles increases.

C The mass of each particle increases.

D The volume of each particle increases.

0625/23/O/N/24 Q12

13 Air in a sealed syringe is slowly compressed by moving the piston. The temperature of the air stays the same.



Which statement about the air is correct?

A The pressure of the air decreases because its particles now travel more slowly.

B The pressure of the air decreases because the area of the syringe walls is now smaller.

C The pressure of the air increases because its particles now hit the syringe walls more frequently.

D The pressure of the air increases because its particles now travel more quickly.

0625/23/O/N/24 Q13

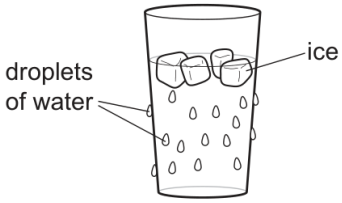
2.2 Thermal properties and temperature

21 MCQ + 5 theory · Paper 2 & Paper 4

Date: _____ Score: _____

9 A man puts some ice into a glass of water on a warm day.

After a short time, he notices that the ice disappears and that water droplets appear on the outside of the glass.



Which two changes of state are taking place?

A condensation and freezing

B condensation and melting

C boiling and melting

D freezing and evaporation

0625/21/M/J/24 Q9

12 During evaporation of a liquid, the most energetic particles escape. The temperature of the remaining liquid changes.

Which row identifies where these particles escape from and describes the temperature change?

	place from which the particles escape	temperature of remaining liquid
A	body of the liquid	decreases
B	body of the liquid	increases
C	surface of the liquid	decreases
D	surface of the liquid	increases

0625/21/M/J/24 Q12

Date: _____ Score: _____

13 Wet clothes are put out on a line to allow the water in the clothes to evaporate.

Which type of weather would cause the water to evaporate most quickly?

A a cold day with no wind

B a cold day with wind

C a hot day with no wind

D a hot day with wind

0625/21/M/J/24 Q13

13 A very hot mug of coffee is near a large unheated swimming pool holding 2.0×10^6 kg of water.

Which statement is correct?

A If the mug of coffee is tipped in the pool, the coffee will lose much more internal energy than the water in the pool gains.

B The internal energy of a substance depends only on its temperature.

C The internal energy of the water in the pool is lower than the internal energy of the mug of coffee.

D When the temperature of the mug of coffee falls, its internal energy will decrease.

0625/21/O/N/24 Q13

14 A student measures the mass of water in an open container over two hours. The container is kept in a warm room. The results are in the table.

time / hours	mass of water / g
0.0	33.9
0.5	30.6
1.0	27.6
1.5	24.9
2.0	22.5

Why does the mass of the water change?

A The water evaporates.

B The water freezes.

C The water condenses.

D The water boils.

0625/21/O/N/24 Q14

15 A 1 kg block of aluminium requires more thermal energy to raise its temperature by 1 °C than a 1 kg block of copper requires.

Why is this?

A Aluminium is a better conductor of thermal energy than copper.

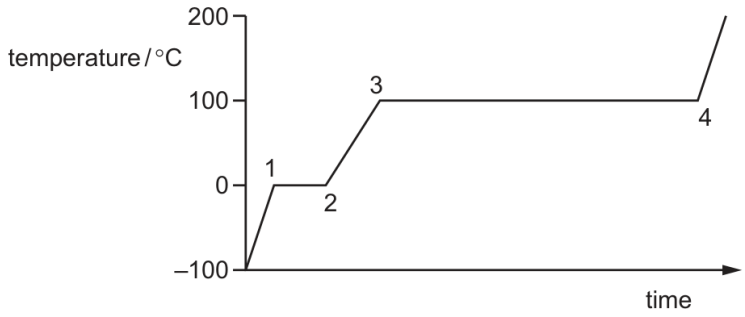
B Aluminium is a poorer conductor of thermal energy than copper.

C Aluminium has a higher specific heat capacity than copper.

D Aluminium has a lower specific heat capacity than copper.

0625/21/O/N/24 Q15

16 A block of ice is at a temperature of –100 °C. Energy is supplied at a constant rate. The graph shows how its temperature changes.



At which points have the ice completely changed state to water and all the water completely changed state to steam?

	completely changed to water	completely changed to steam
A	1	3
B	1	4
C	2	3
D	2	4

0625/21/O/N/24 Q16

9 Which statement correctly describes a change of state?

A A gas condenses to form a liquid.

B A liquid melts to form a solid.

C A solid condenses to form a liquid.

D A solid boils to form a gas.

0625/22/F/M/24 Q9

13 A circular, flat-bottomed dish contains 100 cm^3 of water.

Students are asked how to increase the rate at which the water evaporates.

Their suggestions are listed.

student 1 Pour the water into a flat-bottomed dish of larger diameter.

student 2 Cool the dish and water.

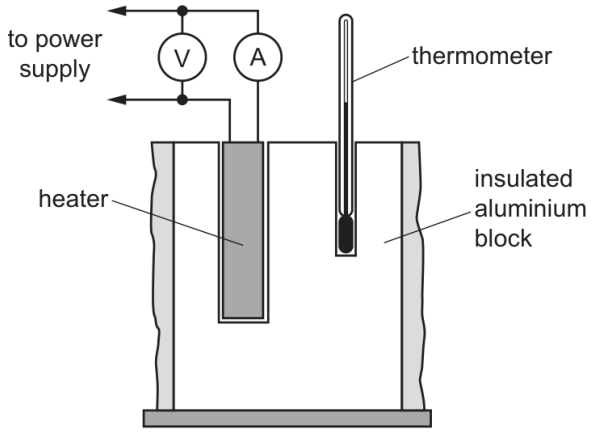
student 3 Create a current of air over the dish with a fan.

Which suggestions will increase the rate of evaporation?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

0625/22/F/M/24 Q13

15 The diagram shows the equipment used in an experiment on the heating of aluminium.



The table gives the results for the experiment.

mass of block	0.80 kg
voltmeter reading	12 V
ammeter reading	5.0 A
time	300 s

The specific heat capacity of aluminium is 900 J / (kg °C).

What is the maximum possible temperature rise in the block?

A 9 °C **B** 20 °C **C** 25 °C **D** 225 °C

0625/22/F/M/24 Q15

Date: _____ Score: _____

9 What are the correct terms for each change of state?

	liquid to solid	liquid to gas
A	solidification	condensation
B	melting	condensation
C	solidification	evaporation
D	melting	evaporation

0625/22/M/J/24 Q9

12 Which effect is caused by thermal expansion?

A a metal surface heating up in direct sunlight

B ice-cream melting on a hot day

C a railway track buckling on a hot day

D ice forming on a pond on a cold day

0625/22/M/J/24 Q12

13 The temperature of the water at the bottom of a waterfall is greater than the temperature of the water at the top.

The energy in the gravitational potential store of the water at the top is transferred to the thermal store at the bottom.

The specific heat capacity of water is 4200 J / (kg °C).

What is the temperature difference for a waterfall of height 21 m?

A 0.0050 °C **B** 0.049 °C **C** 20 °C **D** 200 °C

0625/22/M/J/24 Q13

Date: _____ Score: _____

14 An ice cube is placed in a beaker and is heated.

The ice melts to form water, which evaporates at first and then boils.

The steam condenses on a cold window in the room.

Which process involves a transfer of energy from the ice, water or steam to the surroundings?

A melting

B evaporating

C boiling

D condensing

0625/22/M/J/24 Q14

14 Which row gives the correct name for each change of state shown?

	gas to liquid	liquid to solid	solid to liquid
A	condensation	melting	solidification
B	condensation	solidification	melting
C	evaporation	melting	solidification
D	evaporation	solidification	melting

0625/22/O/N/24 Q14

15 In an experiment to measure specific heat capacity, a block of aluminium is heated and its rise in temperature is measured.

The internal energy gained by the block is ΔE. The mass of the block is m. The rise in temperature of the block is ΔT.

Which expression gives the specific heat capacity of aluminium?

A $\frac{m}{\Delta E \Delta T}$ **B** $\frac{m\Delta T}{\Delta E}$ **C** $\frac{\Delta E}{m\Delta T}$ **D** $\frac{\Delta E \Delta T}{m}$

0625/22/O/N/24 Q15

Date: _____ Score: _____

13 The liquid level in a thermometer rises when the thermometer is placed in hot water.

What causes this?

A The liquid contracts.

B The liquid evaporates.

C The liquid expands.

D The liquid freezes.

0625/23/M/J/24 Q13

14 Which statement describes what happens as ice at 0 °C melts to become water?

A Energy is absorbed and the temperature remains constant.

B Energy is absorbed and the temperature rises.

C Energy is released and the temperature remains constant.

D Energy is released and the temperature rises.

0625/23/M/J/24 Q14

15 Which row explains how increasing the surface area of a fixed volume of liquid water and blowing air over the surface speeds up evaporation?

	increasing the surface area	blowing air over the surface
A	increases the total number of water molecules present	increases the energy in the kinetic store of the liquid water
B	increases the total number of water molecules present	removes water molecules from above the surface of the water
C	increases the number of water molecules close to the surface	increases the energy in the kinetic store of the liquid water
D	increases the number of water molecules close to the surface	removes water molecules from above the surface of the water

0625/23/M/J/24 Q15

14 Which statement about the evaporation of a liquid is correct?

A The least energetic particles escape from the surface and the temperature of the liquid decreases.

B The least energetic particles escape from the surface and the temperature of the liquid increases.

C The most energetic particles escape from the surface and the temperature of the liquid decreases.

D The most energetic particles escape from the surface and the temperature of the liquid increases.

0625/23/O/N/24 Q14

Date: _____ Score: _____

15 A solid is heated causing it to expand.

Which effect does this have on its mass and on its density?

	mass	density
A	decreases	decreases
B	decreases	stays constant
C	stays constant	decreases
D	stays constant	stays constant

0625/23/O/N/24 Q15

* 0000800000010 *

4 **(a)** Describe an experiment to determine the specific heat capacity of aluminium. You may draw a diagram.

Include in your answer:

- the measurements made
- any equations needed.

.....

.....

.....

..... [4]

(b) An aluminium dish is initially at room temperature. Boiling water is poured into the aluminium dish as shown in Fig. 4.1.

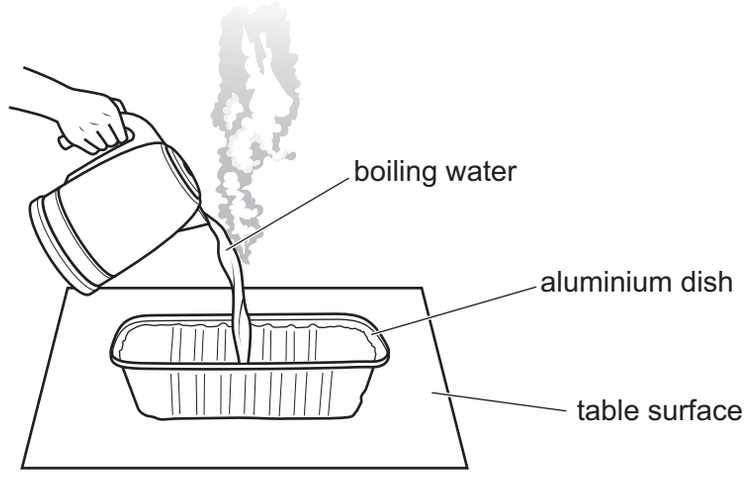


Fig. 4.1

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(i) Explain why, after a short time, the dish and the water are the same temperature.

.....

.....

.....

..... [3]

(ii) Explain, in terms of its particles, why the aluminium expands as the boiling water is poured into the dish.

.....

.....

..... [2]

(iii) The water in the dish evaporates.

Explain, in terms of the water molecules, what is meant by evaporation.

.....

.....

..... [2]

[Total: 11]

4 **(a)** Define specific heat capacity.

.....

.....

..... [2]

(b) A volume of 0.0024 m^3 of oil is heated in a pan for 7.0 min. The temperature of the oil increases from 20 °C to 180 °C.

The density of the oil is 910 kg / m^3. The specific heat capacity of the oil is 2000 J / (kg °C).

(i) Calculate the mass of oil in the pan.

mass = ... [2]

(ii) Calculate the energy required to increase the temperature of the oil.

energy = ... [2]

(iii) Calculate the power required to supply the energy calculated in **(b)(ii)**.

power = ... [2]

[Total: 8]

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* 0019655392606 *

4 **(a)** State **two** ways that evaporation differs from boiling.

1 .....

2 .....

[2]

(b) Fig. 4.1 shows part of a container used to store a mixture of liquid and gaseous oxygen.

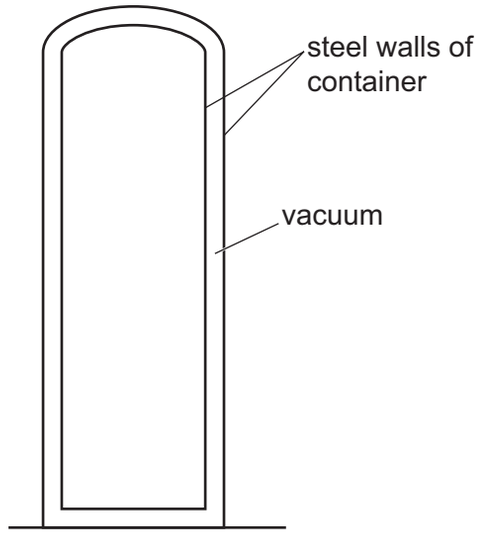

Fig. 4.1

The temperature of the liquid oxygen is –160 °C.

(i) Determine the temperature of the liquid oxygen in K.

temperature = ... K [1]

(ii) The container is made of steel and there is a vacuum between the inner and outer walls. The outer wall of the container is at room temperature.

State **two** methods of thermal energy transfer that a vacuum prevents.

1 .....

2 .....

[2]

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(c) Describe, in terms of particles, how a gas exerts a pressure on the walls of its container.

.....

.....

.....

..... [3]

[Total: 8]

* 0000800000006 *

4 Fig. 4.1 shows a pressure cooker on an electric heating element. The cooker has a tight-fitting lid.

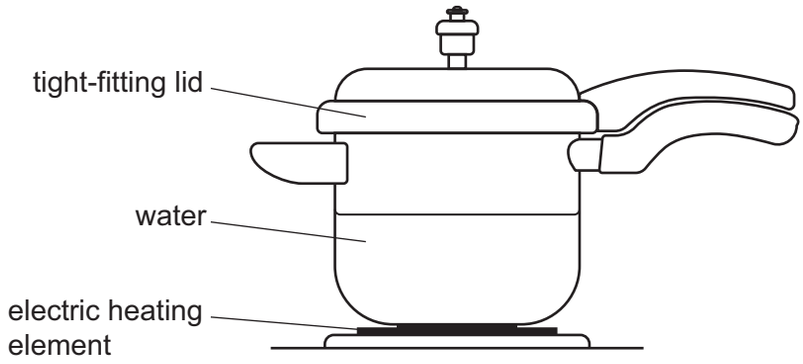

Fig. 4.1

(a) The pressure cooker is half-full of water. As the water is heated some water evaporates before the water boils.
Describe **two** differences between evaporation and boiling of the water in the cooker.

.....

.....

..... [2]

(b) As the water is heated, the pressure of the gas inside the cooker increases.
Explain this increase in pressure in terms of particles.

.....

.....

.....

.....

..... [4]

[Total: 6]

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4 Fig. 4.1 shows a stainless-steel saucepan being heated on an electric cooker. The saucepan contains water.

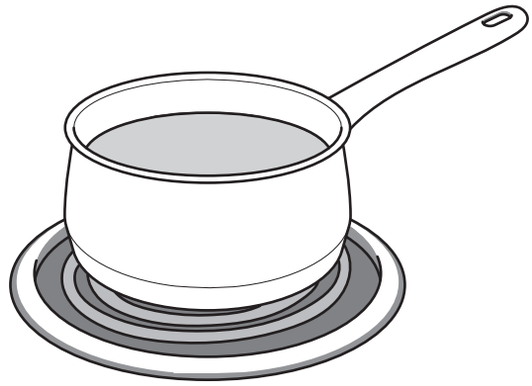

Fig. 4.1

(a) State what happens to the water particles as the water temperature increases.

..... [1]

(b) The saucepan contains 250 cm^3 of water. The specific heat capacity of water is 4200 J / (kg °C). The density of water is 1000 kg / m^3.

(i) Show that the mass of the water in the saucepan is 0.25 kg.

[2]

(ii) Calculate the energy required to increase the water temperature from 20 °C to 65 °C.

energy = ... [3]

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(iii) The heater supplies enough power to heat the water in 39 s.
A student measures the time taken to heat the water as 115 s.

Suggest why the actual time taken to heat the water is longer. Assume that the student takes accurate measurements.

.....

..... [1]

(c) The stainless-steel saucepan is replaced with an aluminium saucepan of the same mass. It contains the same volume of water.

The specific heat capacity of stainless steel is 500 J / (kg °C).
The specific heat capacity of aluminium is 890 J / (kg °C).

Explain how using an aluminium saucepan will affect the time taken to heat the water.

.....

.....

..... [2]

[Total: 9]

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[Turn over

2.3 Transfer of thermal energy

8 MCQ + 4 theory · Paper 2 & Paper 4

Date: _____ Score: _____

15 A solar water heater is designed to absorb energy from sunlight.

Which surface texture and colour would be best for the solar water heater?

A dull black

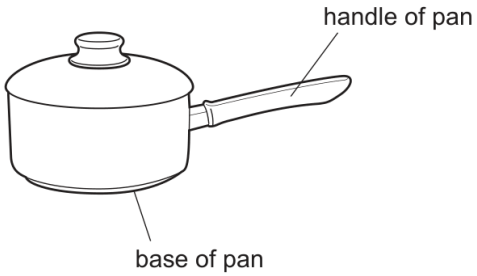
B shiny black

C dull white

D shiny white

0625/21/M/J/24 Q15

10 The diagram shows a pan used for cooking food.



Which row is correct for the materials used to make the base of the pan and the handle of the pan?

	base of pan	handle of pan
A	good thermal conductor	good thermal conductor
B	good thermal conductor	poor thermal conductor
C	poor thermal conductor	good thermal conductor
D	poor thermal conductor	poor thermal conductor

0625/22/F/M/24 Q10

Date: _____ Score: _____

11 Solar heating panels consist of pipes carrying water that absorb radiation from the Sun.

Which texture and colour is the surface of the pipes so that the temperature of the water rises at the quickest rate?

A dull black

B dull white

C shiny black

D shiny white

0625/22/F/M/24 Q11

15 A teacher shows his class a polystyrene cup. The polystyrene is a thick plastic with lots of tiny air bubbles in it.

He asks the class why the cup is so good at keeping a hot drink warm. Three suggestions are made.

1 It contains air which is a poor thermal conductor.

2 The air is trapped in tiny bubbles so very little convection is possible.

3 The plastic is a poor thermal conductor.

Which suggestions are correct?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

0625/22/M/J/24 Q15

16 Two samples of the same material have the same mass but different surface areas.

Each sample is heated to the same temperature and then left to cool to room temperature.

Each sample is allowed to cool to the same final temperature.

Which row correctly compares the decrease in internal energy and the initial rate of cooling for each sample?

	decrease in internal energy	initial rate of cooling
A	X loses more internal energy than Y	X cools down faster than Y
B	Y loses more internal energy than X	Y cools down faster than X
C	X and Y lose the same quantity of energy	X cools down faster than Y
D	X and Y lose the same quantity of energy	Y cools down faster than X

0625/22/M/J/24 Q16

16 Which statement about the transfer of thermal energy is correct?

A All metals conduct thermal energy equally well.

B Convection can only occur in solids or liquids.

C Convection occurs in liquids because hot liquid is more dense than cold liquid.

D The radiation that transfers thermal energy is a type of electromagnetic radiation.

0625/22/O/N/24 Q16

16 The table shows the state of materials P, Q, R and S, and whether each one contains free electrons.

	gas	liquid	solid	contains free electrons
P	✓	✗	✗	✗
Q	✗	✓	✗	✗
R	✗	✗	✓	✓
S	✗	✗	✓	✗

Using the information in the table, which comparison of the thermal conduction in the materials is correct?

A It is better in P than in Q and S.

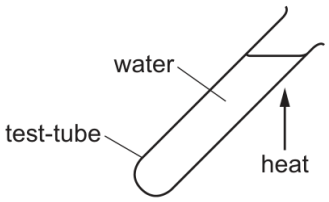
B It is better in Q than in R and P.

C It is better in R than in Q and P.

D It is better in S than in R and Q.

0625/23/M/J/24 Q16

16 A glass test-tube containing water is heated at the top. The water at the top boils, but the water at the bottom remains cold.



Which row explains why the water at the bottom of the test-tube remains cold?

	glass	water
A	good thermal conductor	good thermal conductor
B	good thermal conductor	poor thermal conductor
C	poor thermal conductor	good thermal conductor
D	poor thermal conductor	poor thermal conductor

0625/23/O/N/24 Q16

3 Fig. 3.1 shows a small block of ice floating in a beaker of warm water.

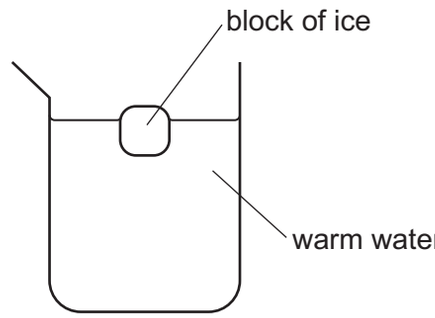


Fig. 3.1

(a) State **one** way in which the motion of the particles in ice differs from the motion of the particles in water.

.....

..... [1]

(b) Energy is transferred from the water to the block of ice.

(i) State the name of the thermal process that transfers energy from the water to the ice.

..... [1]

(ii) Initially, there is 0.34 kg of water in the beaker. The specific heat capacity of water is 4200 J / (kg °C).

Calculate the energy transferred from this water as its temperature decreases from 28 °C to 10 °C.

energy transferred = ... [2]

(iii) The temperature of the water near the ice decreases first.

Explain how convection causes the temperature of all the water in the beaker to decrease.

.....

.....

.....

..... [3]

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(iv) State what happens to the internal energy of the water as the temperature of the water decreases.

Describe the change in terms of the energy of the particles.

.....

.....

..... [2]

[Total: 9]

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5 On a sunny day, the temperatures of a black tarmac road and the air above the road increase.

(a) Explain why the surface temperature of the tarmac increases.

.....

.....

..... [2]

(b) State the method of thermal energy transfer from the tarmac to the air **immediately** above the road.

..... [1]

(c) State the main method of thermal energy transfer from the air immediately above the road to the rest of the air.

..... [1]

(d) Explain why the surface temperature of the tarmac is higher than the surrounding air temperature.

.....

.....

..... [2]

[Total: 6]

5 Fig. 5.1 shows two containers, each filled with hot water.

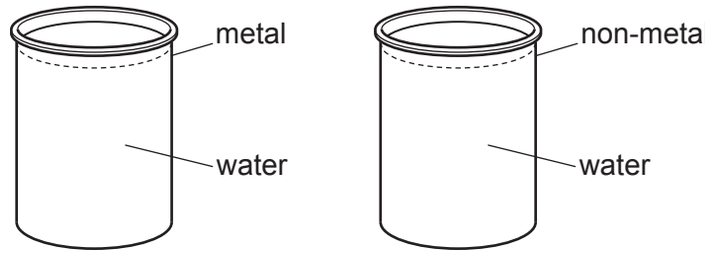

Fig. 5.1

The outer surface of the metal container is hot.

(a) Explain how electrons transfer thermal energy through the metal of the container.

.....

.....

.....

..... [3]

(b) The outer surface of the non-metal container is much cooler than the outer surface of the metal container.

Explain why a non-metal conducts thermal energy less well than a metal.

.....

..... [1]

(c) Explain, in terms of particles, why gases are poor thermal conductors compared to non-metal solids.

.....

.....

..... [2]

[Total: 6]

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3 Fig. 3.1 shows a portable shower used on a campsite. The bag is filled with water. The water is heated using infrared radiation from the Sun.

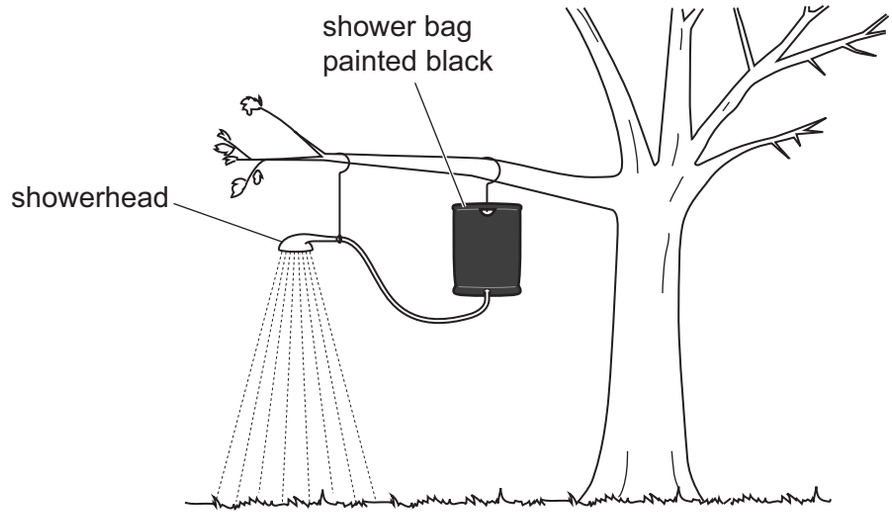

Fig. 3.1

(a) (i) Explain why the shower bag is painted black.

.....

..... [1]

(ii) Explain a **disadvantage** of radiation from the Sun being the only source to heat the water.

.....

..... [1]

(b) Solar energy is a renewable energy resource.

State **two other** renewable energy resources.

1 .....

2 .....
[2]

* 0000800000008 *

(c) During the day, the Sun shines on the shower bag and some of the energy in the infrared radiation from the Sun transfers to the thermal energy stores of the water.

The water absorbs 60% of the energy incident on the bag. The temperature of the water rises from 10 °C to 43 °C.

The mass of the water in the bag is 40 kg. The specific heat capacity of water is 4200 J / (kg °C).

Calculate the energy incident on the shower bag during the day.

Show your working.

energy = ... [4]

[Total: 8]

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3 WAVES

64 questions · Paper 2 & Paper 4

Vocabulary · Topic 3 WAVES

词汇表（中英对照）

wave	波	**ultraviolet**	紫外线
wavelength	波长	**X-rays**	X 射线
frequency	频率	**gamma rays**	伽马射线
amplitude	振幅	**sound**	声音
wavefront	波前	**ultrasound**	超声波
wave speed	波速	**echo**	回声
transverse wave	横波	**critical angle**	临界角
longitudinal wave	纵波	**total internal reflection**	全反射
reflection	反射	**refractive index**	折射率
refraction	折射	**lens**	透镜
diffraction	衍射	**converging lens**	凸透镜
dispersion	色散	**diverging lens**	凹透镜
electromagnetic spectrum	电磁波谱	**focal length**	焦距
radio waves	无线电波	**real image**	实像
microwaves	微波	**virtual image**	虚像
infrared	红外线	**magnifying glass**	放大镜
visible light	可见光		

3.1 General properties of waves

14 MCQ + 2 theory · Paper 2 & Paper 4

Date: _____ Score: _____

16 Which statement about waves is correct?

A Waves do not transfer either energy or matter.

B Waves transfer both energy and matter.

C Waves transfer energy without transferring matter.

D Waves transfer matter without transferring energy.

0625/21/M/J/24 Q16

17 Water waves are reflected at a plane surface.

Which property of the waves is changed by the reflection?

A direction of propagation

B frequency

C speed

D wavelength

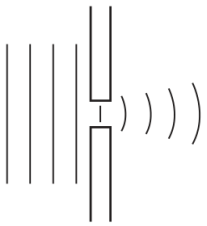
0625/21/O/N/24 Q17

Date: _____ Score: _____

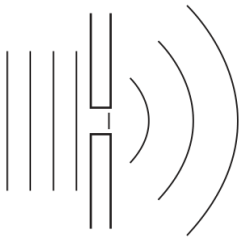
19 Plane water waves approach a narrow gap in a barrier.

Which diagram shows the diffraction pattern that would occur?

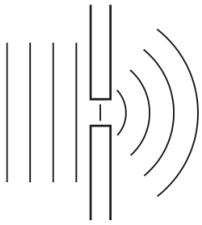
A



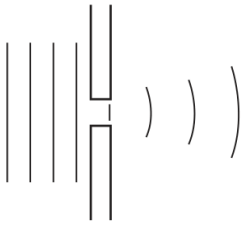
B



C

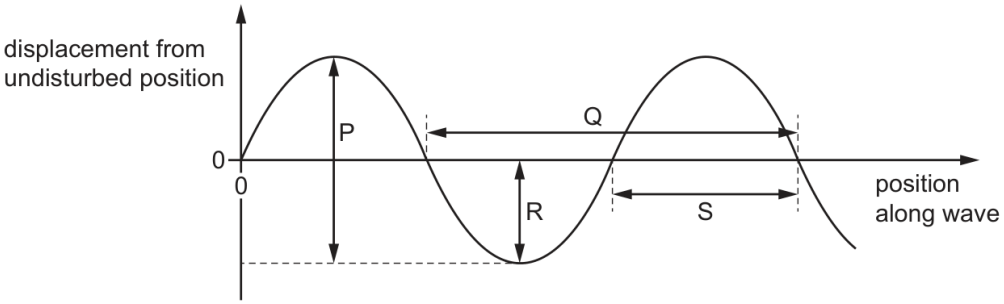


D



0625/21/O/N/24 Q19

16 The diagram shows a wave.



Which dimensions describe the properties of the wave?

A P = amplitude, S = speed

B P = speed, Q = wavelength

C R = amplitude, Q = wavelength

D R = amplitude, S = wavelength

0625/22/F/M/24 Q16

17 In which type of wave is the direction of vibration parallel to the direction of travel?

A electromagnetic waves

B seismic P-waves

C seismic S-waves

D water waves

0625/22/M/J/24 Q17

18 Waves in a ripple tank are diffracted as they pass through a narrow gap.

What can be done to make the spreading due to diffraction greater?

A Decrease the frequency of the waves and keep the speed constant.

B Decrease the speed of the waves and keep the frequency constant.

C Increase the frequency of the waves and keep the speed constant.

D Increase the frequency of the waves and decrease the speed.

0625/22/M/J/24 Q18

17 What is the number of wavefronts per second that pass a fixed point?

A the amplitude of the wave

B the frequency of the wave

C the speed of the wave

D the wavelength of the wave

0625/22/O/N/24 Q17

18 An earthquake under the ocean can produce a dangerous water wave called a tsunami.

The table gives typical data for a tsunami.

ocean depth / m	velocity / km / h	wavelength / km
7000	948	285
2000	504	151

A student suggests three conclusions about the data.

1 The deeper the ocean, the faster the wave.

2 The longer the wavelength, the faster the wave.

3 The faster the wave, the higher the frequency.

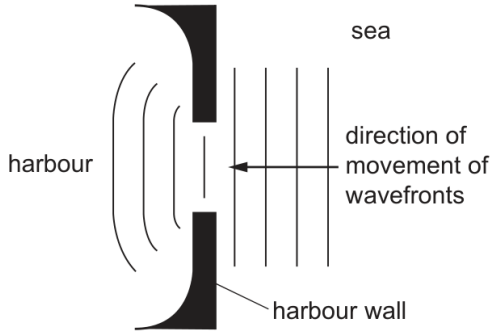
Which conclusions are correct?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

0625/22/O/N/24 Q18

Date: _____ Score: _____

19 The diagram shows what happens to wavefronts at sea when they enter a harbour.



What causes the change in the shape of the wavefronts?

A diffraction

B dispersion

C reflection

D refraction

0625/22/O/N/24 Q19

17 A seismic wave has a frequency of 5.0 Hz and a speed of 10 km / s.

What is the wavelength of the wave?

A 0.20 km **B** 0.50 km **C** 2.0 km **D** 50 km

0625/23/M/J/24 Q17

Date: _____ Score: _____

18 Sound waves will diffract when going through a gap or passing the edge of a barrier.

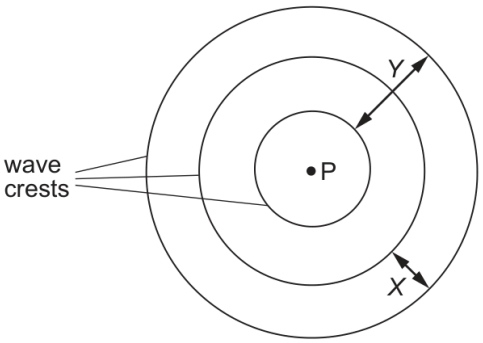
Which size of wavelength gives the most diffraction in each case?

	passing through a gap	passing the edge of a barrier
A	small wavelength	small wavelength
B	small wavelength	large wavelength
C	wavelength similar to gap size	small wavelength
D	wavelength similar to gap size	large wavelength

0625/23/M/J/24 Q18

17 A vertical stick is dipped up and down in water at P.

In two seconds, three wave crests spread out on the surface of the water.



Which statement is correct?

A Distance *X* is the amplitude of the waves.

B Distance *Y* is the wavelength of the waves.

C Each circle represents a wavefront.

D The frequency of the waves is 3 Hz.

0625/23/O/N/24 Q17

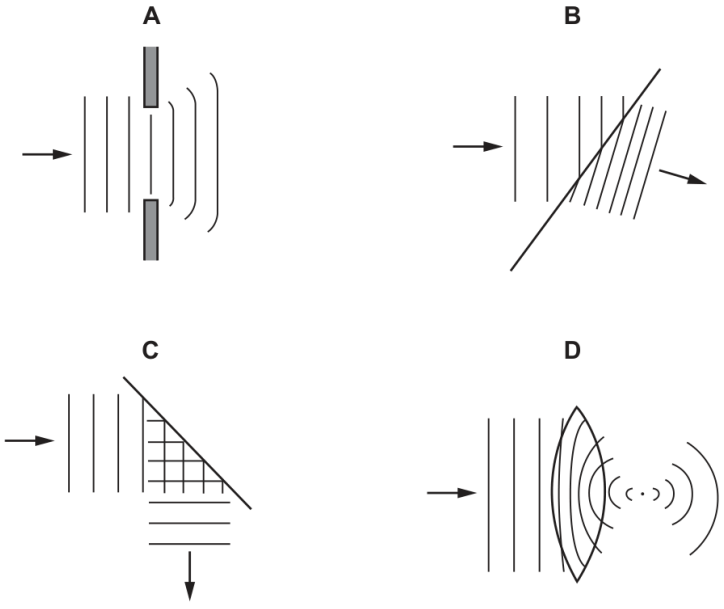
18 Seismic waves from the epicentre of an earthquake take 100 s to reach a point 300 km away. When they arrive, they cause the ground to vibrate with a frequency of 2.0 Hz.

What is the wavelength of these seismic waves?

A 0.67 km **B** 1.5 km **C** 3.0 km **D** 6.0 km

0625/23/O/N/24 Q18

19 Which diagram illustrates diffraction of water waves?



0625/23/O/N/24 Q19

5 **(a)** Describe how a longitudinal wave differs from a transverse wave.

.....

.....

..... [2]

(b) Fig. 5.1 represents a seismic wave produced by an earthquake.

Fig. 5.1

(i) State whether this seismic wave is a P-wave (primary) or an S-wave (secondary). Justify your choice.

.....

..... [1]

(ii) The wave represented in Fig. 5.1 has a wavelength of 1.2×10^4 m.

Calculate the actual distance between point J and point K.

distance = ... [2]

(iii) The wave in **(ii)** travels through the ground at a speed of 4600 m/s.

As the wave passes a certain point, the ground completes 5 oscillations.

Calculate the time that it takes for the wave to pass. Show your working.

time = ... [3]

[Total: 8]

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6 A student plays the violin near the doorway to a large room. Fig. 6.1 shows a young teacher standing where he can hear the sound but cannot see the student.

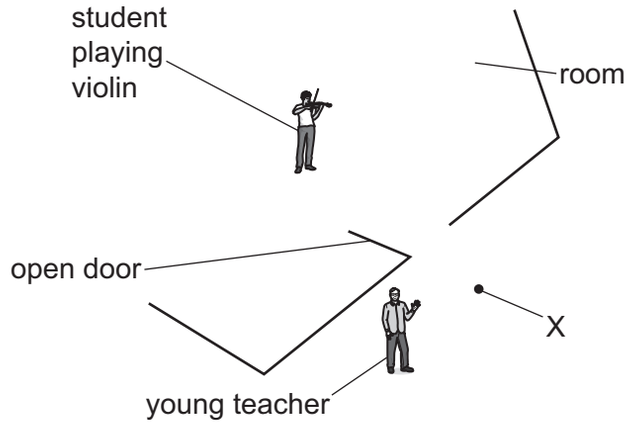

Fig. 6.1

(a) (i) State the wave effect that allows the young teacher to hear sounds from the violin at the position he is standing in Fig. 6.1.

..... [1]

(ii) Calculate the frequency of sound with a wavelength of 0.75 m.
The speed of sound in air is 340 m / s.

frequency = ... [2]

(iii) A violin produces sounds in the frequency range 200 Hz–3800 Hz. The width of the open doorway is 0.75 m.

Explain why the young teacher hears the frequency calculated in **(a)(ii)** clearly but finds a frequency of 3500 Hz much harder to hear.

.....

.....

..... [2]

(b) A plane mirror is placed at point X so that the teacher can see the student.
On Fig. 6.1:

- draw a light ray from the violin to point X and from point X to the teacher
- draw and label the mirror
- add an arrow to the ray to show how the teacher sees the student.

Use a ruler and sharp pencil for this drawing. [3]

[Total: 8]

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3.2 Light

23 MCQ + 5 theory · Paper 2 & Paper 4

Date: _____ Score: _____

17 The angle of incidence of a ray of light incident on a plane mirror is gradually increased.

To the nearest degree, what is the maximum possible angle between the incident and reflected rays?

A 0° **B** 45° **C** 90° **D** 180°

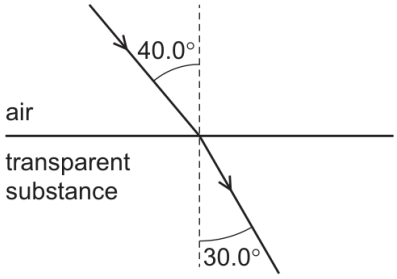
0625/21/M/J/24 Q17

18 Which conditions are necessary for light to be totally internally reflected?

	incident light	angle of incidence
A	is in the less dense medium	less than the critical angle
B	is in the less dense medium	greater than the critical angle
C	is in the more dense medium	less than the critical angle
D	is in the more dense medium	greater than the critical angle

0625/21/M/J/24 Q18

19 The diagram shows a ray of light passing from air into a transparent substance.



What is the refractive index of the transparent substance?

A 1.33 **B** 1.29 **C** 0.778 **D** 0.750

0625/21/M/J/24 Q19

20 The diagram shows a ray of light in an optical fibre.

Which statement correctly explains the condition for the maximum transmission of light by the optical fibre?

A The glass must slow the light as little as possible to make the critical angle for the fibre as large as possible.

B The glass must slow the light as little as possible to make the critical angle for the fibre as small as possible.

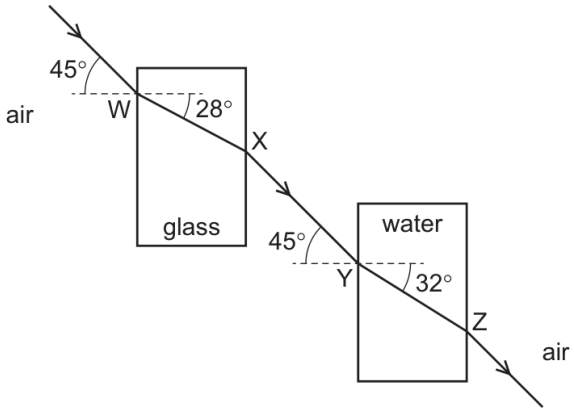
C The glass must slow the light as much as possible to make the critical angle for the fibre as large as possible.

D The glass must slow the light as much as possible to make the critical angle for the fibre as small as possible.

0625/21/M/J/24 Q20

Date: _____ Score: _____

18 A light ray is passed through air, glass and water.

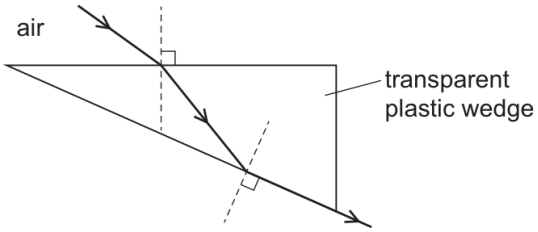


At which points does the light speed decrease?

A W and X **B** W and Y **C** X and Z **D** Y and Z

0625/21/O/N/24 Q18

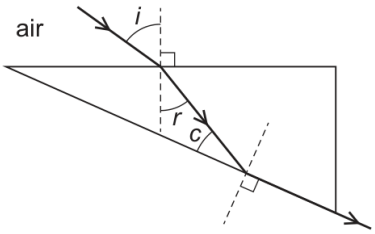
20 A ray of light enters a transparent plastic wedge from air and is incident on the lower surface at the critical angle, as shown.



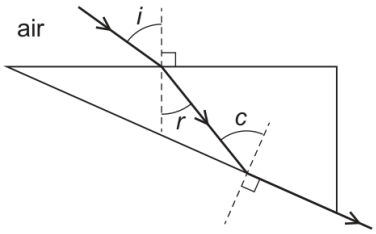
The angle of incidence is *i*, the angle of refraction is *r*, and the critical angle is *c*.

In which ray diagram are the angles labelled correctly?

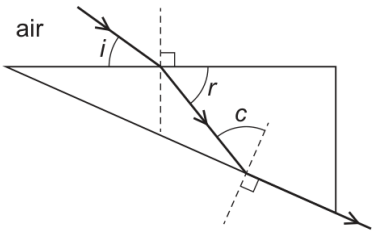
A



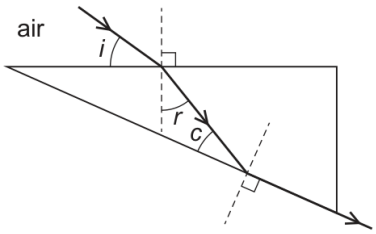
B



C



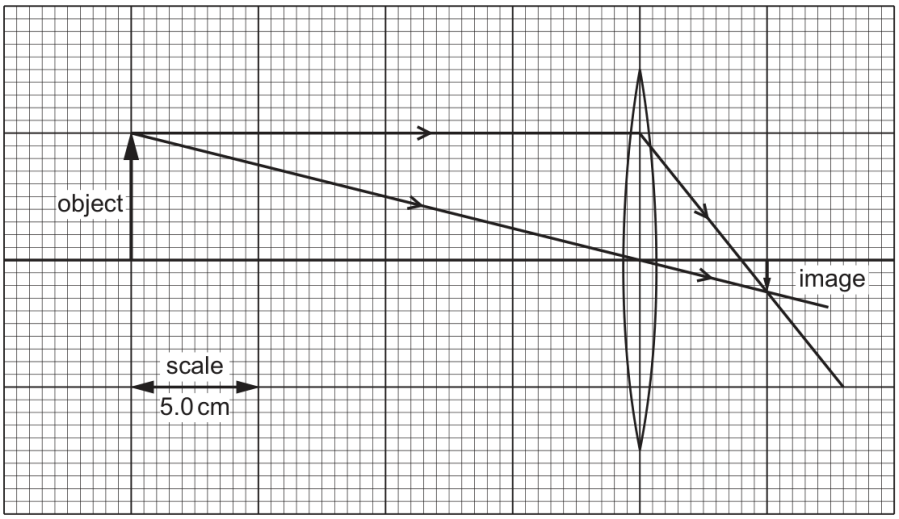
D



0625/21/O/N/24 Q20

Date: _____ Score: _____

21 An object is placed 20 cm in front of a thin converging lens. The scale diagram shows how the lens forms a real, inverted image.



Which row gives the focal length of the lens and the distance of the image from the lens?

	focal length of lens / cm	distance of image from lens / cm
A	4.0	5.0
B	5.0	4.0
C	8.0	10.0
D	10.0	8.0

0625/21/O/N/24 Q21

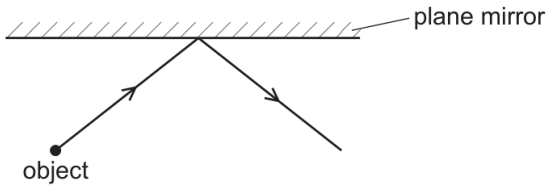
17 Which row describes some characteristics of an image formed in a plane mirror?

	image type	image size
A	real	diminished
B	real	same size as the object
C	virtual	diminished
D	virtual	same size as the object

0625/22/F/M/24 Q17

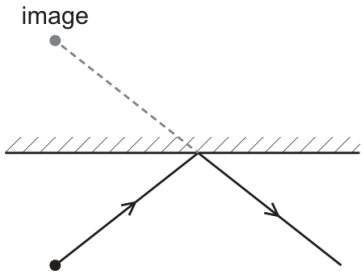
19 An object is placed in front of a plane mirror.

The ray diagram shows the reflection of one ray of light from the object by the mirror.

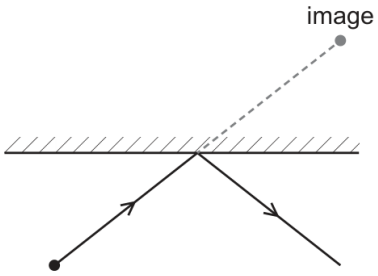


Where does the mirror form an image of the object?

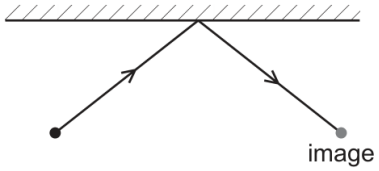
A



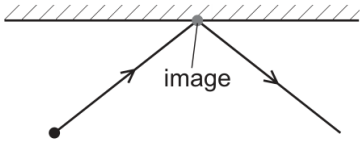
B



C

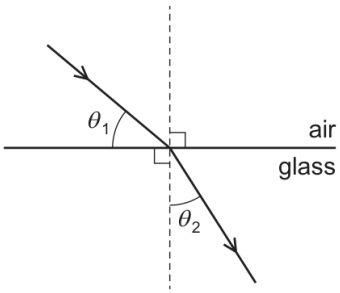


D



0625/22/F/M/24 Q19

20 Light refracts as it enters glass from air, as shown.



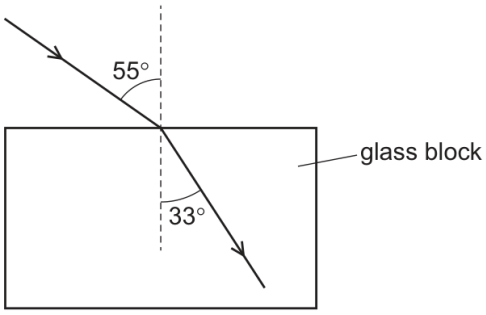
Which expression is equal to the refractive index of glass?

A $\dfrac{\sin\theta_1}{\sin\theta_2}$ **B** $\dfrac{\sin\theta_1}{\sin(90^\circ-\theta_2)}$ **C** $\dfrac{\sin(90^\circ-\theta_1)}{\sin\theta_2}$ **D** $\dfrac{\sin(90^\circ-\theta_1)}{\sin(90^\circ-\theta_2)}$

0625/22/F/M/24 Q20

21 Light travelling at a speed of 3.0×10^8 m/s strikes the surface of a glass block and undergoes refraction as it enters the block.

The diagram shows a ray of this light before and after it enters the block.



What is the speed of light in the glass?

A 1.8×10^8 m/s

B 2.0×10^8 m/s

C 4.5×10^8 m/s

D 5.0×10^8 m/s

0625/22/F/M/24 Q21

22 Which row about converging lenses and diverging lenses is correct?

	converging lenses	diverging lenses
A	always produce inverted images	can focus a parallel beam of light
B	can be used as magnifying glasses	do not have a principal focus
C	can be used to correct short-sightedness	are usually thinner than converging lenses
D	can produce real and virtual images	can be used to correct short-sightedness

0625/22/F/M/24 Q22

Date: _____ Score: _____

19 Diagram 1 shows the page of a book in front of a plane mirror.

An eye is looking at the image of the page.

Diagram 2 shows a large letter G on the page facing the mirror.

diagram 1

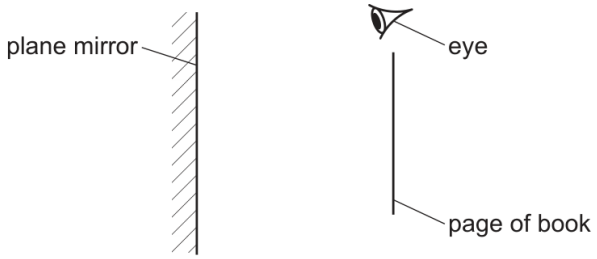


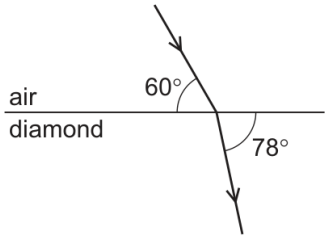
diagram 2

What is the appearance of the image of G seen by the eye?

A **B** **C** **D**

0625/22/M/J/24 Q19

20 The diagram shows a ray of light passing from air into diamond.



What is the refractive index of the diamond?

A 0.89 **B** 1.1 **C** 2.4 **D** 2.5

0625/22/M/J/24 Q20

Date: _____ Score: _____

21 The ray diagrams show the formation of an image by two different converging lenses.

diagram 1

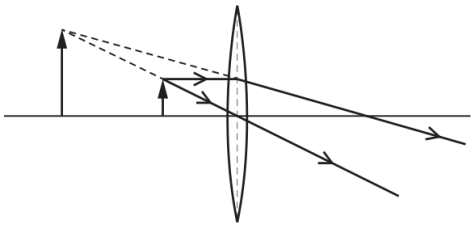
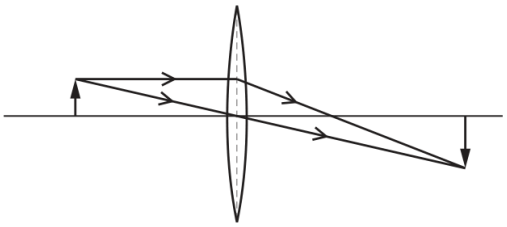


diagram 2



Which row describes the images produced?

	diagram 1	diagram 2
A	real	real
B	real	virtual
C	virtual	real
D	virtual	virtual

0625/22/M/J/24 Q21

22 White light enters a glass prism. The light leaving the other side of the prism is separated into colours.

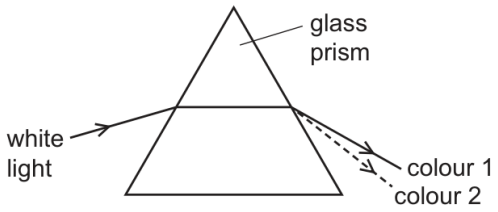


diagram 1

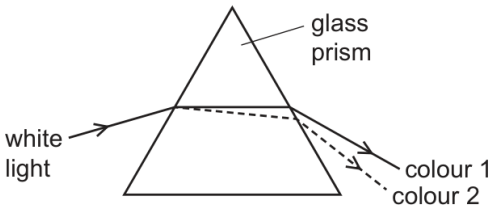


diagram 2

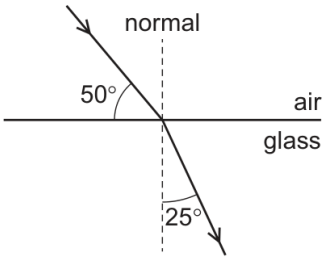
Which row correctly describes what happens?

	path taken by the light	colour 1	colour 2
A	diagram 1	red	violet
B	diagram 1	violet	red
C	diagram 2	red	violet
D	diagram 2	violet	red

0625/22/M/J/24 Q22

Date: _____ Score: _____

20 A ray of light passes from air into glass.



What are the angles of incidence and refraction of the ray?

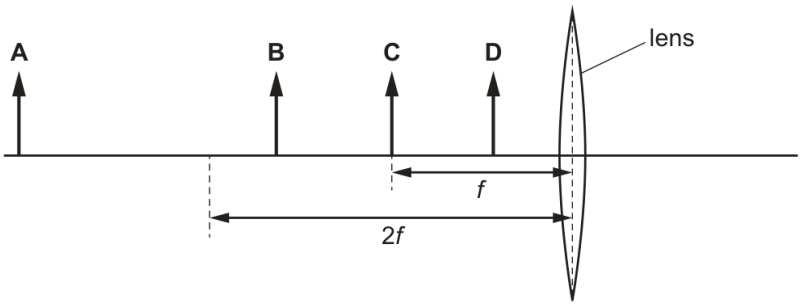
	angle of incidence / °	angle of refraction / °
A	40	25
B	40	65
C	50	25
D	50	65

0625/22/O/N/24 Q20

21 An object is placed in front of a converging lens. The lens has a focal length *f*.

The lens produces a real, enlarged image of the object.

In which labelled position is the object placed?



0625/22/O/N/24 Q21

Date: _____ Score: _____

19 A boy is having his eyes tested. A letter is printed on a card placed over his head. He sees the card in a plane mirror placed in front of him.

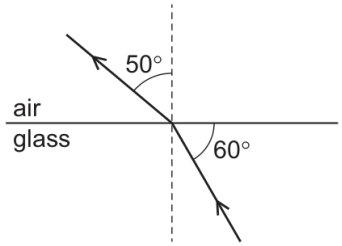
He sees the letter ‘R’ in the mirror.

How is it printed on the card?

A R **B** Я **C** Я **D** Я

0625/23/M/J/24 Q19

20 A ray of light is refracted as it enters air from glass, as shown.



What is the speed of light in the glass?

A 2.0×10^8 m/s

B 2.2×10^8 m/s

C 2.3×10^8 m/s

D 2.7×10^8 m/s

0625/23/M/J/24 Q20

21 A converging lens is being used as a magnifying glass.

Which statement is correct?

A The image is further away from the lens than the object is.

B The image is inverted.

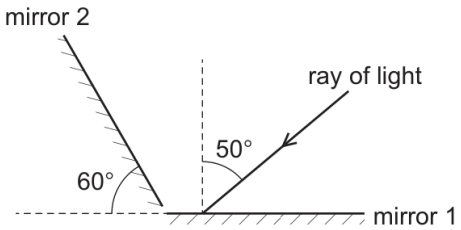
C The image is real.

D The object must be placed at the principal focus of the lens.

0625/23/M/J/24 Q21

20 A single ray of light is reflected by two mirrors.

The diagram shows the ray of light incident on mirror 1.



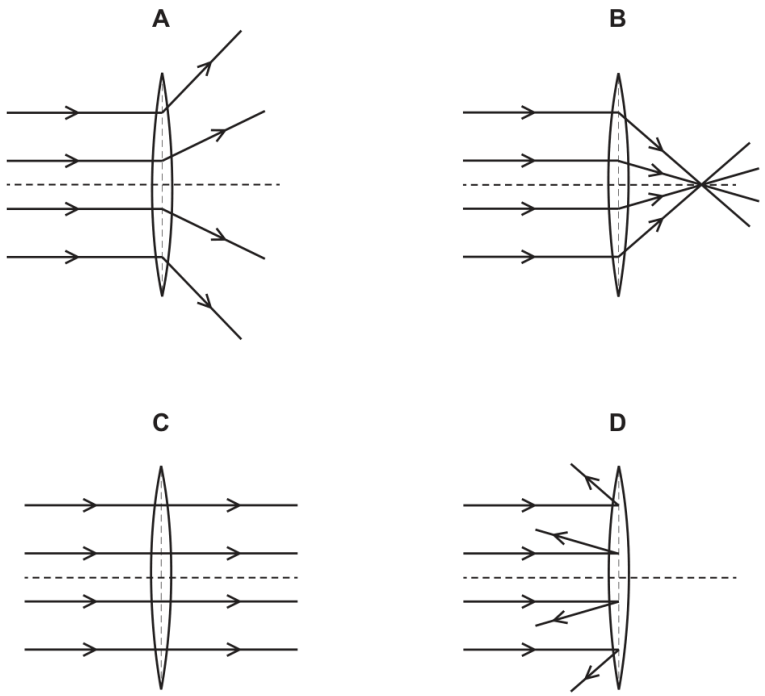
What is the angle of reflection in mirror 2?

A 50° **B** 60° **C** 70° **D** 80°

0625/23/O/N/24 Q20

Date: _____ Score: _____

21 Which diagram shows the effect of a converging lens on a parallel beam of light?



0625/23/O/N/24 Q21

4 The lens in a magnifying glass is a converging lens.

(a) Fig. 4.1 shows the lens of the magnifying glass, its two focal points, F_1 and F_2, and its principal axis.

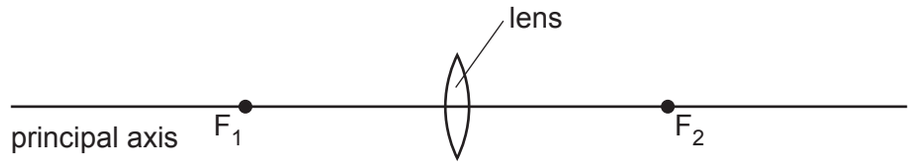


Fig. 4.1

(i) State what is meant by 'focal point'.

.....

.....

..... [2]

(ii) A student using the magnifying glass sees a magnified image of an object.

On Fig. 4.1, mark:

- a point X on the principal axis for a possible position of the object
- a point E for a possible position of the student's eye.

[1]

(iii) Underline **two** words in the list that describe the image produced in **(a)(ii)**.

inverted **real** **upright** **virtual**

[1]

(b) The refractive index of the glass used to make the lens is 1.5.

(i) The speed of light in air is 3.0×10^8 m/s.

Calculate the speed of light in the glass.

speed in glass = ... [2]

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(ii) State what happens to the wavelength of light as it passes into the lens.

.....

..... [1]

(c) Converging lenses are used in spectacles (glasses) to correct one problem with vision.

State the name of the problem and explain how a converging lens is used to correct it. You may draw a diagram.

name of problem: .....

.....

.....

.....

.....

[3]

[Total: 10]

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6 Fig. 6.1 shows a full-scale diagram of an object O and its image I produced by a converging lens. The lens and its position on the principal axis are not shown.

Fig. 6.1

(a) On Fig. 6.1, draw:

- a single ray to locate the position of the centre of the converging lens
- a line to represent the position of the lens and label the line L.

[2]

(b) Determine the focal length of the lens by drawing another ray on Fig. 6.1.

focal length = ... [2]

(c) The object is moved 2.0 cm closer to the lens.

State **two** changes to the characteristics of the image.

1 .....

2 .....

[2]

[Total: 6]

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5 Fig. 5.1 shows a ray of yellow light incident on a glass prism ABC.

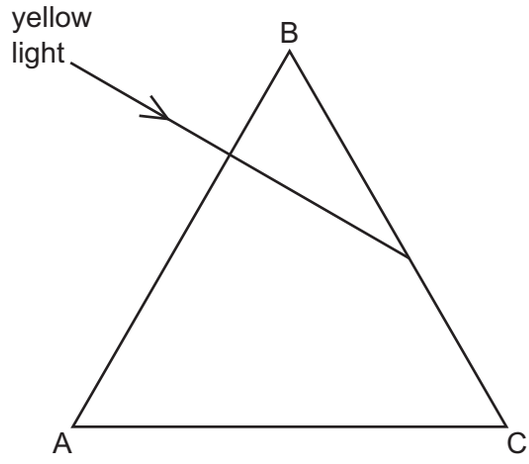

Fig. 5.1

(a) Explain why the ray does **not** change direction when it enters the prism at face AB.

.....

..... [1]

(b) The critical angle for the glass is 42°.

(i) Calculate the refractive index of the glass.

Show your working.

refractive index = ... [2]

(ii) On Fig. 5.1, continue the path of the light through the prism and after it leaves the prism. [3]

(c) Internet data can be transferred using infrared waves in optical fibres.

State **two** advantages of using optical fibres to transmit data.

1 .....

2 .....
[2]

[Total: 8]

6 Fig. 6.1 shows a thin converging lens used to produce a magnified image of an object AB.

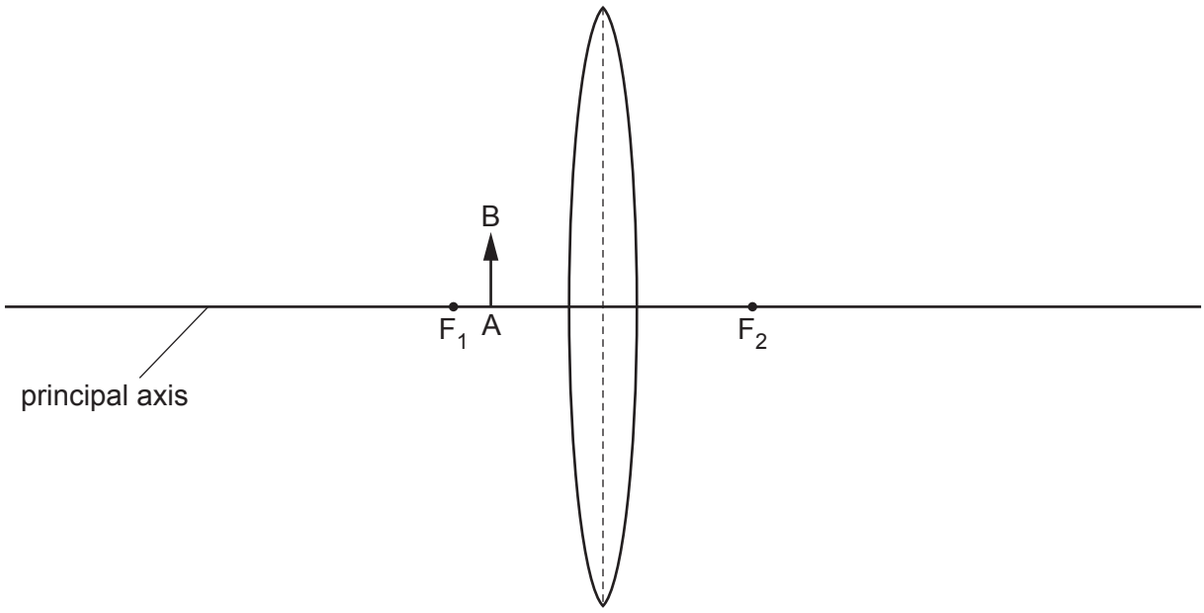

Fig. 6.1

(a) Explain the meaning of the terms principal focus and focal length.

principal focus .....

.....

focal length .....

.....

[2]

(b) On Fig. 6.1, draw the magnified image of AB. Show your working. [4]

[Total: 6]

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4 **(a)** Fig. 4.1 shows a ray of light as it enters the side of a plastic block. The ray of light passes from air into the plastic.

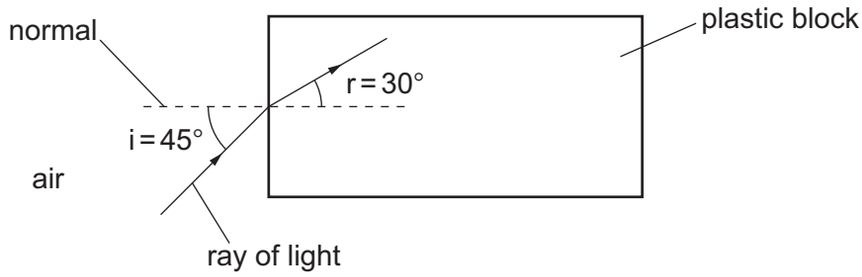

Fig. 4.1

(i) State how the speed, wavelength and frequency of the wave in the plastic block compare with their values in the air.

speed: .....

wavelength: .....

frequency: .....

[2]

(ii) Show that the refractive index of the plastic is 1.4.

Show your working.

[1]

(iii) Calculate the critical angle for the plastic.

critical angle = ... [2]

(b) Fig. 4.2 shows the same plastic as in **(a)** used to make an optical fibre. A ray of light is passing along the fibre.

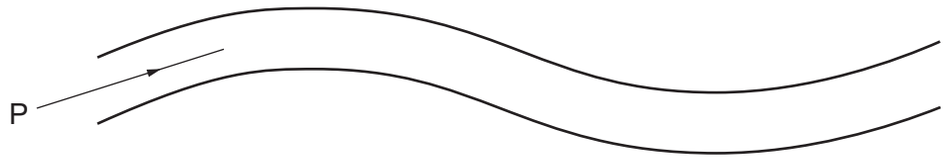

Fig. 4.2

(i) Carefully continue the ray of light P until it reaches the other end of the fibre. [2]

(ii) State **two** uses for optical fibres.

1 .....

2 .....
[2]

[Total: 9]

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3.3 Electromagnetic spectrum

8 MCQ + 1 theory · Paper 2 & Paper 4

Date: _____ Score: _____

21 Which region of the electromagnetic spectrum is used for detecting fake bank notes?

A radio

B microwaves

C ultraviolet

D X-rays

0625/21/M/J/24 Q21

22 Radio waves are electromagnetic waves.

What is the wavelength of a radio wave of frequency 2.0×10^5 Hz?

A 0.00067 m **B** 0.0015 m **C** 670 m **D** 1500 m

0625/21/O/N/24 Q22

23 Thermal radiation is part of the electromagnetic spectrum.

What is the name of this region of the spectrum?

A gamma rays

B infrared

C ultraviolet

D visible light

0625/22/M/J/24 Q23

Date: _____ Score: _____

24 A remote-controlled vehicle is travelling on the surface of a planet. The vehicle senses an obstacle ahead. It sends a radio message to the control room from where it is being controlled. The control room is 2.4×10^6 km away from the vehicle. The control room sends a message back to the vehicle telling it to stop.

What is the minimum time that elapses between the vehicle sensing the obstacle and receiving the message back from the control room?

A 8.0 ms **B** 16 ms **C** 8.0 s **D** 16 s

0625/22/M/J/24 Q24

22 What is the speed of electromagnetic waves in a vacuum?

A 300 m / s

B 300 km / s

C 300 000 km / s

D 300 000 000 km / s

0625/22/O/N/24 Q22

22 Which colours of visible light are in the correct order of increasing wavelength?

A red → yellow → blue

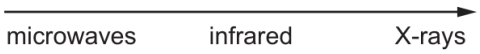
B orange → green → violet

C indigo → green → red

D blue → green → violet

0625/23/M/J/24 Q22

23 The diagram shows three types of electromagnetic radiation listed in a particular order. The electromagnetic radiation is travelling in a vacuum.



Which quantities increase in magnitude going from left to right across the list?

A both speed and frequency

B frequency only

C speed only

D neither speed nor frequency

0625/23/M/J/24 Q23

22 Which three regions of the electromagnetic spectrum are used in the applications listed?

- sterilising water
- optical fibres
- satellite television signals
- mobile (cell) phones

A radio, infrared, visible

B ultraviolet, infrared, microwaves

C visible, radio, ultraviolet

D X-rays, visible, infrared

0625/23/O/N/24 Q22

5 **(a)** **(i)** Table 5.1 shows applications of regions of the electromagnetic spectrum.

Complete the second column of the table with the region of the electromagnetic spectrum used for each application.

Choose from the regions in this list:

gamma rays **infrared** **microwaves** **radio waves** **ultraviolet**

Each region may be used once, more than once or not at all.

Table 5.1

application	region of electromagnetic spectrum
cancer treatment	gamma rays
Bluetooth data connection	
optical fibres	
security marking	
sterilising food	
wireless internet	

[3]

(ii) State the approximate speed of radio waves in air.

speed = ... m/s [1]

(b) Fig. 5.1 shows successive crests of a wave after a plane wave has passed through a gap.

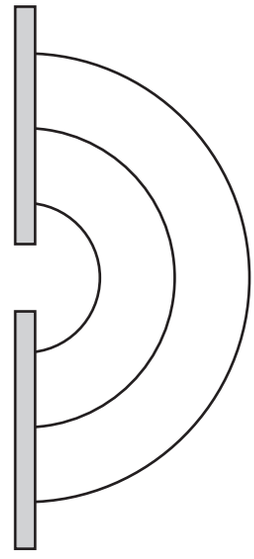


Fig. 5.1

(i) On Fig. 5.1 draw **three** successive crests before the wave reaches the gap. [2]

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(ii) Fig. 5.2 shows a much wider gap. A plane wave of the same wavelength as in **(b)(i)** is incident on the gap from the left side of the barrier.

Fig. 5.2

On Fig. 5.2, draw **three** successive crests of the wave after the wave has passed through the gap. [2]

[Total: 8]

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[Turn over

3.4 Sound

7 MCQ + 4 theory · Paper 2 & Paper 4

Date: _____ Score: _____

22 A sound wave travels at 330 m / s. The distance between the centre of a compression and the centre of the nearest rarefaction in the sound wave is 2.5 cm.

What is the frequency of the sound wave?

A 66 Hz **B** 130 Hz **C** 6600 Hz **D** 13 000 Hz

0625/21/M/J/24 Q22

23 Which waves are used in the medical scanning of soft tissue?

A gamma rays

B infrared

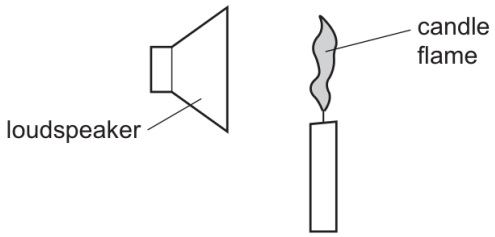
C microwaves

D ultrasound

0625/21/M/J/24 Q23

23 A candle flame is placed in front of a loudspeaker.

The loudspeaker produces a sound wave that causes air particles to vibrate. The vibrating air particles make the candle flame vibrate in the same direction as the air particles.



Which row shows the direction of vibration of the candle flame and the nature of sound waves?

	direction of vibration	nature of sound waves
A	↕	longitudinal
B	↕	transverse
C	↔	longitudinal
D	↔	transverse

0625/21/O/N/24 Q23

18 The range of audible frequencies for elephants is 10 Hz to 12 000 Hz, and for horses is 55 Hz to 33 000 Hz.

Which statement about the range of audible frequencies for a healthy human ear is correct?

A The human ear can hear all sounds heard by both elephants and horses.

B The human ear can hear all sounds heard by elephants but not all those heard by horses.

C The human ear can hear all sounds heard by horses but not all those heard by elephants.

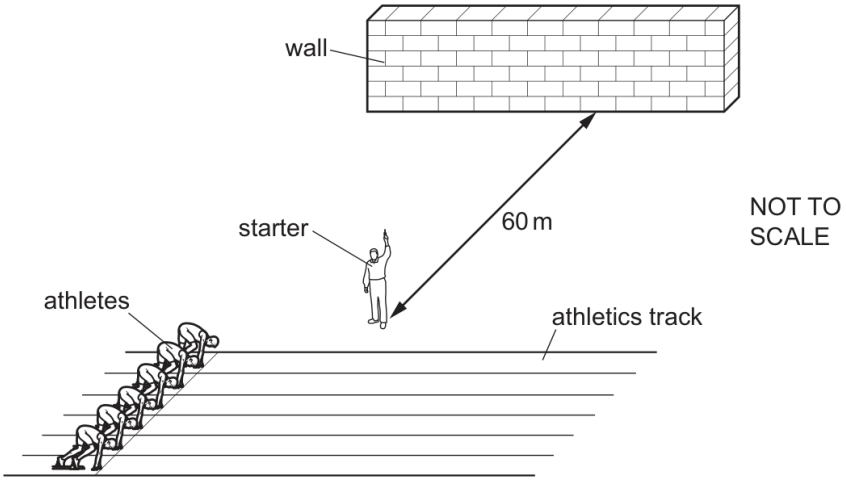
D The human ear cannot hear all sounds heard by elephants nor those heard by horses.

0625/22/F/M/24 Q18

Date: _____ Score: _____

23 The starter in an athletics race fires the starting pistol. There is a wall 60 m from the starter.

Take the speed of sound as 330 m / s.



How long after firing the pistol does the starter hear the echo of the sound from the wall?

A 0.18 s **B** 0.36 s **C** 2.8 s **D** 5.5 s

0625/22/O/N/24 Q23

24 Which row gives the typical values of the speed of sound at room temperature in the materials stated?

	speed of sound / m / s		
	air	water	iron
A	340	1500	5100
B	340	5100	1500
C	5100	1500	340
D	3.0×10^8	3.0×10^8	3.0×10^8

0625/23/M/J/24 Q24

23 A man stands 110 m from a high wall. He makes a short, sharp sound and then hears an echo from the wall.

The speed of sound in air is 330 m / s.

How long after making the sound does the man hear the echo?

A 0.33 s **B** 0.67 s **C** 1.5 s **D** 3.0 s

0625/23/O/N/24 Q23

5 A loudspeaker produces a sound wave in air. The distance between the centre of a compression and the centre of a neighbouring rarefaction is 0.10 m.

(a) Calculate the wavelength of the sound wave.

wavelength = ... [1]

(b) State a typical value for the speed of sound in air.

..... [1]

(c) **(i)** Calculate the frequency of the sound from the loudspeaker.

frequency = ... [2]

(ii) Explain whether the sound from the loudspeaker is audible to a human with normal hearing.

.....

..... [1]

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(d) Another loudspeaker produces a sound of wavelength 0.40 m. Sound from the loudspeaker reaches a sound absorbing surface with a gap of width 0.80 m at the centre.

Fig. 5.1 shows the arrangement.

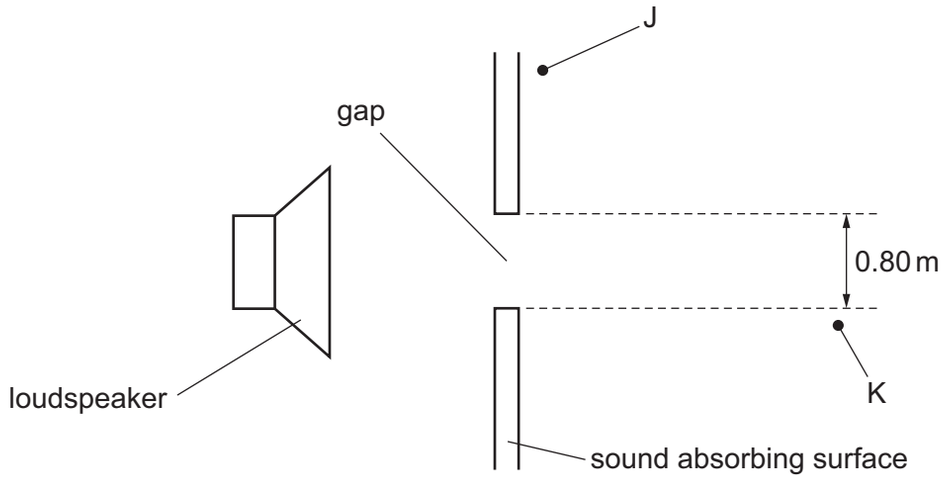

Fig. 5.1

Explain whether it is possible to detect sound from the loudspeaker at either point J or at point K.

point J .....

.....

.....

point K .....

.....

.....

[4]

[Total: 9]

* 0019655392610 *

6 **(a)** A sound wave travels through air. Fig. 6.1 shows a pressure–time graph for the air at one place.

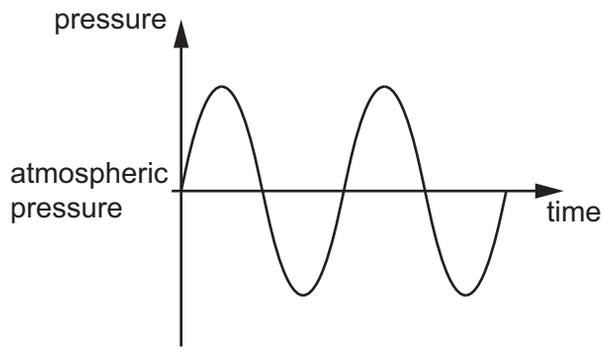

Fig. 6.1

(i) On Fig. 6.1:

- label **one** point C to indicate a compression
- label **one** point R to indicate a rarefaction.

[2]

(ii) Explain why this graph **cannot** be used to find the wavelength of the sound wave.

.....

…..... [1]

(iii) The sound becomes louder and of lower pitch.

State what happens to:

the amplitude of the sound

.....

the frequency of the sound.

.....
[1]

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(b) A sound of frequency 13 kHz is transmitted through water.
The speed of sound in water is 1500 m / s.

Calculate the wavelength of this sound in water.

wavelength = ... [3]

(c) State the approximate speed of sound in air.

speed = ... [1]

[Total: 8]

* 0000800000010 *

8 A fisherman uses high frequency sound waves to locate fish in the sea. Fig. 8.1 shows the sound waves emitted from the boat.

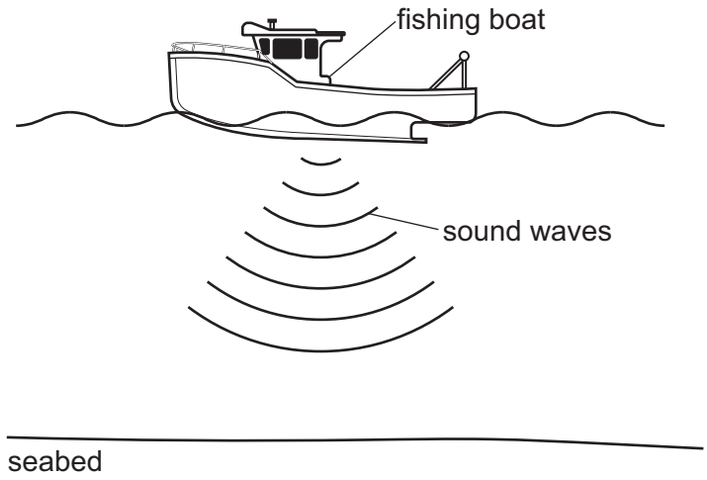

Fig. 8.1 (not to scale)

(a) State the name of sound waves which have a frequency greater than 20 kHz.

..... [1]

(b) High frequency sound waves travel from the boat through the sea water.
The speed of sound in water is 1500 m / s. The seabed is 22 m below the boat.

Calculate the time taken for the boat to receive the reflected wave from the seabed after the sound is emitted.

time = ... [3]

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(c) Fig. 8.2 shows a fish below the boat.

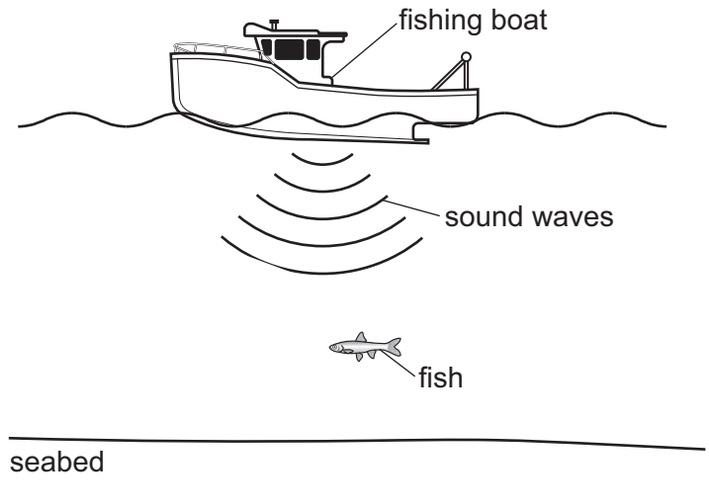

Fig. 8.2 (not to scale)

Describe and explain how the reflected sound wave received by the boat from the fish differs from the reflected sound wave received from the seabed.

.....

.....

..... [2]

[Total: 6]

8 Fig. 8.1 shows images produced during two different medical scanning procedures.

ultrasound scan of a fetus

X-ray scan of a hand

Fig. 8.1

(a) (i) Define ultrasound.

.....

..... [1]

(ii) State how the speed of sound in liquid compares to the speed of sound in air.

..... [1]

(iii) X-rays are part of the electromagnetic spectrum.

State the speed of X-rays in a vacuum.

..... [1]

(b) Describe **three** similarities or differences between the use of ultrasound and X-rays in medical scanning procedures.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

[Total: 6]

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4 ELECTRICITY AND MAGNETISM

75 questions · Paper 2 & Paper 4

Vocabulary · Topic 4 ELECTRICITY AND MAGNETISM

词汇表（中英对照）

electric charge	电荷	**magnet**	磁铁
current	电流	**magnetic field**	磁场
potential difference (p.d.)	电势差	**magnetic field lines**	磁感线
electromotive force (e.m.f.)	电动势	**pole**	磁极
resistance	电阻	**electromagnet**	电磁铁
voltage	电压	**solenoid**	螺线管
circuit	电路	**electromagnetic induction**	电磁感应
series	串联	**generator**	发电机
parallel	并联	**transformer**	变压器
resistor	电阻器	**step-up**	升压
diode	二极管	**step-down**	降压
thermistor	热敏电阻	**motor**	电动机
LDR	光敏电阻	**commutator**	换向器
ammeter	电流表	**brushes**	电刷
voltmeter	电压表	**earthing**	接地
fuse	保险丝		

4.1 Simple phenomena of magnetism

8 MCQ + 1 theory · Paper 2 & Paper 4

Date: _____ Score: _____

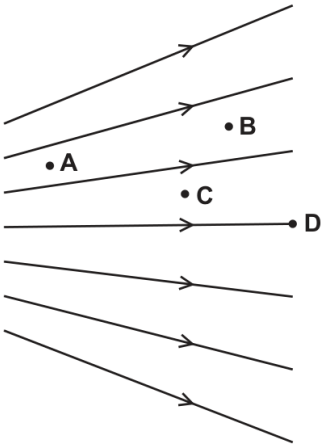
24 Which row describes suitable materials for use in a temporary magnet and in a permanent magnet?

	temporary magnet	permanent magnet
A	soft iron	soft iron
B	soft iron	steel
C	steel	soft iron
D	steel	steel

0625/21/M/J/24 Q24

25 A magnetic field is represented in the diagram by magnetic field lines.

At which point is the magnetic field strongest?

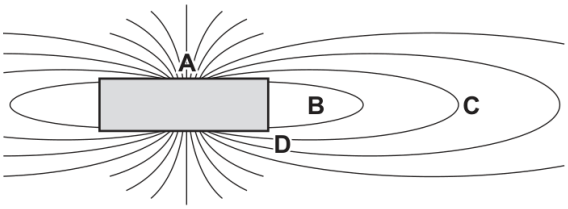


0625/21/M/J/24 Q25

Date: _____ Score: _____

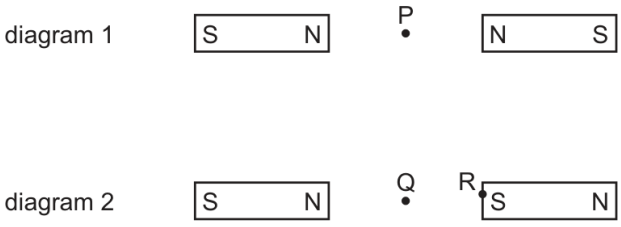
24 The diagram shows part of the magnetic field around a strong magnet.

In which position does a magnetic pole experience the strongest force?



0625/21/O/N/24 Q24

26 The diagrams show two arrangements of a pair of identical bar magnets.



Three points, P, Q and R, are shown.

At which point is the magnetic field due to the magnets weakest and at which point is it strongest?

	weakest magnetic field	strongest magnetic field
A	P	Q
B	P	R
C	Q	P
D	Q	R

0625/22/F/M/24 Q26

25 Which statement about the direction of a magnetic field at a point is correct?

A It is the direction of the force on an N pole placed at that point.

B It is the direction of the force on an S pole placed at that point.

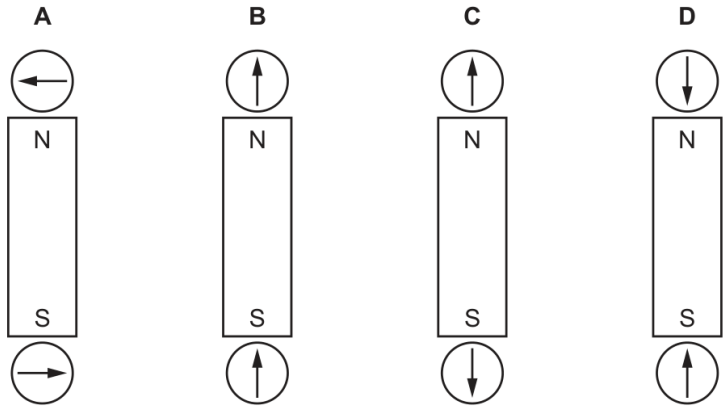
C It is the direction of the force on a positive charge placed at that point.

D It is the direction of the force on a negative charge placed at that point.

0625/22/M/J/24 Q25

25 Two small compasses are placed close to a strong bar magnet.

In which directions do the compass needles point?



0625/23/M/J/24 Q25

26 Students are asked for a statement about magnetic fields.

Three statements are listed.

1 A magnetic field is described as a region in which a stationary electric charge experiences a force.

2 The direction of a magnetic field at a point is the direction of the force on an S pole at that point.

3 The relative strength of a magnetic field is shown by the spacing of the magnetic field lines.

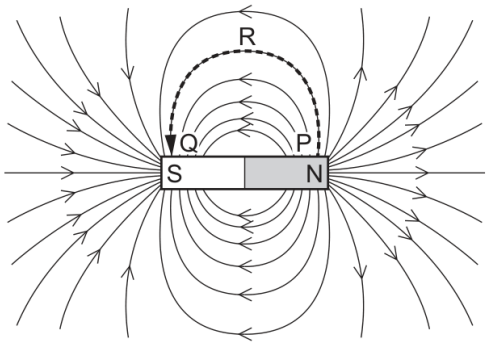
Which statements are correct for magnetic fields?

A 1, 2 and 3 **B** 1 and 2 only **C** 2 and 3 only **D** 3 only

0625/23/M/J/24 Q26

Date: _____ Score: _____

24 The diagram shows the magnetic field of a bar magnet.



How does the strength of the magnetic field vary with position along the dotted path PRQ?

A It is constant along the path PRQ.

B It is strong at P and reduces in strength along the path PRQ.

C It is weak at P and increases in strength along the path PRQ.

D It is strong at P, decreases to a minimum at R and then increases towards Q.

0625/23/O/N/24 Q24

* 0019655392712 *

7 **(a)** Fig. 7.1 shows three bars of steel, A, B and C.

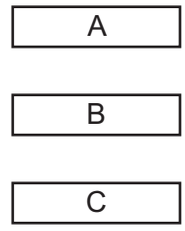


Fig. 7.1

A student is given the three pieces of steel. Two of the pieces are magnetised and one piece is unmagnetised.

Describe and explain how the student determines which piece is unmagnetised using only the three pieces of steel.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Fig. 7.2 shows a circuit diagram of a step-down transformer.

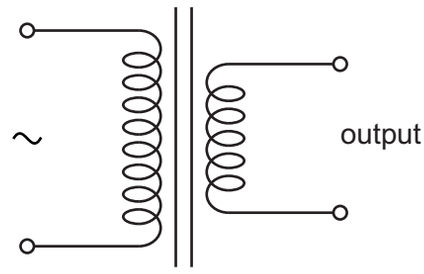


Fig. 7.2

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(i) The mains voltage supplied to the transformer is 240 V. The output power of the transformer is 45 W. The transformer is 100% efficient.

Calculate the input current to the transformer.

input current = ... [3]

(ii) Draw a labelled diagram of a step-down transformer. On the labels, state a suitable material for each of the components.

[3]

[Total: 10]

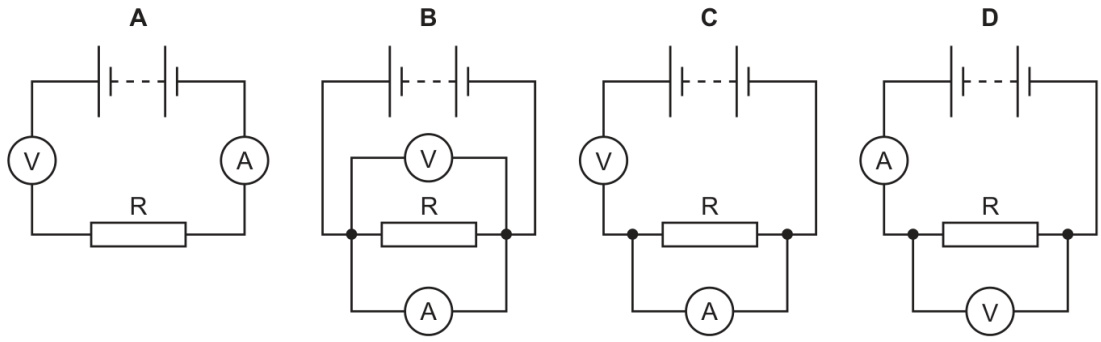
4.2 Electrical quantities

20 MCQ + 5 theory · Paper 2 & Paper 4

Date: _____ Score: _____

26 A student determines the resistance of resistor R. She uses a circuit including a voltmeter and an ammeter.

Which circuit does she use?



0625/21/M/J/24 Q26

25 There is a current of 0.60 A in a closed circuit.

How much charge passes a point in the circuit in 2.5 min?

A 0.24 C **B** 1.5 C **C** 90 C **D** 5400 C

0625/21/O/N/24 Q25

26 Which two physical quantities have the unit J / C?

A charge and energy

B charge and potential difference (p.d.)

C electromotive force (e.m.f.) and p.d.

D e.m.f. and charge

0625/21/O/N/24 Q26

Date: _____ Score: _____

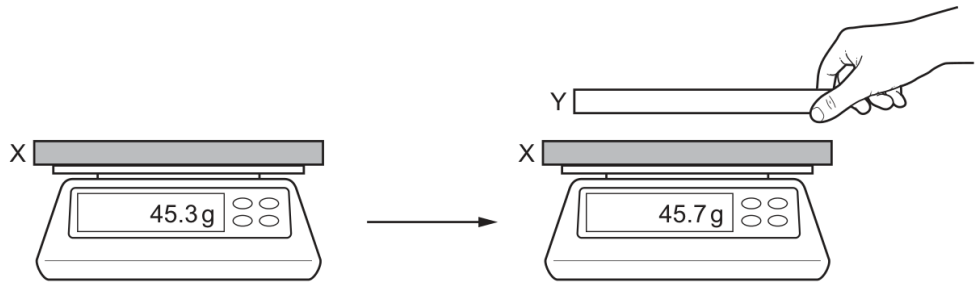
27 A copper wire is placed in an electrical circuit and its resistance is measured. The wire is then replaced with a second copper wire with twice the length and a quarter of the diameter.

What is the ratio of the resistance for the two wires?

A 1 : 2 **B** 1 : 4 **C** 1 : 8 **D** 1 : 32

0625/21/O/N/24 Q27

23 A charged rod X is placed on a balance and another rod Y is brought close to it, as shown.



Which combination of charges would cause the change in the balance reading shown?

	X	Y
A	negative charge	negative charge
B	negative charge	positive charge
C	negative charge	no charge
D	positive charge	no charge

0625/22/F/M/24 Q23

24 What is the unit of potential difference (p.d.)?

A ampere

B coulomb

C coulomb / second

D volt

0625/22/F/M/24 Q24

Date: _____ Score: _____

27 A nichrome wire, P, has a length of 1.0 m and a diameter of 1.0 mm.

A second nichrome wire, Q, has a length of 2.0 m and a diameter of 2.0 mm.

Which statement correctly compares the resistance of Q with that of P?

A The resistance of Q is one-quarter the resistance of P.

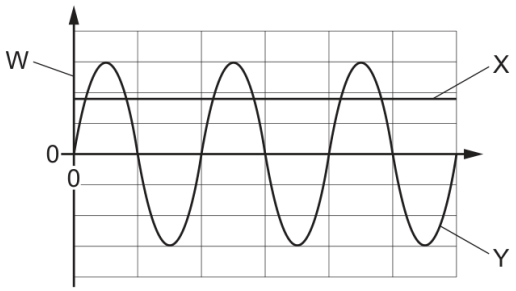
B The resistance of Q is one-half the resistance of P.

C The resistance of Q is the same as the resistance of P.

D The resistance of Q is four times the resistance of P.

0625/22/F/M/24 Q27

26 A student draws graphs of an a.c. supply and a d.c. supply.



Which row shows the correct labels for W, X and Y?

	label of W	a.c. supply	d.c. supply
A	time	X	Y
B	time	Y	X
C	voltage	X	Y
D	voltage	Y	X

0625/22/M/J/24 Q26

27 A resistor transfers 100 J of energy when a charge of 10 C flows through it.

What is the potential difference across the resistor?

A 0.10 V **B** 10 V **C** 10 W **D** 1000 W

0625/22/M/J/24 Q27

28 Wire X and wire Y are made from the same metal.

The table gives information about the two wires.

	length / m	cross-sectional area / mm^2	resistance / Ω
wire X	1.0	0.40	2.0
wire Y	1.3	0.30	

What is the resistance of wire Y?

A 0.31 Ω **B** 0.78 Ω **C** 2.1 Ω **D** 3.5 Ω

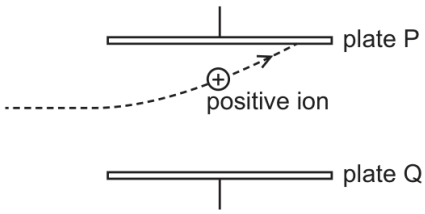
0625/22/M/J/24 Q28

25 A positive ion in a vacuum moves at a steady speed in a straight line.

There is a potential difference between the two parallel metal plates P and Q so that they are oppositely charged.

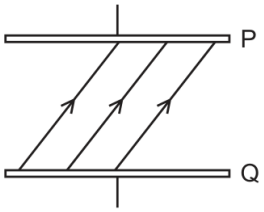
When the positive ion enters the region between plates P and Q, it changes direction as shown.

The diagram shows the view from above the ion.

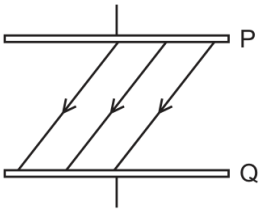


Which diagram shows the direction of the electric field between the plates?

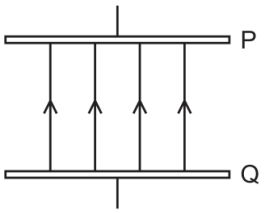
A



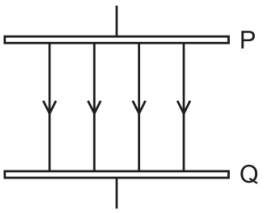
B



C



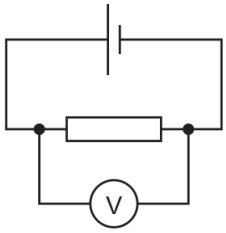
D



0625/22/O/N/24 Q25

Date: _____ Score: _____

26 In the circuit shown, the voltmeter reads 2.0 V. A charge of 5.0 C passes through the resistor in a certain time.

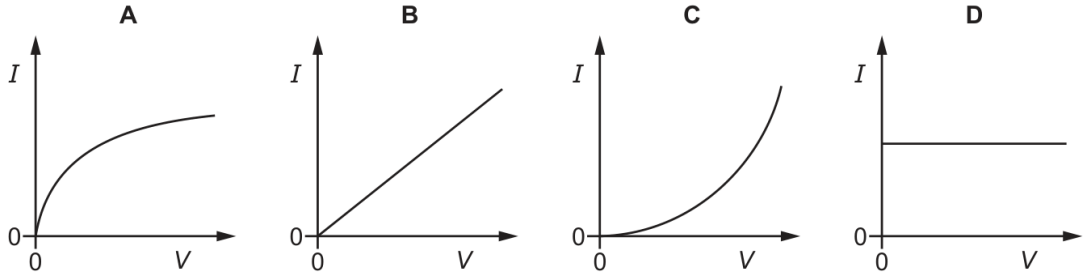


How much energy is supplied to the resistor in this time?

A 0.40 J **B** 2.5 J **C** 10 J **D** 20 J

0625/22/O/N/24 Q26

27 Which diagram shows the current–voltage (I–V) graph for a filament lamp?



0625/22/O/N/24 Q27

27 A student rubs a plastic rod with a cloth.

The rod becomes electrically charged and is attracted by a negatively charged object.

Which statement about the rod is correct?

A It is negatively charged because it has gained electrons.

B It is negatively charged because it has lost electrons.

C It is positively charged because it has gained electrons.

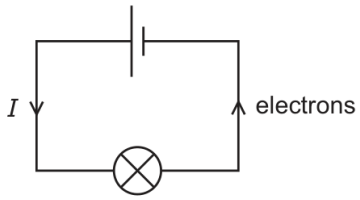
D It is positively charged because it has lost electrons.

0625/23/M/J/24 Q27

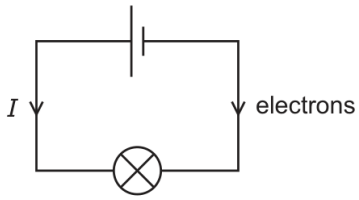
Date: _____ Score: _____

28 Which circuit shows the directions of the conventional current I and the flow of electrons?

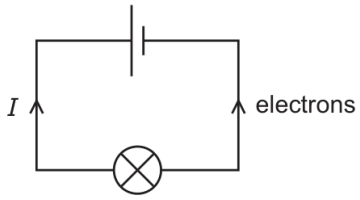
A



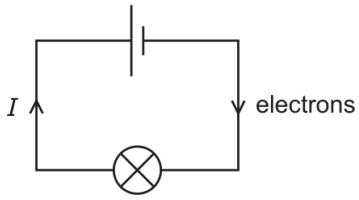
B



C



D



0625/23/M/J/24 Q28

29 A cell has an electromotive force (e.m.f.) of 1.5 V.

How much energy is transferred by the cell when it drives 6.0 C of charge round a complete circuit?

A 4.0 J **B** 4.0 W **C** 9.0 J **D** 9.0 W

0625/23/M/J/24 Q29

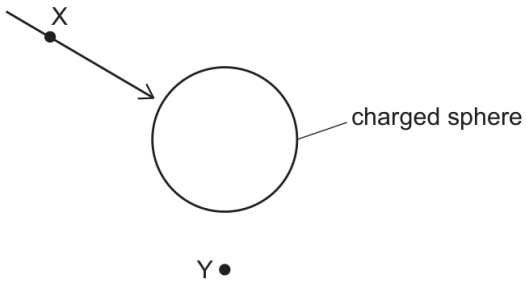
30 A metal resistance wire of length l and cross-sectional area A has resistance R.

What is the resistance of a metal wire of the same material that has length $2l$ and cross-sectional area $\frac{A}{2}$?

A $\frac{R}{4}$ **B** R **C** $2R$ **D** $4R$

0625/23/M/J/24 Q30

25 The diagram shows a charged sphere and the direction of the electric field at point X due to the sphere.



An electron is placed at point Y.

Which diagram shows the direction of the force acting on the electron due to the charged sphere?

A

B

C

D

0625/23/O/N/24 Q25

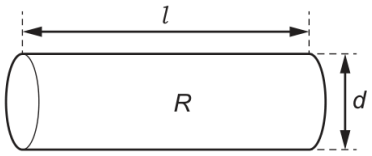
26 A battery moves 3.0 C of charge round a complete circuit. The battery transfers 12 J of energy during this process.

What is the electromotive force (e.m.f.) of the cell?

A 0.25 V **B** 4.0 V **C** 12 V **D** 36 V

0625/23/O/N/24 Q26

27 A cylinder of conducting putty has length l, diameter d and resistance R. The putty is now moulded into a cylinder of diameter $2d$ that has the same volume.



By which factor does the resistance of the putty cylinder decrease?

A 2 **B** 4 **C** 8 **D** 16

0625/23/O/N/24 Q27

6 Fig. 6.1 shows an isolated metal sphere suspended by an insulating thread from the ceiling.

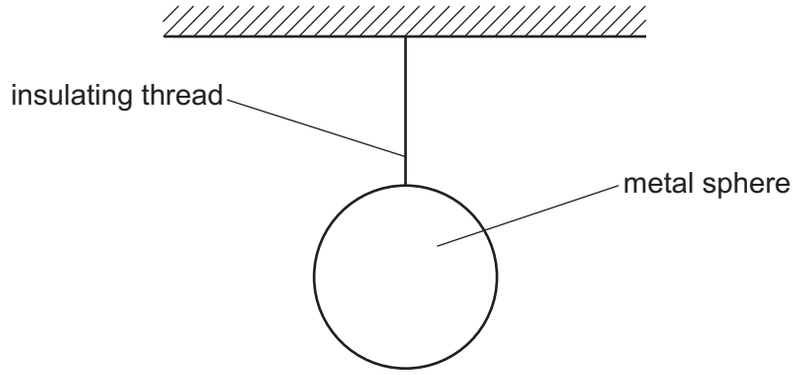


Fig. 6.1

The sphere is negatively charged.

(a) The charge on the sphere produces an electric field in the surroundings.

(i) State what is meant by 'electric field'.

.....

..... [1]

(ii) Draw on Fig. 6.1 to show the pattern and direction of the electric field produced by the charge on the sphere. Draw at least **four** lines. [3]

(b) The magnitude of the charge on the sphere is 3.5×10^{-10} C.

An earthed metal wire is touched against the surface of the sphere and the sphere is discharged.

(i) State what happens in the wire as the sphere is discharged.

.....

.....

..... [2]

(ii) It takes a time of 0.14 ns for the sphere to discharge completely.

Calculate the average current in the earthed wire as the sphere discharges.

average current = ... [3]

[Total: 9]

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[Turn over

7 The electromotive force (e.m.f.) of a battery is 7.5 V.

(a) Define the term electromotive force.

.....

.....

..... [2]

(b) The battery is connected in series with a variable resistor and a 30 Ω resistor. The battery is made using 1.5 V cells.

(i) Draw a circuit diagram that shows all the 1.5 V cells connected to produce an e.m.f. of 7.5 V, the variable resistor and the 30 Ω resistor.

[3]

(ii) The resistance of the variable resistor can be varied from 0 Ω to a maximum resistance of 150 Ω.

Using the axes in Fig. 7.1, draw a graph to show how the current in the circuit varies with the resistance of the variable resistor as it increases from 0 Ω to 150 Ω.

Determine and label the value of the maximum current on the *y*-axis.

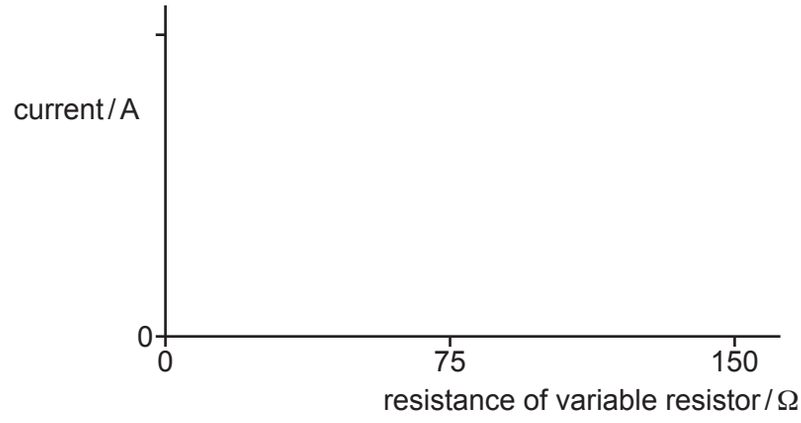

Fig. 7.1

[4]

[Total: 9]

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7 A washing machine has an electric motor and an electric heater. Fig. 7.1 shows a simplified circuit diagram for the washing machine.

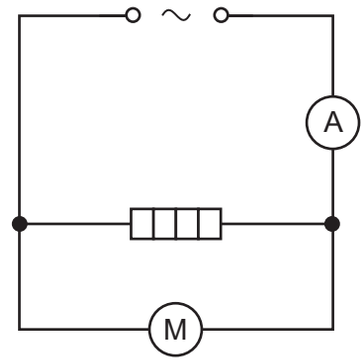

Fig. 7.1

The heater has a resistance of 25 Ω and the power supply has an electromotive force (e.m.f.) of 230 V.

(a) State the meaning of electromotive force.

.....

..... [2]

(b) State the potential difference (p.d.) across the heater.

p.d. = ... [1]

(c) Calculate the current in the heater.

Current = ... [2]

(d) The current in the motor is 1.6 A.
Determine the reading on the ammeter in Fig. 7.1. Explain your answer.

Ammeter reading .....

Explanation .....

.....
[2]

[Total: 7]

7 Fig. 7.1 shows two charged metal plates. X marks the position of the centre of the space between the plates.

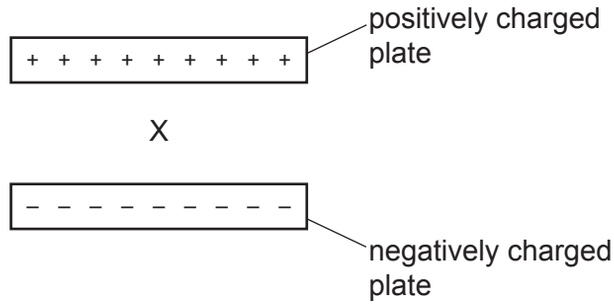

Fig. 7.1

(a) **(i)** On Fig. 7.1, draw at least **four** field lines to show the pattern and the direction of the electric field between the two charged plates. [2]

(ii) Describe the effect on a negatively charged particle placed at X.

.....

..... [1]

(b) During a thunderstorm, an electric field is set up between a cloud and the ground. Charges on the cloud and on the ground are shown in Fig. 7.2.

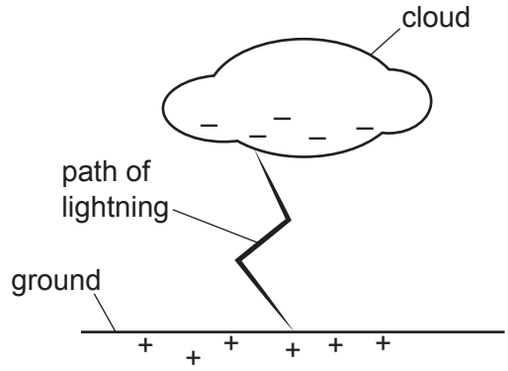

Fig. 7.2

The lightning shown in Fig. 7.2 discharges a current of 28 000 A for 0.0012 s.

(i) Calculate the charge that flows from the cloud to the ground.

charge = ... [2]

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(ii) The lightning transfers 1.2×10^8 J of energy.

Calculate the potential difference between the base of the cloud and the ground.

potential difference = ... [2]

[Total: 7]

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5 Fig. 5.1 shows a metal sphere S. The sphere has been charged with a negative charge.

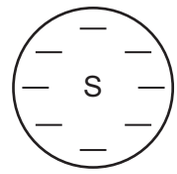

Fig. 5.1

(a) (i) There is an electric field around sphere S.

On Fig. 5.1, draw **four** field lines to show the pattern of the field and indicate the direction of the field with arrows on the lines. [2]

(ii) Fig. 5.2 shows a position X next to sphere S.

A small negatively charged particle is placed at position X.

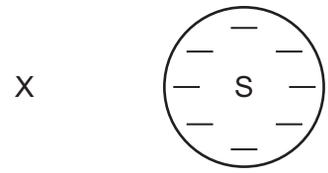

Fig. 5.2

State the direction of the force on the negatively charged particle at X due to the electric field around sphere S.

..... [1]

(iii) The negatively charged particle at X is released from rest.

Describe the motion of the small negatively charged particle due to the electric field around sphere S.

.....

.....

..... [2]

(b) Fig. 5.3 shows sphere S being spray painted. Sphere S is negatively charged. As the paint particles exit the wide nozzle of the paint sprayer, they become charged with a positive charge.

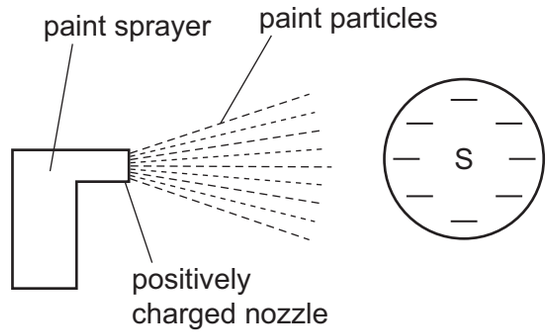

Fig. 5.3

(i) Explain why the paint particles spread out when they leave the nozzle.

.....

..... [1]

(ii) The sphere can be painted by hand using a paintbrush.

Suggest and explain **one** advantage to using charged paint from a spray gun to paint sphere S.

advantage .....

explanation .....

.....

.....

[2]

[Total: 8]

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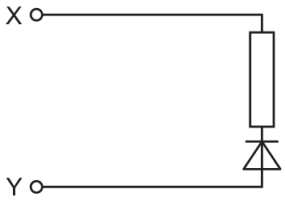
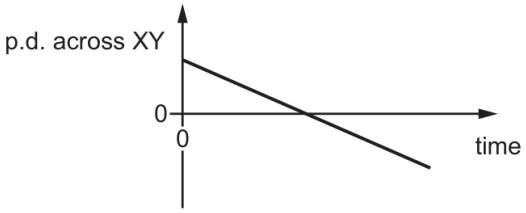
4.3 Electric circuits

14 MCQ + 4 theory · Paper 2 & Paper 4

Date: _____ Score: _____

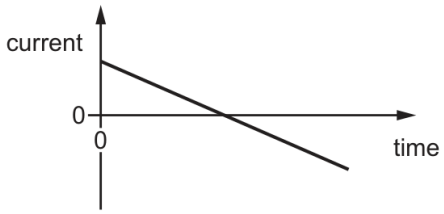
27 A diode and a resistor are connected across a variable d.c. supply. Terminal X is initially positive and at a maximum value. The potential difference (p.d.) across XY is adjusted so that it changes, as shown by the graph.



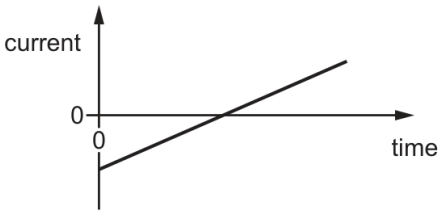


Which graph shows how the current in the circuit (from X to Y) varies during the same time interval?

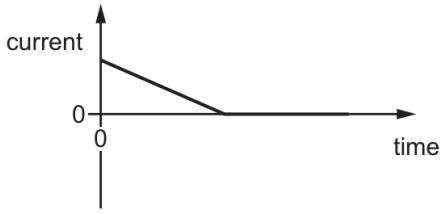
A



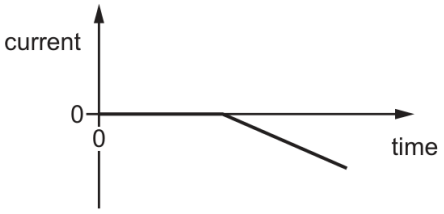
B



C



D

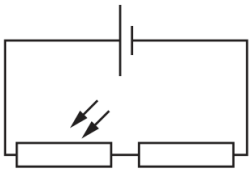


0625/21/M/J/24 Q27

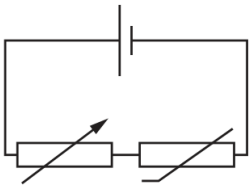
Date: _____ Score: _____

28 Which circuit contains a diode connected in series with a fixed resistor?

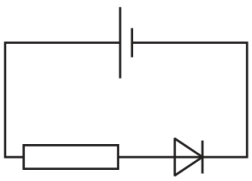
A



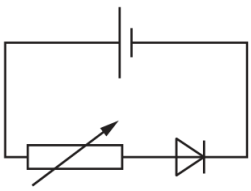
B



C

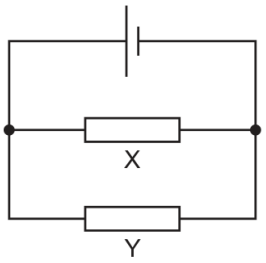


D



0625/21/O/N/24 Q28

29 In the circuit shown, resistor X has twice the resistance of resistor Y.



Which statement about the circuit is correct?

A The current in the cell is greater than the current in X.

B The current in X is greater than the current in Y.

C The potential difference (p.d.) across the cell is greater than the p.d. across X.

D The p.d. across X is greater than the p.d. across Y.

0625/21/O/N/24 Q29

30 A 4 Ω resistor and an 8 Ω resistor are connected in series with a power supply.

A voltmeter connected across the 8 Ω resistor reads 36 V.

What is the voltmeter reading when it is connected across the 4 Ω resistor?

A 12 V **B** 18 V **C** 72 V **D** 108 V

0625/21/O/N/24 Q30

28 A student has two resistors.

One resistor has a resistance of 2.0 Ω and the other has a resistance of 40 kΩ.

Which statement is correct?

A When connected in parallel, the combined resistance is 1.9 Ω.

B When connected in parallel, the combined resistance is 2.0 Ω.

C When connected in series, the combined resistance is 1.9 Ω.

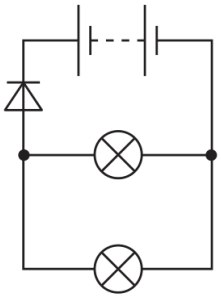
D When connected in series, the combined resistance is 42 Ω.

0625/22/F/M/24 Q28

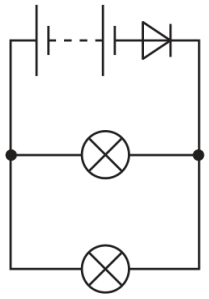
Date: _____ Score: _____

29 In which circuit do both lamps light?

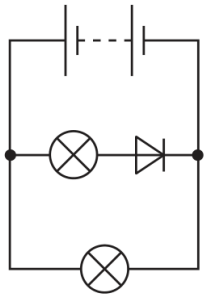
A



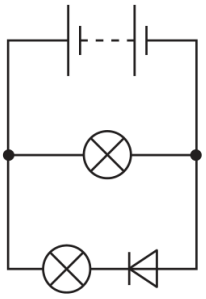
B



C

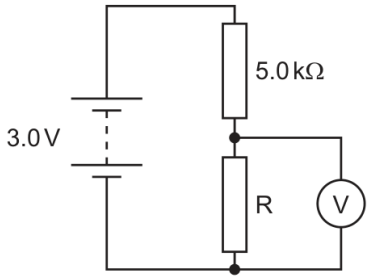


D



0625/22/F/M/24 Q29

29 The diagram shows two resistors used as a potential divider.



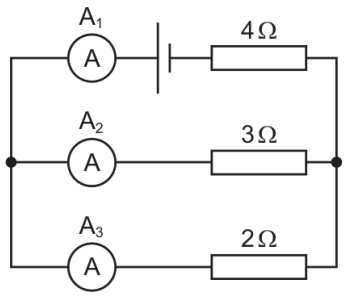
The reading on the voltmeter is 1.8 V.

What is the resistance of the resistor R?

A 3.0 kΩ **B** 3.3 kΩ **C** 7.5 kΩ **D** 13 kΩ

0625/22/M/J/24 Q29

28 The diagram shows an electrical circuit.



Which statement is correct?

A The current reading on ammeter A_1 is the highest.

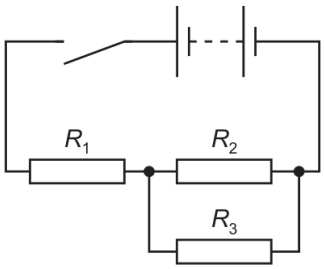
B The current reading on ammeter A_2 is the highest.

C The current reading on ammeter A_3 is the highest.

D The current readings on all three ammeters A_1, A_2 and A_3 are the same.

0625/22/O/N/24 Q28

29 The diagram shows an arrangement of resistors in series and parallel.



Which equation is used to calculate the combined resistance of the resistors?

A $R_1 + \dfrac{R_2 + R_3}{R_2 \times R_3}$

B $R_1 + \dfrac{R_2 \times R_3}{R_2 + R_3}$

C $R_1 + \dfrac{1}{R_2 + R_3}$

D $R_1 \times \dfrac{R_2 + R_3}{R_2 \times R_3}$

0625/22/O/N/24 Q29

30 An electrical engineer has made a pair of components P and Q.

The resistance of P decreases as the temperature rises. The resistance of Q increases as the temperature rises.

The total resistance of P and Q in series remains constant when the temperature changes. The two components are connected in series with a power source supplying a constant current.

Which statement is correct?

A If a temperature change causes the potential difference (p.d.) across P to double, the p.d. across Q will always halve.

B The difference between the p.d. across P and the p.d. across Q does not change with temperature.

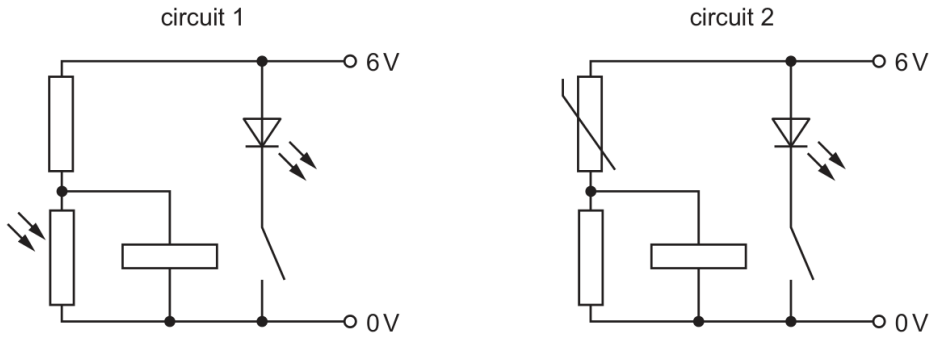
C The p.d. across component P increases as the temperature rises.

D The sum of the p.d. across P and the p.d. across Q does not change with temperature.

0625/22/O/N/24 Q30

31 The diagram shows two circuits.

Each relay is normally open when there is no current in the coil.



Which environmental conditions will cause the light-emitting diodes in circuit 1 and circuit 2 both to emit light?

	temperature	light intensity
A	high	high
B	high	low
C	low	high
D	low	low

0625/23/M/J/24 Q31

28 Which statement describes how a diode operates in an electrical circuit?

A When the temperature increases, its resistance increases.

B When the current becomes too high, it melts and cuts off the current.

C It only allows current in one direction.

D It uses a low-current circuit to switch on a circuit carrying a much larger current.

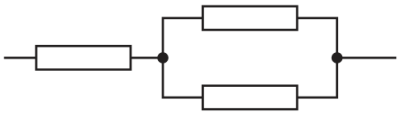
0625/23/O/N/24 Q28

Date: _____ Score: _____

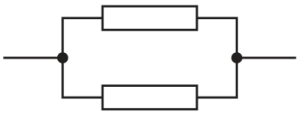
29 A number of identical resistors of resistance 10 Ω are arranged in circuits.

Which arrangement of resistors will give an overall resistance of 25 Ω?

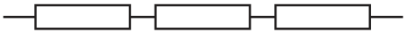
A



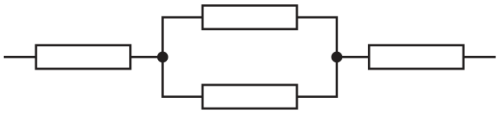
B



C



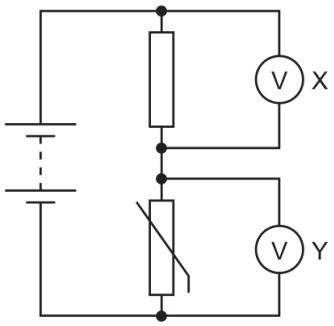
D



0625/23/O/N/24 Q29

30 The circuit diagram shows a thermistor in a potential divider.

Two voltmeters X and Y are connected in the circuit.



When the temperature of the thermistor increases, its resistance decreases.

As the thermistor becomes warmer, what happens to the reading on each voltmeter?

	reading on voltmeter X	reading on voltmeter Y
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

0625/23/O/N/24 Q30

* 0000800000014 *

6 A potential divider is made by connecting a light-dependent resistor (LDR) and a thermistor in series. Fig. 6.1 shows the potential divider, a voltmeter and a direct current (d.c.) power supply connected into a circuit.

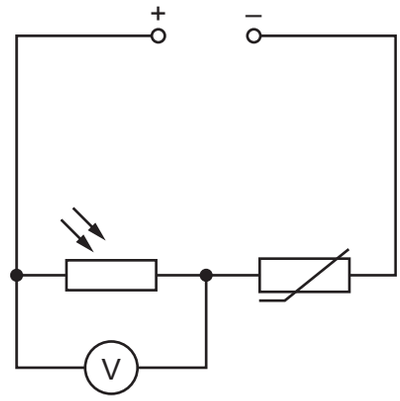


Fig. 6.1

The voltmeter measures the potential difference (p.d.) across the LDR.

(a) Define potential difference (p.d.).

.....

.....

..... [2]

(b) The electromotive force (e.m.f.) of the supply is *E*.

Describe how the p.d. across the thermistor can be determined using the reading on the voltmeter.

.....

..... [1]

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(c) The resistance of the LDR decreases and the resistance of the thermistor increases.

(i) State what has happened to the light intensity incident on the LDR and the temperature of the thermistor.

intensity of incident light on LDR: .....

temperature of thermistor: .....

[1]

(ii) Explain what happens to the reading on the voltmeter.

.....

.....

.....

..... [3]

[Total: 7]

7 **(a)** Draw the circuit symbol for a potential divider.

[1]

(b) Fig. 7.1 shows a circuit.

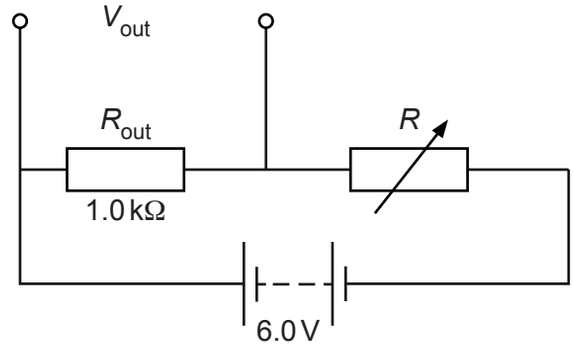

Fig. 7.1

(i) Calculate the value of V_{out} when the value of R is 3.0 kΩ.

V_{out} = ... [2]

(ii) The value of R is adjusted until the current in the circuit is 1.7 mA.

Calculate the charge that flows through the circuit in 300 s.

charge = ... [2]

[Total: 5]

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[Turn over

* 0019655392714 *

8 **(a)** Fig. 8.1 shows a circuit. The circuit is designed to switch on a night light when the surroundings are dark.

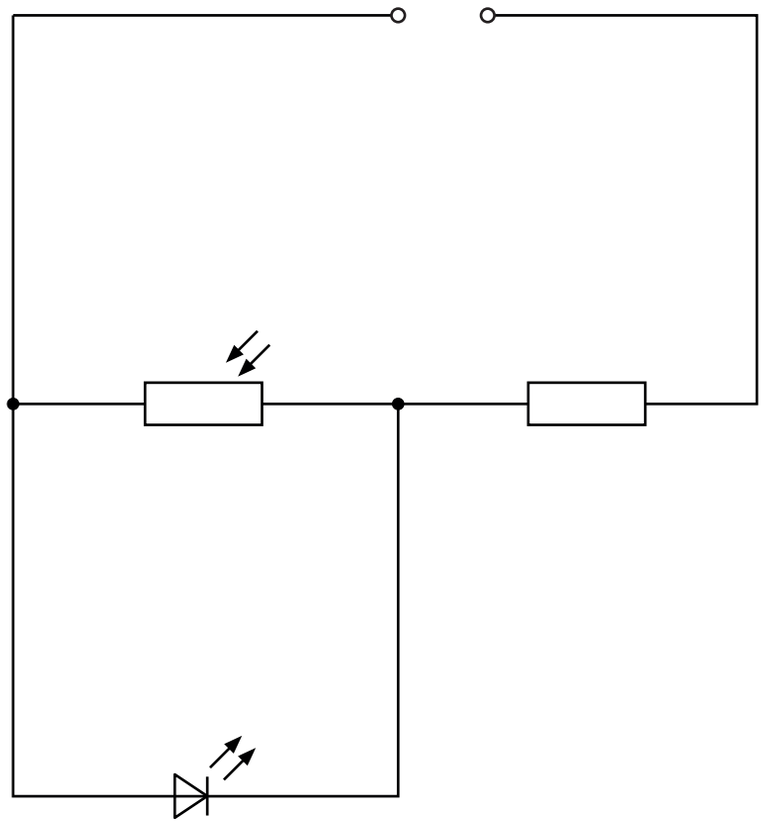

Fig. 8.1

(i) On Fig. 8.1, draw the circuit symbol for a voltmeter used to measure the potential difference (p.d.) across the light-dependent resistor (LDR). [1]

(ii) The surroundings change from light to dark.

1. State the effect of this change on the resistance of the LDR.

..... [1]

2. State and explain the effect of this change on the p.d. across the light-emitting diode (LED).

.....

.....

..... [2]

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(b) Fig. 8.2 shows another circuit. Lamps A and B are identical filament lamps.

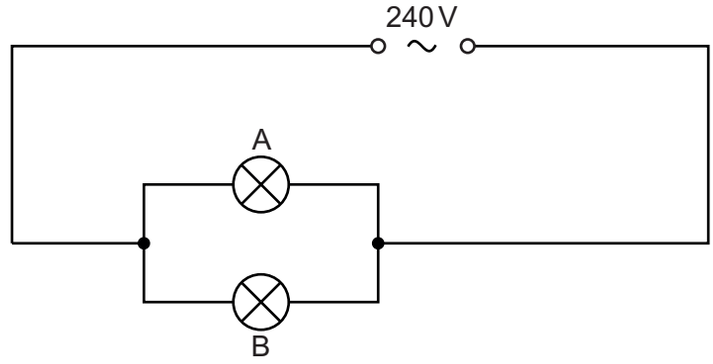

Fig. 8.2

The current supplied by the power supply is 0.50 A.

Calculate the resistance of lamp A.

resistance = ... [3]

[Total: 7]

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6 **(a)** A car windscreen is covered in condensation (small droplets of water). Thermal energy is used to remove the droplets of water. The thermal energy is provided by three resistors on the windscreen.

Fig. 6.1 shows two possible circuits for the three resistors.

The three resistors are identical.

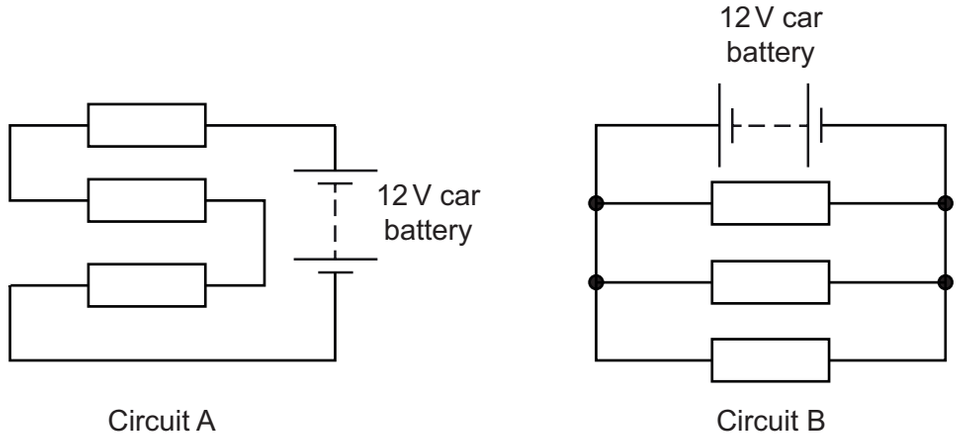

Fig. 6.1

(i) Describe **two** advantages of using Circuit B.

1 .....

.....

2 .....

.....

[2]

(ii) Describe, in terms of the water particles, the process by which the water droplets are removed from the car windscreen using the heater.

.....

.....

.....

..... [2]

* 0000800000014 *

(b) Fig. 6.2 shows a circuit containing two resistors, P and Q. The circuit is powered by a 12 V battery.

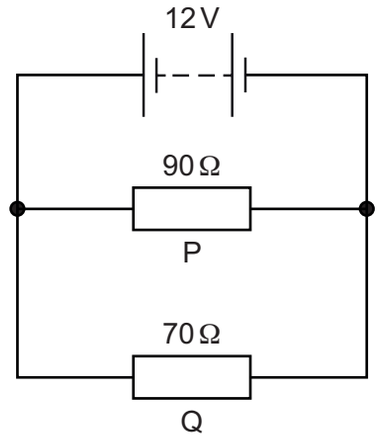

Fig. 6.2

(i) Calculate the current in resistor Q.

current = ... [2]

(ii) Calculate the energy transferred electrically when the current calculated in **(b)(i)** is present in resistor Q for 5 minutes.

energy = ... [3]

(iii) Energy is transferred from the battery by the electrical current.

State the energy store in the battery.

..... [1]

(iv) Calculate the total resistance of the circuit.

total resistance = ... [2]

[Total: 12]

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4.4 Electrical safety

1 MCQ + 0 theory · Paper 2 & Paper 4

Date: _____ Score: _____

30 Two electrical appliances are connected to the mains supply.

The cable connected to one appliance includes an earth wire.

The cable connected to the second appliance does **not** need an earth wire.

What is a reason for this difference?

A One appliance has a metal case, but the other appliance does not.

B One appliance is fitted with a fuse, but the other appliance is not.

C One appliance is fitted with a switch, but the other appliance is not.

D One appliance needs more current than the other appliance.

0625/22/M/J/24 Q30

4.5 Electromagnetic effects

18 MCQ + 4 theory · Paper 2 & Paper 4

Date: _____ Score: _____

28 When a conductor in a complete circuit cuts across a magnetic field, a current is induced in the conductor.

Which statement about the induced current is correct?

A The induced current is in the same direction as the motion of the conductor.

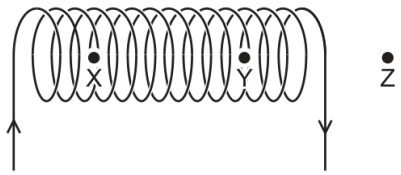
B The induced current is in the opposite direction to the motion of the conductor.

C The direction of the induced current is in the same direction as the magnetic field.

D The direction of the induced current opposes the change causing it.

0625/21/M/J/24 Q28

29 The diagram shows a solenoid carrying an electric current.

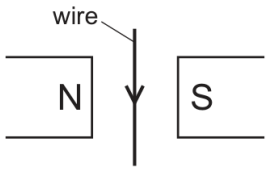


Which row compares the strength of the magnetic field due to the solenoid at points Y and Z with the strength of the magnetic field at point X?

	magnetic field strength at Y	magnetic field strength at Z
A	equal to X	less than X
B	equal to X	greater than X
C	greater than X	less than X
D	greater than X	greater than X

0625/21/M/J/24 Q29

30 A conducting wire is placed between the poles of a magnet. When an electric current in the wire is in the direction shown, then the force on the wire acts out of the page.



Three statements of different conditions and how the wire is affected are given.

1 When the current is towards the top of the page and the direction of the magnetic field is unchanged, the force produced acts into the page.

2 When the current is towards the bottom of the page and the magnetic field is reversed, the force produced acts into the page.

3 When the current in the wire is alternating, the wire vibrates into and out of the page.

Which statements are correct?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

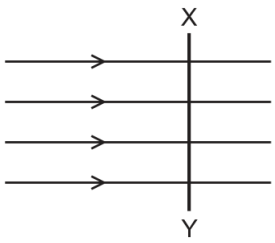
0625/21/M/J/24 Q30

31 Which types of electric current are in the primary coil and the secondary coil of a step-up transformer?

	primary coil	secondary coil
A	a.c.	a.c.
B	a.c.	d.c.
C	d.c.	a.c.
D	d.c.	d.c.

0625/21/M/J/24 Q31

31 A metal rod, XY, is placed in a magnetic field so that it is perpendicular to the field, as shown.



In which direction is the rod moved so that there is an induced current from Y to X?

A left to right

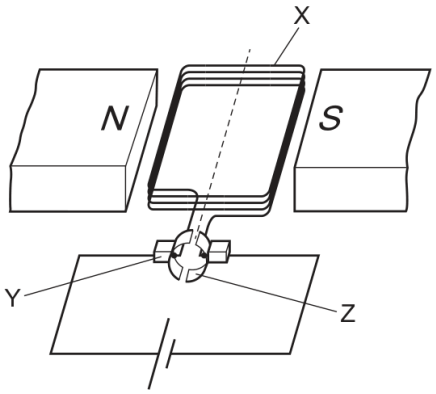
B right to left

C into the paper, away from the observer

D out of the paper, towards the observer

0625/21/O/N/24 Q31

32 The diagram shows a basic electric motor.



What are the names of the features?

	X	Y	Z
A	axle	battery	brushes
B	coil	brushes	commutator
C	generator	wires	transformer
D	solenoid	connectors	switch

0625/21/O/N/24 Q32

33 Two transformers, P and Q, each have 2000 turns on their primary coils and 1000 turns on their secondary coils. Both transformers are 100% efficient.

The secondary of transformer P is connected to the primary of transformer Q.

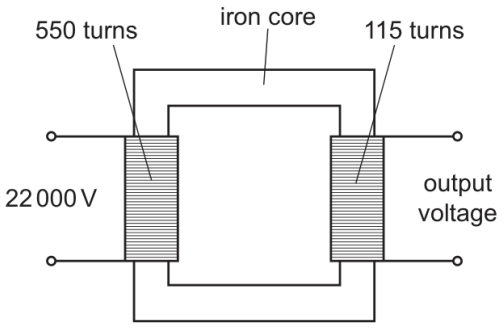
Q provides a motor with a current of 0.2 A.

What is the current being supplied to the primary of transformer P?

A 0.8 A **B** 0.4 A **C** 0.1 A **D** 0.05 A

0625/21/O/N/24 Q33

25 The diagram shows a transformer.



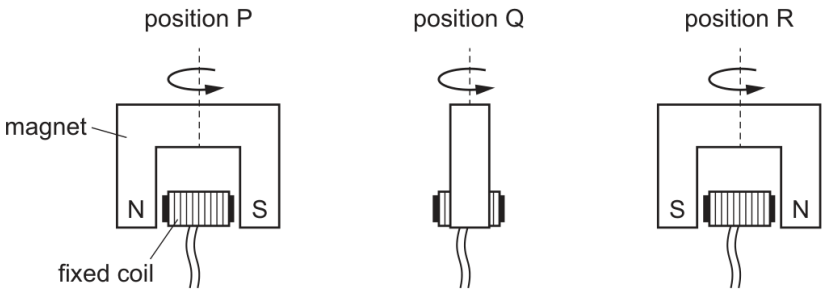
What is the output voltage?

A 0.35 V **B** 2.9 V **C** 4600 V **D** 105 000 V

0625/22/F/M/24 Q25

31 The diagram shows a simple a.c. generator. The magnet rotates around a fixed coil wound round a fixed iron core. The electromotive force (e.m.f.) output is a sine wave, a series of crests, zeros and troughs.

The magnet is shown in three positions, P, Q and R, during its rotation. The magnetic field in the coil is zero at Q but changing most rapidly.



Which statement about the induced e.m.f. in the coil is correct?

A The magnitude of the e.m.f. at position Q is larger than that at position P.

B The magnitude of the e.m.f. at position Q is zero.

C The magnitude of the e.m.f. at position Q is smaller than at position R, but not zero.

D The magnitude of the e.m.f. at position R is larger than at position P.

0625/22/M/J/24 Q31

32 A 24 V power supply and a 100% efficient transformer are used to light a 6.0 V, 12 W lamp.

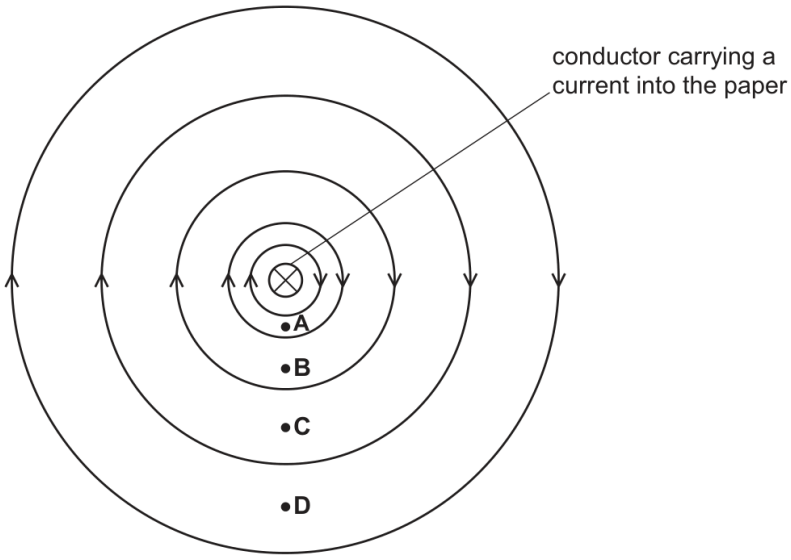
What is the current in the power supply?

A 0.50 A **B** 2.0 A **C** 3.0 A **D** 8.0 A

0625/22/M/J/24 Q32

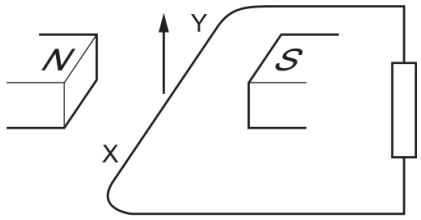
24 The diagram shows the magnetic field around a conductor which is carrying a current.

Where is the strength of the field greatest?



0625/22/O/N/24 Q24

31 The diagram shows a wire XY moving upwards in a magnetic field.



A current is induced in the wire.

The current-carrying wire XY experiences a force because of its interaction with the magnetic field.

Which row gives the direction of the induced current and the direction of the force experienced by the wire XY?

	direction of induced current	direction of the force
A	from X to Y	downwards
B	from X to Y	upwards
C	from Y to X	downwards
D	from Y to X	upwards

0625/22/O/N/24 Q31

32 Diagram 1 shows a magnet being pushed into a coil that is connected to a centre-zero sensitive voltmeter.

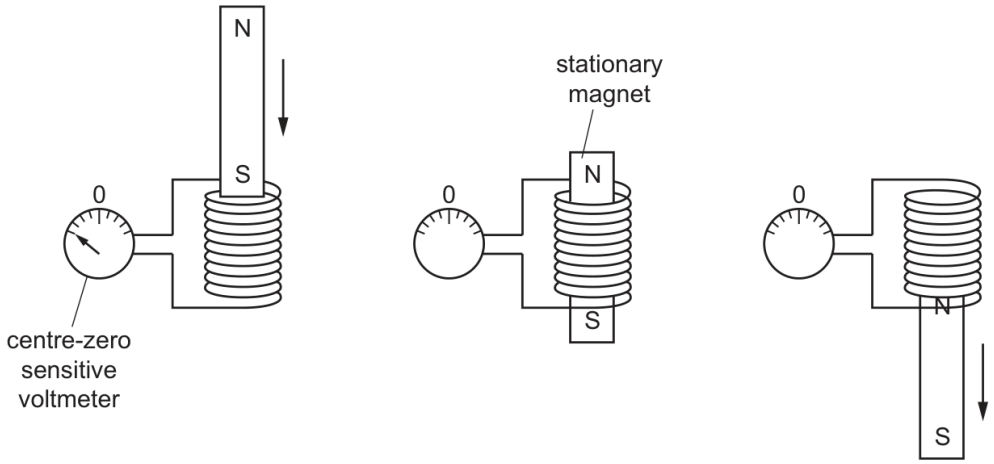


diagram 1 diagram 2 diagram 3

Which row shows the directions of the pointer on the voltmeter when the magnet is as shown in diagrams 2 and 3?

	diagram 2	diagram 3
A		
B		
C		
D		

0625/22/O/N/24 Q32

33 What is the purpose of the brushes and the split-ring commutator in a d.c. motor?

A to change the size of the current

B to keep the coil turning in the same direction

C to keep the speed of rotation constant

D to make the output an alternating voltage

0625/22/O/N/24 Q33

32 A transformer is 100% efficient.

Which pair of equations are both valid for the transformer?

A $I_pV_p = I_sV_s$ $\quad \frac{V_p}{N_p} = \frac{V_s}{N_s}$

B $\frac{V_p}{I_p} = \frac{V_s}{I_s}$ $\quad N_pV_p = N_sV_s$

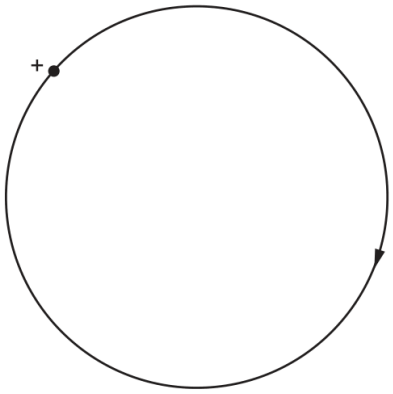
C $\frac{V_p}{I_p} = \frac{V_s}{I_s}$ $\quad \frac{V_p}{N_p} = \frac{V_s}{N_s}$

D $I_pV_p = I_sV_s$ $\quad N_pV_p = N_sV_s$

0625/23/M/J/24 Q32

31 In an experiment, strong magnetic fields provide a force that keeps charged particles moving with a horizontal circular motion.

Positive particles are accelerated clockwise when viewed from above as shown.



In which direction is the magnetic field?

A vertically into the page

B vertically out of the page

C pointing away from the centre of the circular motion

D pointing towards the centre of the circular motion

0625/23/O/N/24 Q31

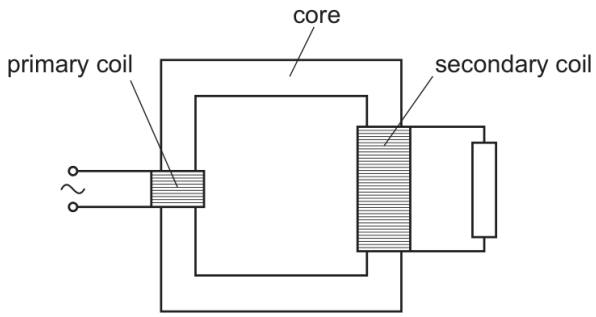
32 When there is a current in a solenoid, a magnetic field is formed around it.

How does the magnetic field change if the current is increased?

	spacing of field lines	strength of field
A	closer together	stronger
B	closer together	weaker
C	further apart	stronger
D	further apart	weaker

0625/23/O/N/24 Q32

33 The diagram shows the structure of a simple transformer.



Which row shows the correct conditions for the efficiency of the transformer to be as high as possible?

	primary and secondary coils	core
A	thick copper wire	soft iron
B	thin copper wire	soft iron
C	thick iron wire	copper
D	thin iron wire	copper

0625/23/O/N/24 Q33

7 A solid bar is inside a copper solenoid. Fig. 7.1 shows that the copper solenoid is connected in series with a battery and a variable resistor.

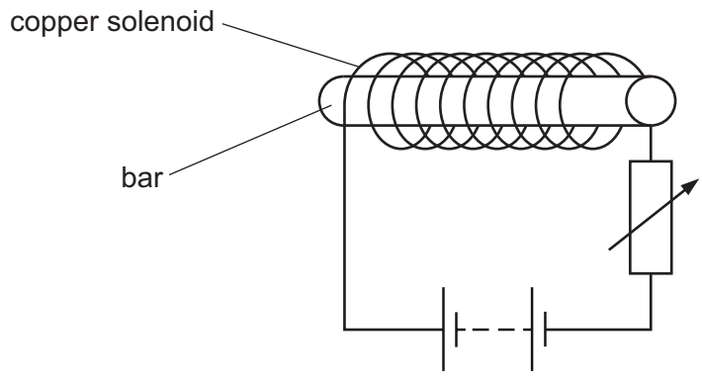


Fig. 7.1

The device shown in Fig. 7.1 is an electromagnet.

(a) Suggest a suitable material for the bar.

..... [1]

(b) The right-hand end of the bar is the S pole.

(i) Fig. 7.2 shows the bar viewed from above.

On Fig. 7.2, draw at least six field lines to show the pattern and direction of the magnetic field surrounding the bar.

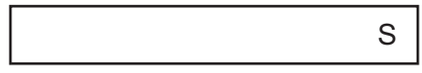


Fig. 7.2

[3]

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(ii) The resistance of the variable resistor increases.

Explain what happens to the magnetic field surrounding the bar and state how the pattern of field lines that represents the field changes.

.....

.....

.....

..... [3]

(c) A square coil of many turns is placed close to the bar. Fig. 7.3 shows the plane of the square coil parallel to the flat circular surface at the right-hand end of the bar.

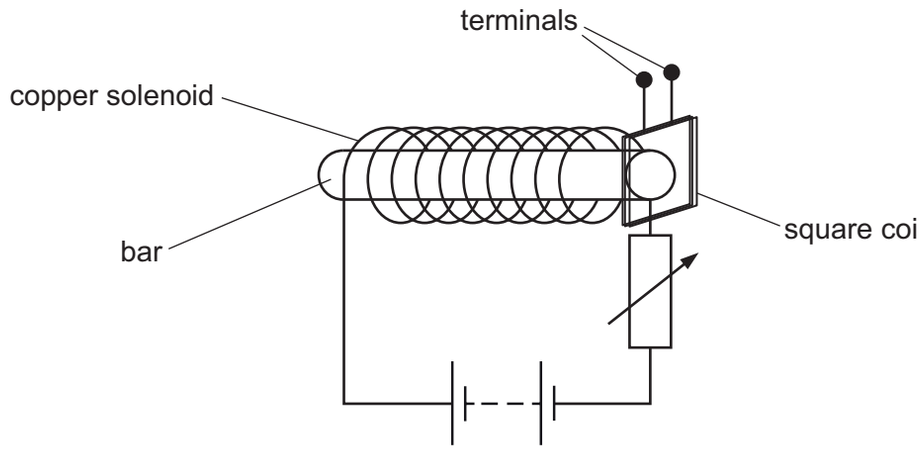

Fig. 7.3

The resistance of the variable resistor is alternately increased and decreased.

Explain what happens in the wires of the square coil.

.....

.....

.....

..... [3]

[Total: 10]

8 **(a)** Fig. 8.1 shows a wire carrying a large current.

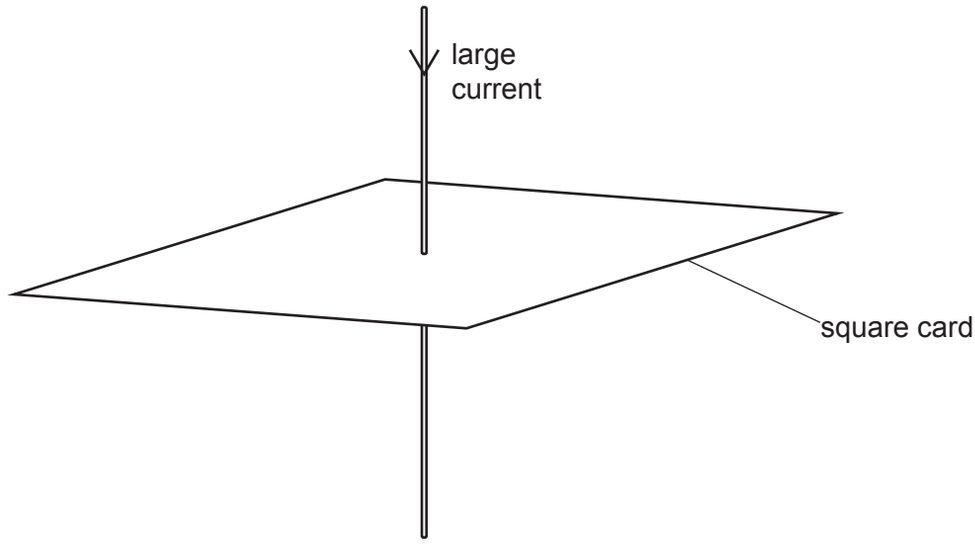

Fig. 8.1

(i) Fig. 8.2 shows the square card viewed from above.

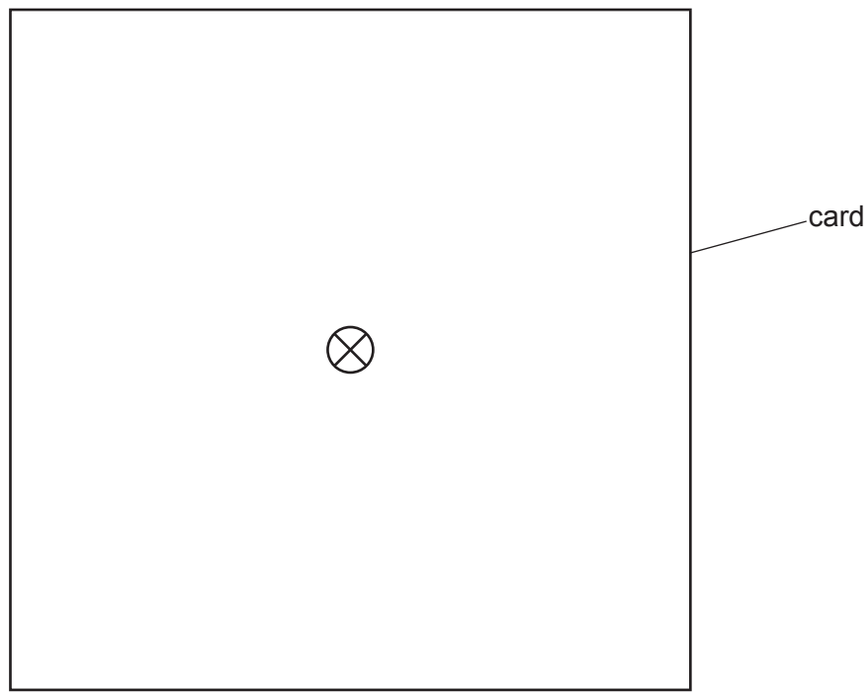

Fig. 8.2

On Fig. 8.2, draw **three** magnetic field lines that indicate the direction of the magnetic field and how its strength varies with distance from the wire. [3]

(ii) The current in the wire increases and the direction of the current is reversed.

State how these changes affect the magnetic field.

.....

..... [2]

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(b) Electricity is transmitted at high voltage.

Explain why a high voltage increases the efficiency of transmission even with thinner wires.

.....

.....

.....

..... [3]

[Total: 8]

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[Turn over

* 0000800000012 *

9 Fig. 9.1 shows a wireless charging plate used to charge the battery in a mobile phone (cell phone). The coil of wire is part of an electric circuit.

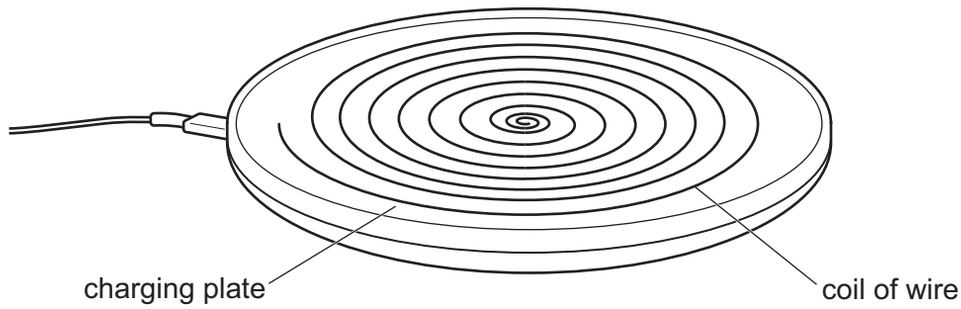

Fig. 9.1

The charging plate is connected to an a.c. power supply. The power supply is turned on.

(a) Describe the magnetic field around the charging plate in terms of its magnitude and direction.

.....

.....

..... [2]

(b) A mobile phone is placed on the charging plate as shown in Fig. 9.2. The coil in the mobile phone is part of a separate circuit that charges the battery.

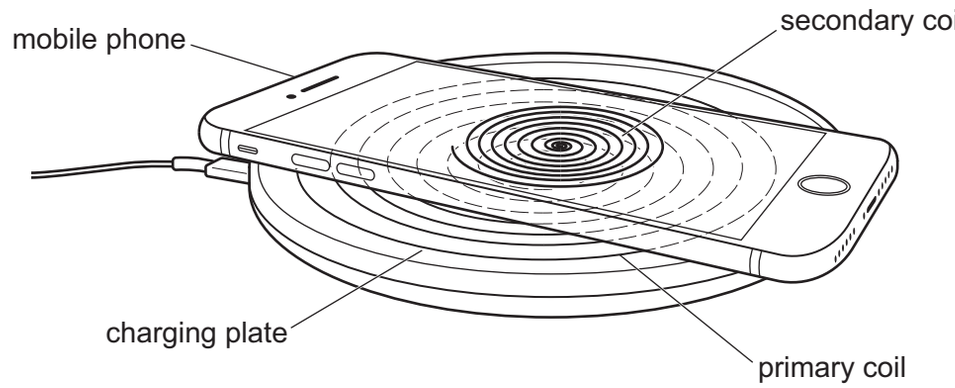

Fig. 9.2

The coil in the charging plate and the coil in the mobile phone act like a transformer.

(i) Explain why there is a current in the secondary coil shown in Fig. 9.2.

.....

.....

..... [2]

(ii) Suggest why the transformer made from the charging plate and mobile phone is not 100% efficient.

.....

..... [1]

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(c) The mobile phone battery can be recharged using this charging plate and stores 4.5×10^4 J of energy when fully recharged. The current in the secondary coil is 0.63 A when the output voltage is 12 V.

(i) Calculate the time taken to fully recharge a completely uncharged battery.

time = ... [2]

(ii) Calculate the charge passing through the battery in 60 s.

charge = ... [2]

[Total: 9]

9 Fig. 9.1 shows a mobile phone (cell phone) being charged on a wireless charging plate.

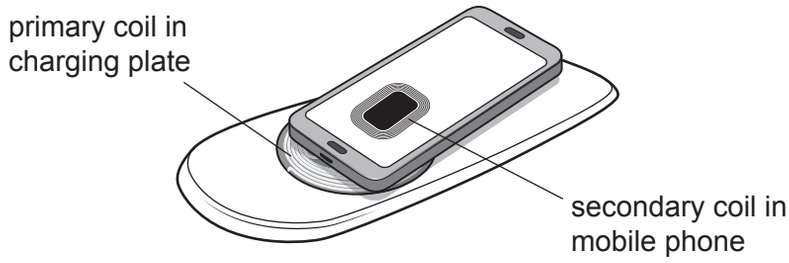

Fig. 9.1

(a) When the charging plate is switched on, there is an alternating current (a.c.) in the primary coil. A secondary coil is in the mobile phone.

Explain how a current is produced in the secondary coil.

.....

.....

..... [3]

(b) The maximum energy stored in the battery of the mobile phone is 0.012 kW h.

(i) Show that this maximum energy is 4.3×10^4 J.

[1]

(ii) The charging plate in Fig. 9.1 has a useful output power of 15 W.
The phone manufacturer claims that the battery can be charged to 50% capacity in less than 30 minutes.

Show that this claim is true.

[3]

[Total: 7]

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[Turn over

Knowledge Recall

知识点默写

Part 1 general physics

Define:

Acceleration(a=Δv/t):_____unit:_____

Force(F=Δp/t）:_____unit:_____

Impulse(=Ft):_____unit:_____

Density(=m/V):_____unit:_____

Pressure(=F/A) :_____unit:_____

gravitational field strength(g=F/m):_____ _____unit:____

Power(=W/t)_____unit:_____

Moment(=Fd):_____unit:__

Momentum(=mv):_____unit:__

Work(=FS)_____unit:__

State what is meant by:

Hooke’s law (F=kx):_____

_____

principle of the conservation of momentum:_____

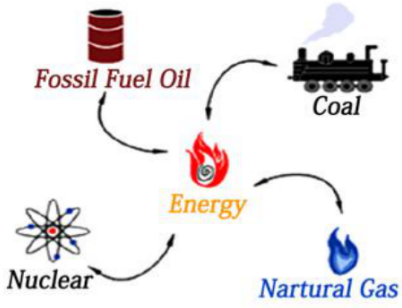
_____ $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

note : v is “+”value in positive direction ,but “—”value in negative direction

- Conditions for a body in _____平衡: i:_____ ii:_____
- _____再生能源&_____不可再生能源

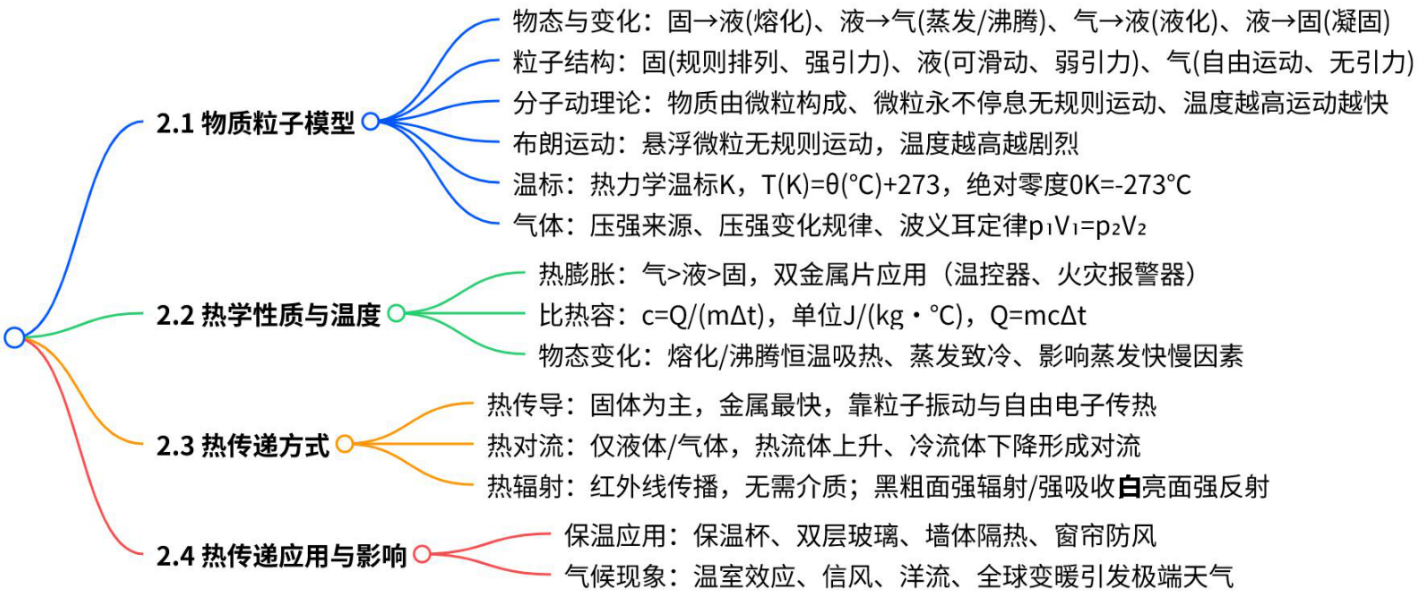
Non-Renewable Energy

不可再生能源都是燃料类型的，用完就没啦：



the Sun is the source of energy for **all** our energy resources

IGCSE Physics----- Thermal Physics Key Summary



1 *Kinetic particle model of matter*

(1)the particle structure of solids, liquids and gases

Solid Fixed shape and volume	**Liquid** Fixed shape changed volume	**Gas** no fixed shape and volume
•Strong attractive force hold Particles close together •_____规则排列 •____振动 in fixed positions	•Weaker attractive forces than solids to hold Particles together. •_____不规则排列 •Particles vibrate but can _____滑动over each other	• no attractive forces between particles •Particles apart to each other • Particles move and_____

(2) evidence for the kinetic molecular model of matter

_____布朗运动:_____

_____

(3) **simple kinetic molecular model of matter**物质的分子动理论模型

i, Matter is made up of very tiny particles (_____分子, _____原子, _____离子**)**

ii, All the particles can move _____随机的 **and** continuously持续的.

iii, The higher the temperature, the _____**the particles move.**

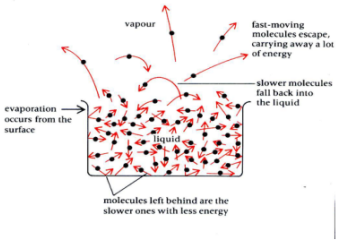
temperature is a measure of _____平均动能 **of molecules.**

- _____**内能--the sum of** _____**分子动能** and_____**分子势能.**

➢ why evaporation makes things cooler?为什么蒸发让物体变冷

_____

_____



➢ **what is the difference between boiling and** _____**蒸发**:

(i) _____

(ii)_____

（4）the _____开尔文温度（热力学温标） **unit:**_____**(K)**

_____绝对零度:_____the point at which molecules have no kinetic energy

absolute zero=0K=_____**°C**

T (K) = θ (°C) + _____

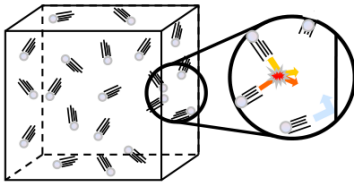
(5) Gases

➢ **Explain gas pressure** in terms of the change of momentum of the particles striking the walls creating a force.

_____

_____

_____



➢ Pressure changes

•**If the volume is constant, then increasing the temperature will increase the pressure.**

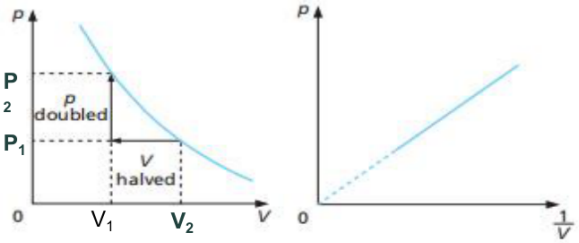
explain:_____.

If the temperature is constant, then decreasing the volume will increase the pressure.

explain:_____

➢ **For a fixed mass of gas, at constant temperature, its pressure is_____反比 to its volume.**

pV = constant or $P_1 \times V_1 = P_2 \times V_2$



2 Thermal properties and temperature

(1) _____**热膨胀** of solids, liquids and gases

_____加热后分子振动更剧烈，间距变大，物体体积变大

expansion is greatest in_____, less in_____ and lowest in_____

(2)_____比热容 $C = \dfrac{Q}{m\Delta t}$

- define :_____ **unit:** _____

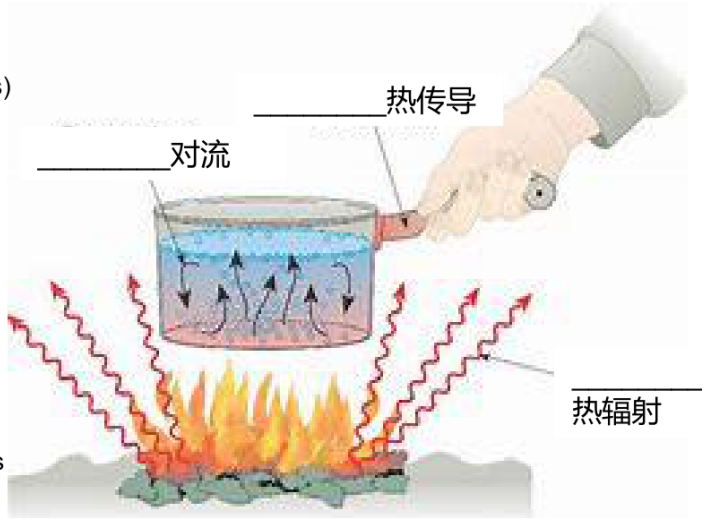
3._____热传递

(1)_____热传导is best in _____（solids，liquids or gas)
energy passed by particle _____振动。

➢ metal is better _____热导体but worse____绝缘体
because energy passed by ion vibration and a better way
---electrons are free to move randomly and collide with atoms
then pass on the energy.

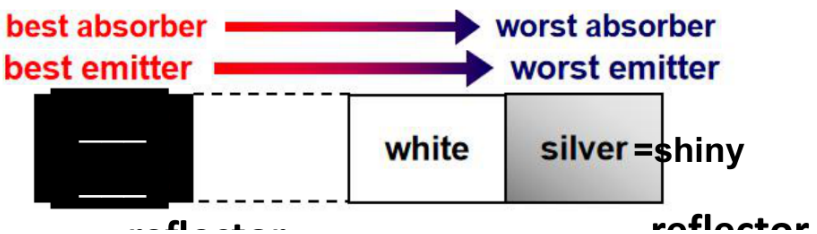
(2)_____对流 occurs in _____and _____
（solids，liquids or gas)

Warm up→ less dense → rises; cool down→ more dense→ sinks



•(3)_____热辐射： All objects **emit** (give out) thermal energy in the form of_____红外线 but very hot objects also give out light waves.

• An emitter will cool down quickly, an absorber will heat up more quickly and a reflector will not heat up quickly



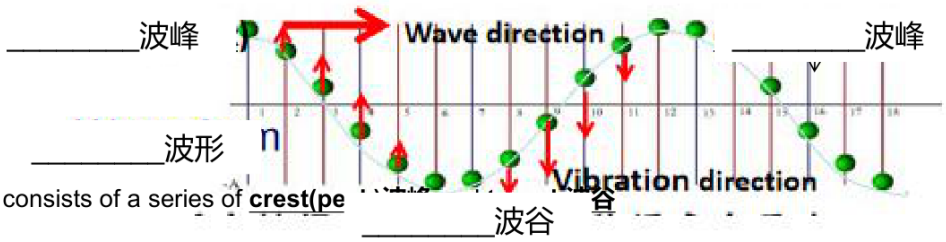
3 waves

1 General properties of waves

(1)two forms:根据传播方向分类

- _____横波:_____

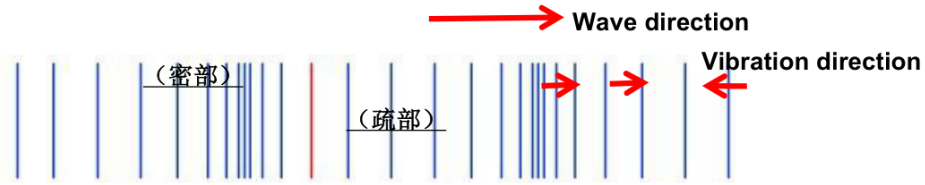
Examples：_____电磁波 ，_____水波



wave form波形 consists of a series of crest(pe ... 谷

- _____纵波:_____

Examples:_____声波

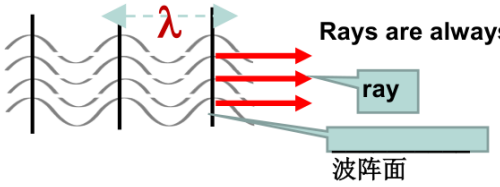


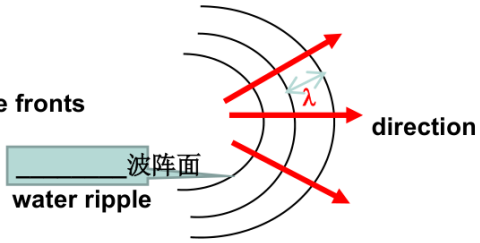
wave form波形 consists of a series of _____密部(partivle close together) and _____疏部(particles furthest away)

- In an earthquake there are two types of seismic waves:
  primary P-waves and secondary S-waves.

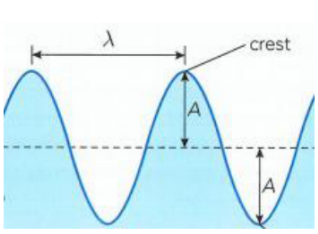
P-waves are_____These waves are called **primary waves** as they travel_____ and so are felt first.
S- waves are_____**called Secondary wave**

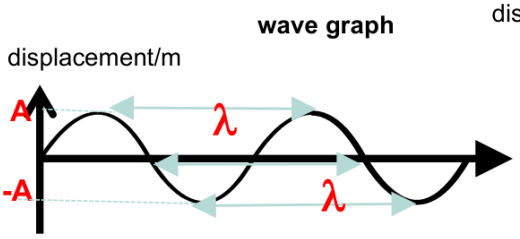
(2) to describe the wave:

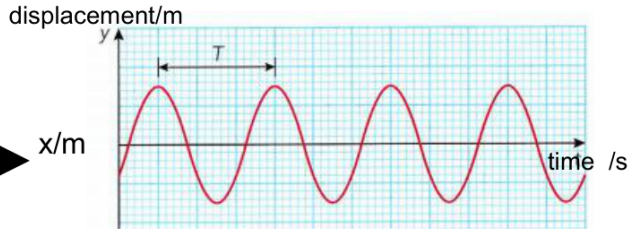




(1)A---amplitude（振幅）:_____
(2)λ----wavelength（波长）：_____unit :m
(3)f---frequency（频率）:_____. **f=1/T unit: hertz (Hz赫兹) = 1/second (1/s)**
(4)T---period（周期）：_____. **T=1/f unit:s**
(5)v---wave speed（波速）v=_____公式 **unit: m/s**

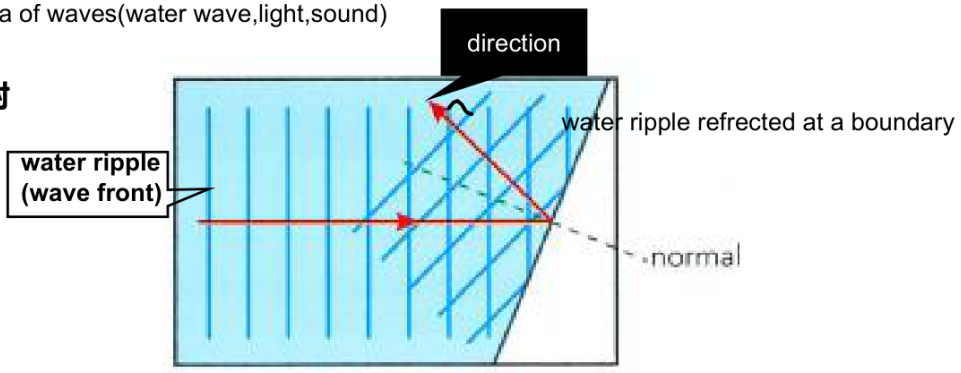






(4) the common phenomena of waves(water wave,light,sound)

➢ _____反射



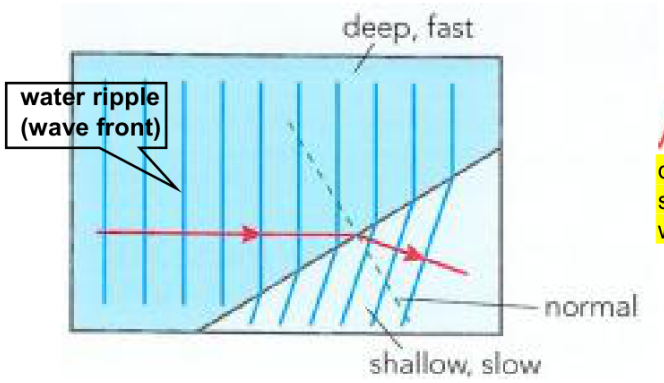
- Waves bounce away from surface at same angle they strike it.
- **the law of reflection反射定律:** _____入射角）= _____反射角

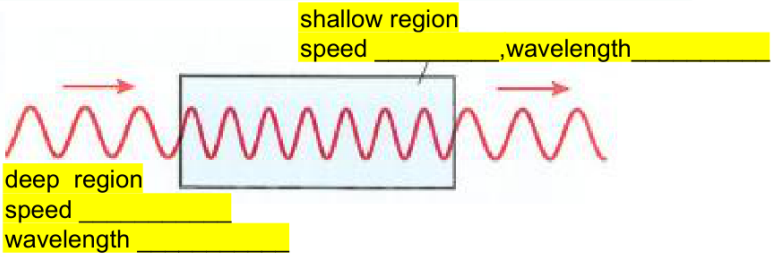
- The plane mirror image像 has these properties:
  - _____
  - _____
  - _____左右颠倒
  - Image is_____虚幻的:

➢ _____折射

_____
_____由于不同介质中的波速不同导致的波的传播方向变化

e.g. water wave move _____慢 in shallow region, but _____快 in deep region,so water wave is refracted when it passing from the deep region to the shallower region.





The speed of light in vacumn(air) is _____, it will _____变慢in some medium(water, glass) .
The more the light is _____, the more it bends _____ the normal line.
the _____ the angle to the normal line.

_____折射率 n = _____ = _____

in medium: n>1
greater n-------> smaller speed---> smaller angle
(means "more refracted=more change in direction")

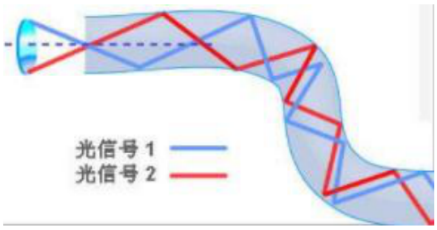
the medium with **greater n (smaller v) is _____光密材料**

➢ _____(TIR)全反射

When light passes from an _____optical dense to a _____dense medium, all the incident light is _____inside the denser medium

Critical angle C（临界角）：_____

$$n = \frac{\sin 90^0}{\sin C} = \frac{1}{\sin C} \text{---} > \sin C = _____$$

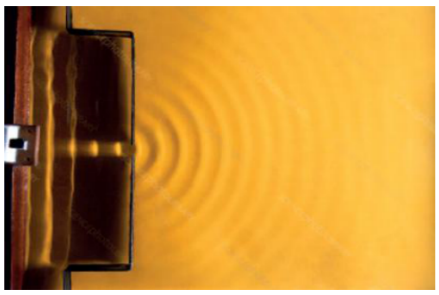


➢ _____衍射

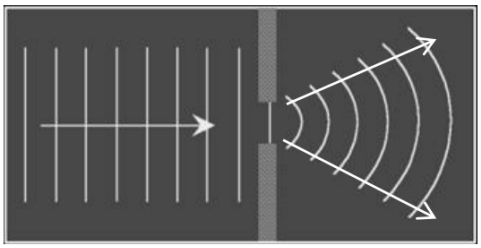
the _____ of waves when they pass through an 狭缝 or round an _____

波长越长衍射现象越容易

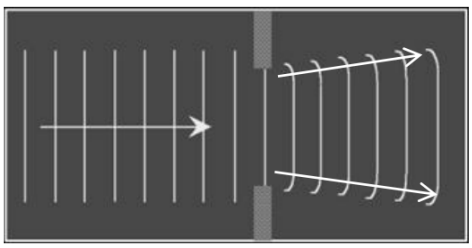
Diffraction water waves in a ripple tank



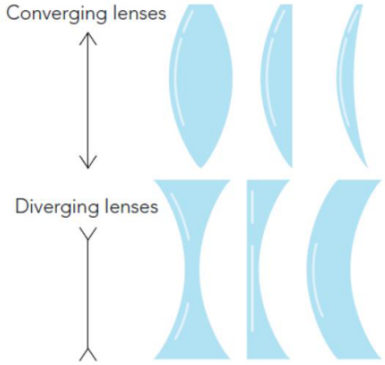
more diffraction: _____



less diffraction: _____



3.2.3 Thin lenses



_____（凸透镜/会聚透镜）

_____凹透镜/发散透镜）

➢ **Physical terms about the lens**

_____焦点）：_____

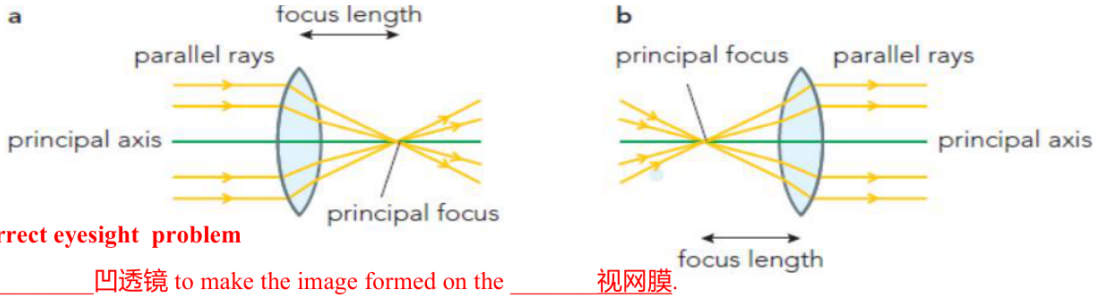
●_____焦距）：_____

●_____主光轴）：line that goes through optical center.

●_____光心）：the center of the lens

●_____实像）：image can be caught on a screen;when object is further away from the optical centre than **one F**

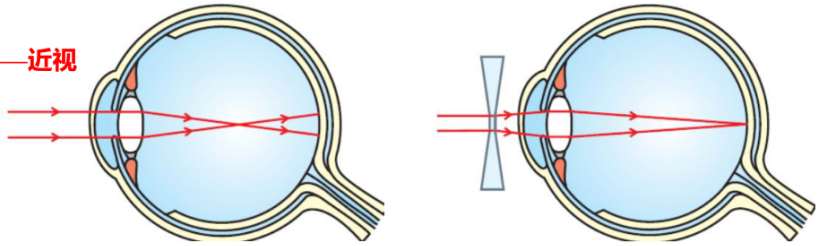
●_____虚像）：image cannot be caught on a screen When the object is closer to the optical centre than **one F**.



➢ **Using lense to correct eyesight problem**

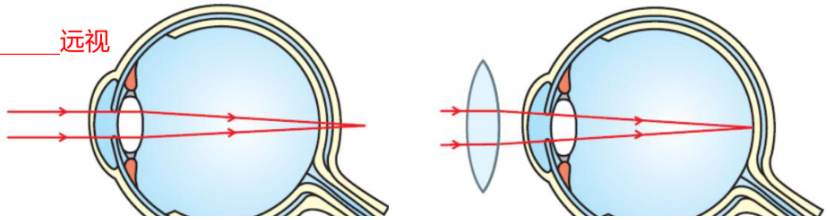
● ---Using _____凹透镜 to make the image formed on the _____视网膜.

_____近视



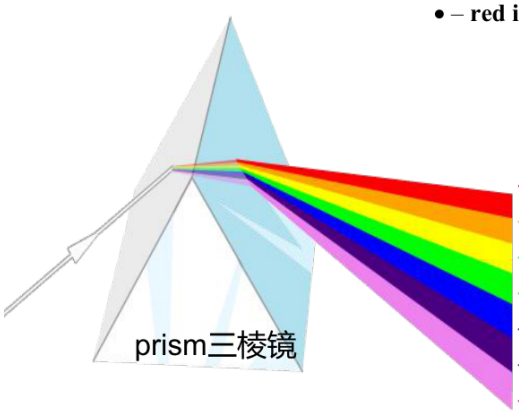
● ---Using_____凸透镜 to make the image formed on the _____视网膜.

_____远视



3.2.4 _____ of light光的色散

- **White light is a mixture of colors**, and the prism refracts each color by a different amount due to different speed of colourful light
- – **red is refracted least,violet most;**



– red-------- wavelength, speed , refracted
– orange
– yellow
– green
– blue
– indigo
– --------shortest , smallest , most

- _____单色光: _____单一频率的光 (used in laser激光)

3.3 _____电磁光谱

fλ=c=_____ in vacumn（真空）;
the greater _____,the smaller _____,

Radiation	Wavelength range / m
无线电波	> _____
微波	_____
红外线	_____
可见光	_____
紫外线	_____
X-光波	_____
γ-光波	_____

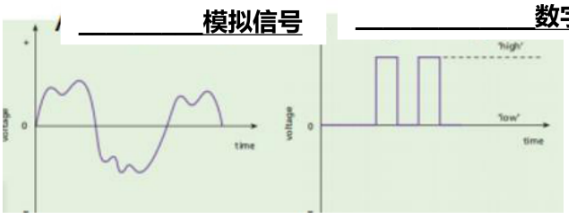
Properties the E.M. waves（电磁波） have in common相同的:
i) they are all transverse wave
ii) they all can travel through vacuum
iii) they have same speed in vacuum
$c=3\times10^8$m/s
iv) they all can transfer energy and information

（1）.typical uses of the different regions of the electromagnetic spectrum including:

- **Radio waves（无线电波）**:_____广播 (TV, AM/FM radio) _____无线电频率识别
- **Micro waves（微波）**: _____手机通信, _____卫星通信, _____无线网络, _____微波炉加热
- **Infrared**: _____遥控器,intruder alarm, _____电烤炉, _____热成像and _____安保系统, _____ 光纤通信
  visible light: vision, photography, illumination
- **UV(ultra violet) rays（紫外光）**: _____**晒黑** and _____**灭菌** , security marking, detecting fake ,bank notes,
- **X-rays（X-射线）** _____医学成像& CT scan,
- **Gamma rays（r-ray）**: _____医疗灭菌,_____癌症治疗

通信信号有两种:

- Analogue uses a whole range of continuous variations to transmit a signal.
- Digital signals use only 2 states, on(high) and off(low).



3.4 Sound

(2) Speed of sound waves:wave speed depends on medium.

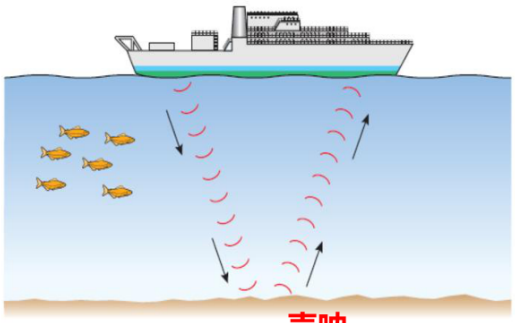
eg:Air : m/s ; Water: m/s ; Concrete : ; Steel: m/s
sound wave move in solid ,then liqiud , in gas.

(4) three properties of sound note:
音调） --increase with frequency
响度） -- incease with amplitude
音色）--determined by wave form
可听到频率）:Humans can hear frequencies between Hz.

(5) 回声: The sound wave is reflected back from the surface.
use: 回声定位.

---Sound with frequency higher than 20kHz, is called 超声波）
---Sound with frequency lower than 20Hz, is called 次声波）

超声波应用



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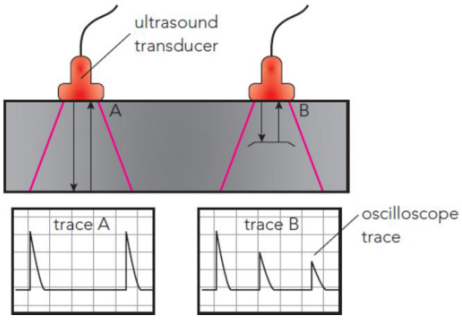


material testing

医学成像（B超）

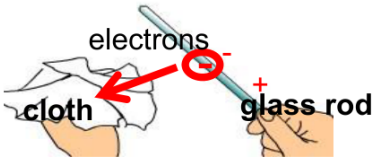
4 Electricity and Magnetism

4.2.1 Electric charge电荷

amount of charge: symble--**Q(q)** ;unit---**C(_____库伦)**

_____(异种电荷吸引 and
_____(同种电荷排斥)

(3) charging a body--the result of **loss or gain in electrons**(movement of electrons).
When you _____摩擦 a glass rod (or bar) with a cloth, electrons from the rod move onto the cloth. So **the cloth _____becomes negatively charged** overall, and the **rod _____ is positively charged**



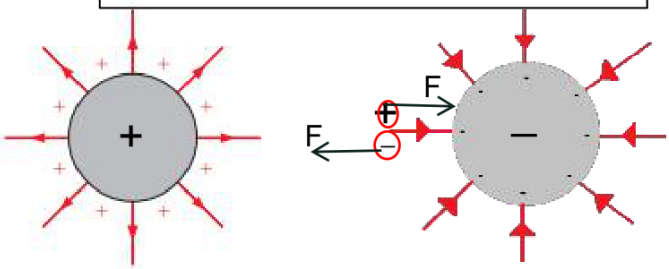
(4) and **conductors**（导体）:

_____绝缘体: A material through which **electrons** _____ , that ***can*** be charged by friction.

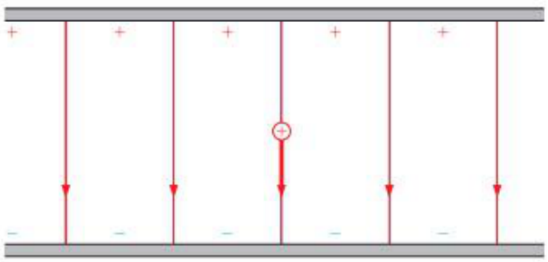
_____**conductors**导体: A material through which electrons_____.

(5)_____**电场):**_____
field lines indicate the **direction of**_____

field lines pattern for _____点电荷



field lines pattern between two charged parallel plates
_____匀强电场



charged conducting sphere=**point charge** concentrated at its centre
inside the conducting sphere,there is no electrical field.

(6) define current:_____ **I=Q/t** **unit:_____安培**

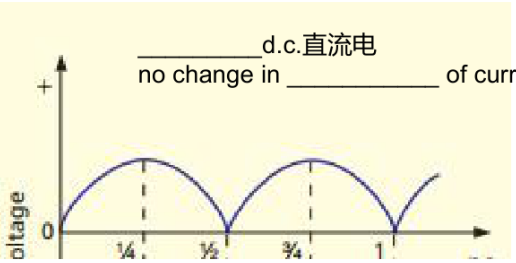
(7) define E.M.F:_____ **unit:_____伏特， 1V=_____**

$$\text{E.m.f(E)} = \frac{\text{W(work done)}}{\text{Q}} = \frac{\Delta\text{E(electrical energy supplied)}}{\text{Q}}$$

(8) define voltage/Potential difference:_____

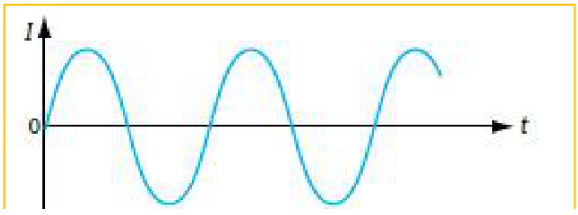
$$\text{V} = \frac{\text{W(work done)}}{\text{Q}} = \frac{\Delta\text{E(electrical energy dissipated)}}{\text{Q}}$$

(9) Resistance $R = \frac{V}{I}$ **unit:_____欧姆**



_____d.c.直流电
no change in _____ of current

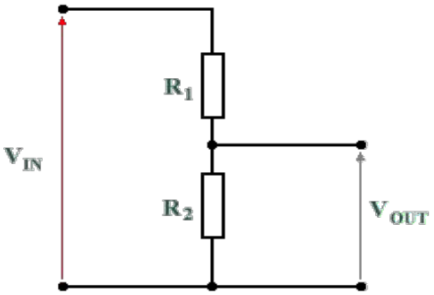
_____a.c.交流电
changed _____



千瓦时1kWh=1000Wx3600s=3.6X10^{6}J
(1h=1hour=3600s)

meaning:energy transferred in _____ at a rate of transfer of _____

use of potential divider circuit **串联分压电路的应用**



$V_1/V_2=R_1/R_2$

$V_{out}= V_{in}R_2/(R_1+R_2)$

V_{out} depends on the ratio of R_1 and R_2

基本关系就是

_____电压和电阻成正比

电压大小取决于电阻R1与R2之比

4.4 Electrical safety

•**Damaged insulation（绝缘层破损）:** contact with the_____**火线** due to gap in the insulation causes _____**触电）** which can cause serious injury or shock.

precausion:

- _____**保险丝）:** A **fuse** protects a circuit. Thin piece of wire which_____It is placed on the _____火线 before the switch.

_____**跳闸开关:** a safety switch that automatically open(trip) electricity from flowing when a current exceeds a certain value.

- **Benefits of _____接地） a Metal Case:**
  - Many electrical appliances, have metal cases, the earth wire creates a safe route for current to flow through, if live wire touches casing Earth terminal connected to metal casing, so the current goes through _____地线instead of causing an _____**触电**.

4 magnetism

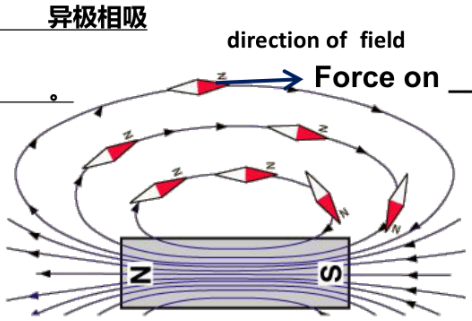
4.1 Simple phenomena of magnetism

4.1 .1 magnetism

(1)**forces between magnets磁铁**

_____**同极相斥 and** _____**异极相吸**

(2) _____**磁场:** _____。



(3) _____**磁性材料（**_____**）** ---such as **i 铁, s 钢, n 镍 and c 钴**

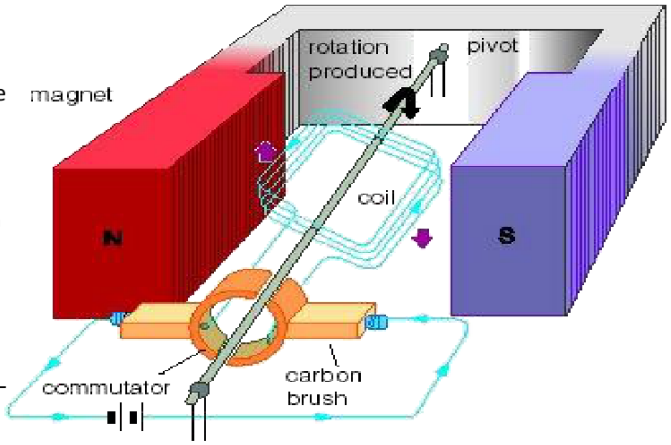
(5) Magnetic Properties of Iron and Steel

- **Iron铁** is _____**软磁铁--- temporary magnetism(**Gets magnetised faster but loses its magnetism as soon as inducing magnet is removed).
  use：_____**变压器铁芯 &**_____**电磁继电器 or** _____**电磁铁**
- **steel钢** is_____**硬磁铁---permanent magnetism(**Slow to be magnetized but retains magnetism for a long time) **use:making magnet**

(6) three ways to increase the strength of an **electromagnets电磁铁 or solenoids螺线管**

- _____
- _____
- _____ .

4.1.3 直流电动机



(1)how a DC motor runs on a direct current.

- When a current-carrying coil is in a magnetic field, magnet exerts force on current
- there are **two** _____ **with** _____**&**_____
  to produce _____

The coil is made of insulated copper wire and is free to _____between the poles of the magnet.

- The _____**开环换向器** is fixed to the coil and rotates with it.
- When the coil overshoots the vertical, **the commutator** ___ _____**, so the forces**_____**and**_____ _____.
- The **brushes are two** _____ **which rub against the commutator and keep the coil** _____ **to battery.**

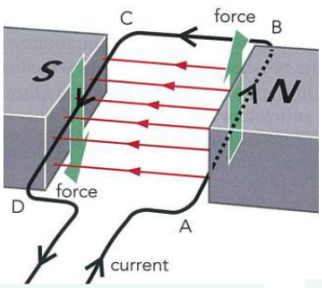
(2)

TURNING EFFECT INCREASED BY:

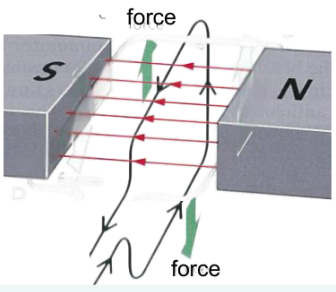
- _____
- _____
- _____

REVERSING ROTATION CAN BE DONE BY:

- _____
- _____



- **moment=**_____
- when the coil plane is_____to magnetic field.



- **moment=**_____

when the coil plane is at_____ to the magnetic field.

4.5 Electromagnetic effects

4.5.1 电磁感应 = _____磁场变化

Show understanding that when a conductor moving across a magnetic field then _____变化的磁场linking with a conductor can_____**感应电动势** in the conductor

(1) Describe an experiment to demonstrate electromagnetic induction

(2)State the factors affecting the **magnitude大小** of an **induced e.m.f感应电动势**.

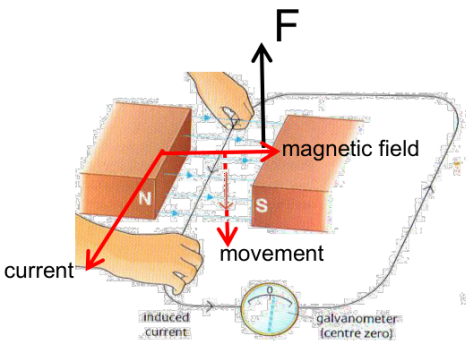
(i) _____

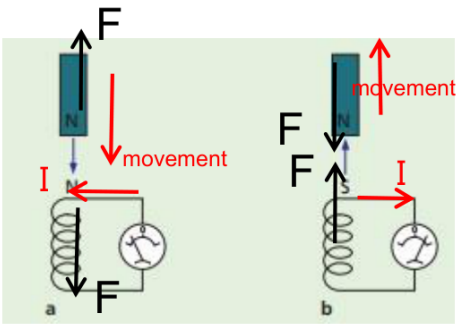
(ii)_____

(iii) _____

(3)Show understanding that the direction of an induced e.m.f (or current)_____

"the force on induced current alway "_____" the _____of magnet or coil





➢ example 1

wire"_____" the magnetic field downward

-------> _____

----------> _____ on induced current upward to _____ the motion downward

➢ example 2

- magnet push towards the coil

  ------> _____

  ------->force on induced current to_____ the magnet moving close

- magnet push away the coil

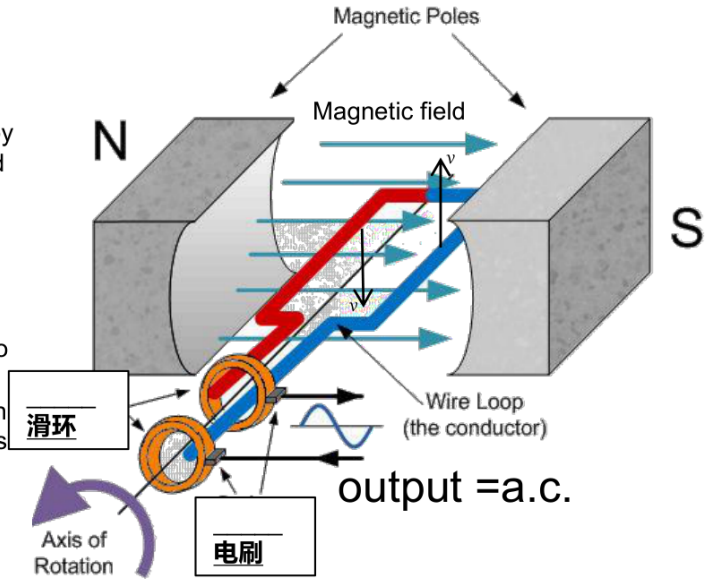
  ------> current induced

  ------->force on induced current to _____ the magnet moving away

4.5.2 The a.c. generator 交流发电机

(1) how to work

- The coil is made of insulated copper wire and is _____by turning the shaft; the _____**滑环** are fixed to the coil and rotate with it.
- The _____**电刷** are two_____which rub against the _____ and keep the coil _____ the outside part of the circuit, usually made of carbon.
- **slip rings used** _____
- When the coil is rotated, it_____magnetic field lines, so an EMF is generated, which makes a current flow.
- Each side of the coil travels upwards then downwards then upwards etc. so the current flows backwards then forwards then backwards etc. so it is an alternating current.



(2)current change with time

- **induced current I = _____**
  when the coil plane is _____ to magnetic field,

- **induced current I=_____**
  when the coil plane is _____垂直 to the field, since it is cutting NO field lines.

(3)**The induecd EMF can be increased by:**

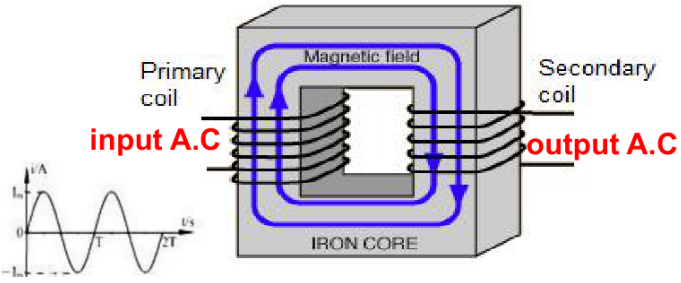
- o _____
- o _____
- o _____

4.5.3 _____变压器

(1) how to work:

.

- _____
  _____
- _____
  _____
- _____
  _____.



- _____**升压变压器)** increases the voltage
- _____**降压变压器)** decreases voltage. for ideal transformer (NO energy loss)

_____ $=I_S/I_P$

input power= output power

. $I_p V_p = I_s V_s$

the higherV ， the smaller I

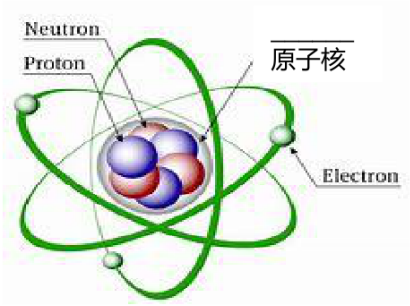
(2)the advantages of **high-voltage transmission高压输电**--- _____when the transmitted _____

explain: transmitted输送 current I=_____ **power lost in a resistor** P = _____

having a lower current will the power loss, so voltage, current get

IG physics Nuclear physics key Summary

1 .The nuclear model of the atom原子的核式结构



_____ 原子核

both_____质子 and_____中子are called _____**核子**

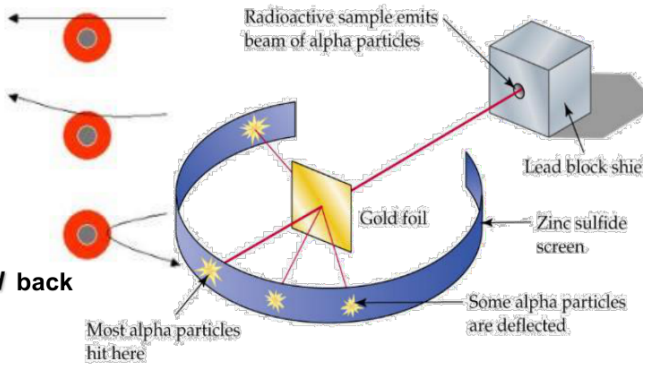
2.Conclusion of _____阿尔法粒子散射实验

- **Most *α particles are not deflected***

Show that：_____

A few *α particles are deflected*偏转 and
Extremely few (less than one in million) *α particles are deflected* back by nearly 180°

Show that：_____



3._____同位素: _____

4 _____核裂变：_____

- **_____核聚变：_____**

for all nuclear reaction, _____**and**_____ are conserved.

_____**核能** released in the form of **γ ray**
due to there is total mass lost after reaction,
the total _____质能守恒.

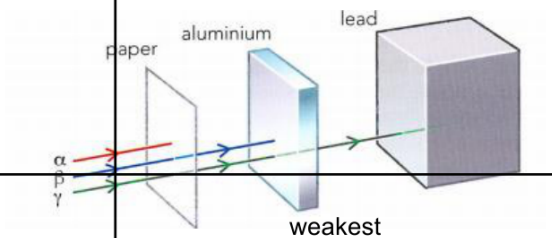
$^{1}_{1}H + {}^{2}_{1}H \rightarrow {}^{3}_{2}He$ + **nuclear energy（核能）γ**

total mass $^{1}_{1}H + {}^{2}_{1}H$ > total mass $^{3}_{2}He$

5._____背景辐射: _____

. It mainly comes from natural sources such (a) _____ (b) _____ (d)_____

- **characteristics（特点） of three kinds of radioactive放射性 emission射线**

	Alpha emission α 射线	**Beta β射线**	**Gamma γ ray γ 射线**
Nature	**Helium nucleus $_2$ He**	**electron**	**Electromagnetic Radiation**
Charge	**+ 2e**	**- e**	**No charge**
Mass	**4u** 4	**1/1840 u**	**Massless**
_____穿透性	run a few centimetres in air stopped by a paper	run a few metres in air stopped by a thin aluminum	run untill stopped by a thick lead 
_____电离	most stongly (due to most +charge)	weak	weakest
deflection in magnetic and electric field电磁场中的偏转	deflected	deflected more (opposite to α)	no deflection
use	_____烟雾报警器利用电离性_____	**measure thickness due to the _____穿透性**	**(1)_____消毒;** **r(2)adiotherapy –γ rays kill_____** **(3)detect faults in manufactured goods**

6._____放射性衰变

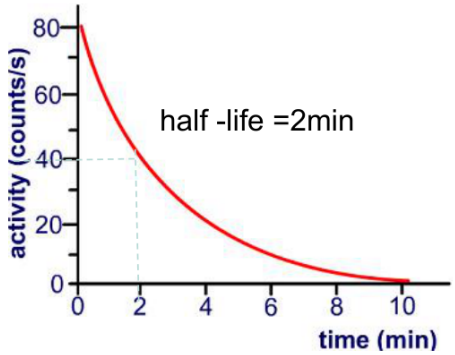
α-decay: equation: ${}^{A}_{Z}X \rightarrow$ ____ $+ \; {}^{4}_{2}\alpha$

β-decay: equation: ${}^{A}_{Z}X \rightarrow$ ____ $+ \; {}^{0}_{-1}e^{-}$

the electron come from inside the nucleus,one neutron has split into a proton and electron

${}^{1}_{0}n \rightarrow$ _____ + _____

7. _____半衰期:_____



IG-Space physics key summary

1 Earth,Sun and Moon

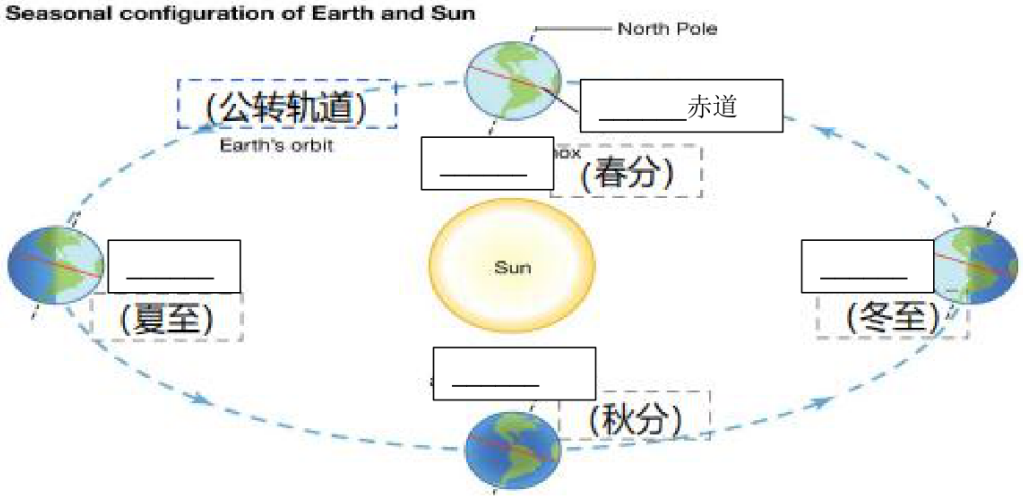
(1）.Explanation of day and night:

The Earth _____转动 on its _____地轴 ,It takes _____hours for Earth to complete one rotation.As the Earth spins on its axis,part of the Earth that faces the Sun has _____白天. And the part that faces a way from the Sun has ___黑夜

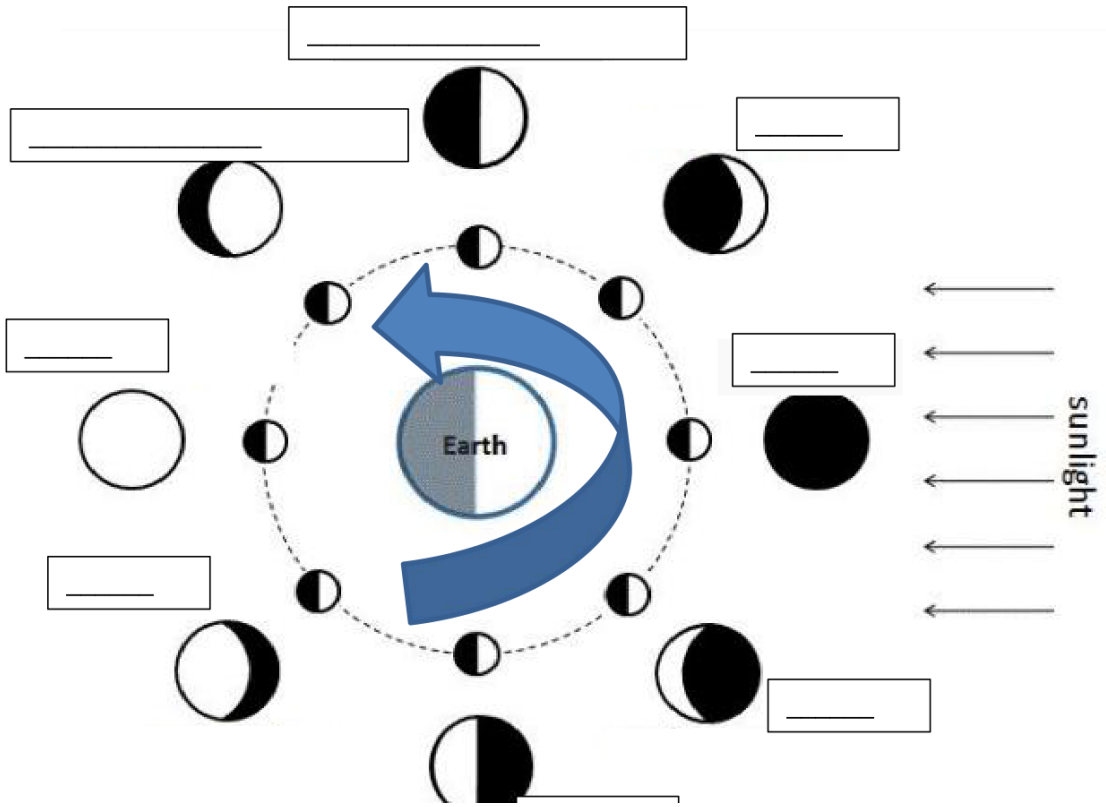
（2）.Explanation of the periodic nature of the seasons.

The Earth _____坏绕 the sun once_____一年. While it orbits the sun, it causes each _____半球 to either face the sun or face away from it.The_____半球 facing the sun has _____ and the _____ pointing _____it has winter.

• The _____地轴倾斜**causes seasons**.



（3)It takes approximately one month for the Moon to orbit the Earth and this is related to the periodic nature of the _____月相周期=27.5day



IG-Space physics key summary

(4). The Solar System :

- one _____恒星： the Sun.
- the eight planets : 轨道半径越大，公转周期越_____,转速越_____(判断变大或者变小）

four _____岩质内行星(体积小，密度大)：_____（水星）_____（金星体表温度最高）, Earth, _____（火星）

four _____气态外行星（体积大，密度小）:_____(木星）, _____（土星卫星最多）, _____天王星 and _____海王星.

- minor planets that orbit the Sun,an object which orbits the Sun but is not large enough or far enough from another object,that include the_____矮行星 such as _____冥王星.

●moons:that orbit the planets. is the_____天然卫星

●Smaller Solar System bodies, including

_____小行星 in the_____小行星带 and_____流星体: rocks that are smaller than the planet which orbit the Sun

_____彗星: aballofice, dust and gas which orbits the Sun in a highly _____ 椭圆轨道.

_____吸积盘: a rotating disc of matter formed by_____沉积物: ____ and _____pulled together by gravity

(5) **.Gravitational field(引力场)：**

. The force that keeps an object in orbit around the Sun is the _____万有引力of the Sun.

- the strength of the gravitational field depends on the _____ and _____

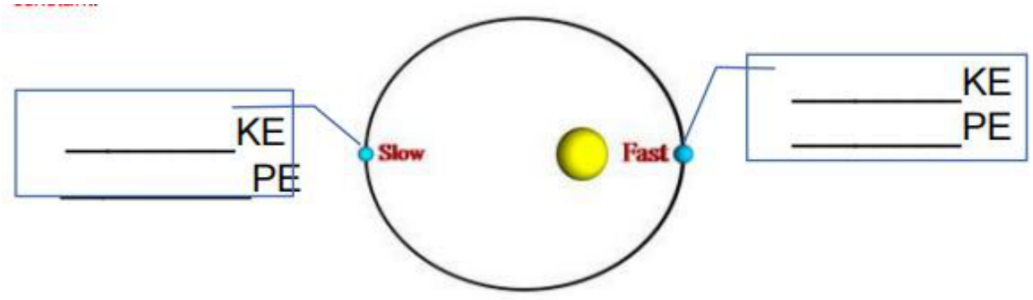
The strength of the Sun's gravitational field _____ and that the orbital speeds of the planets _____as the distance from the Sun _____(判断递增还是递减关系）.

理想圆轨道：orbital speed v = 2πr/T

r : _____环绕半径 T : _____环绕周期

实际_____椭圆轨道: moving fastest at closest to the Sun ,距离越大，速度越小

总能量total energy =KE+PE 不变



IG-Space physics key summary

Stars and the Universe

- The Sun is a star in the_____**星系** known as the _____**银河系.**
- TThe Sun is a star of medium size, consisting mostly of **hydrogen**(_____%) and **helium**(_____%).
  the sun radiates most of its energy in the_____(50%), _____(40%)and _____(10%)
- Stable stars like our sun are powered by the _____**核聚变**. This makes the Sun shine.
- _____光年：_____

Describe the life cycle of a star:恒星的生命周期

(a) a star is formed from_____星际 clouds of _____ and _____ that contain **hydrogen**

(b) a _____原恒星is on interstellar cloud collapsing and increasing in temperature as a result of its internal _____万有引力。

(c) a _____原恒星 becomes a _____ when the _____ is balanced by an _____force due to the high temperature in the centre of the star

(d) all stars eventually run out of _____ as f_____ for the nuclear reaction

根据恒星质量不同演化两条路径

- For more massive stars

more massive stars expand to form _____红超巨星 when most of the hydrogen in the centre of the star has been corverted to helium，then explodes as a _____超新星, forming a _____星云 containing hydrogen and new heavier elements,leaving behind a **neutron star**（中子星）or a **black hole**（黑洞）

- For less massive star(such as SUN)

a ____红巨星from a less massive star forms a _____行星状星云 with a _____白矮星at its centre.

表格

英文术语	中文译名	关键考点
Nebula	星云	恒星诞生的摇篮，主要成分氢和氦
Protostar	原恒星	引力坍缩阶段，未启动核聚变
Main-sequence star	主序星	氢聚变稳定进行，恒星最长的阶段
Red giant/Red supergiant	红巨星 / 红超巨星	氢耗尽后膨胀的阶段，由质量决定类型
Planetary nebula	行星状星云	中小质量恒星抛射的外层气体
White dwarf	白矮星	中小质量恒星的核心残骸，电子简并支撑
Supernova	超新星	大质量恒星的剧烈爆炸，抛射重元素
Neutron star	中子星	大质量恒星核心残骸，中子简并支撑
Black hole	黑洞	超大质量恒星核心坍缩的产物，引力极强

Stable star

IG-Space physics key summary

The universe

- _____红移 as an _____in the observed _____ of electromagnetic radiation emitted from _____远离的stars and galaxies.

- The _____红移 in the light from distant galaxies is evidence that the universe is _____膨胀and supports the_____（宇宙大爆炸理论）

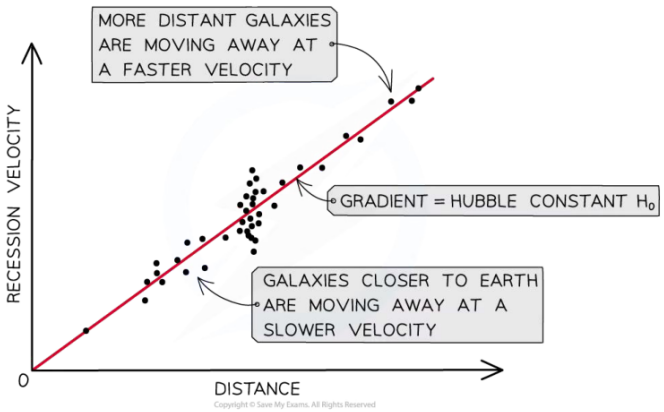
- The _____ of a specific frequeny is observed at all points in space Around us and is known as _____(宇宙微波背景辐射)(CMBR)

The CMBR was produced shortly after the _____ and that this radiation has been expanded into the_____ region of the electromognetic spectrum as the Universe _____. And that this is evidence for the idea that all the matter in the Universe was present at a _____.

- Hubble's law shows that _____

$$H_0 = \frac{v}{d}$$

v: _____
d: _____
H_0:_____



The equation $\frac{1}{H_0} = \frac{d}{v}$ represents an estimate for the _____ of the Universe.